

Review

Atomic Layer Deposition of Two-Dimensional Layered Materials: Processes, Growth Mechanisms, and Characteristics

Jiyu Cai,^{1,3} Xiaoxiao Han,^{1,2,3} Xin Wang,^{1,2,3} and Xiangbo Meng^{1,2,*}

Since the discovery of graphene, there has been an ever-increasing interest in two-dimensional (2D) layered materials with exceptional properties. To this end, a variety of synthesis methods have been developed. However, it is still challenging to produce large-scale high-quality single-crystalline 2D materials. In this regard, atomic layer deposition (ALD) has recently shown great promise and has stimulated more and more research efforts, ascribed to its unique growth mechanism and distinguished capabilities to achieve nanoscale films with excellent uniformity, unrivaled conformality, and atomic-scale controllability. This review comprehensively summarizes recent progress on ALD for 2D atomic sheets, including 25 different materials and more than 80 ALD processes. This work highlights different technical routes to ALD, their precise controllability, and their underlying principles for 2D materials. It is expected that this work will help boost more research efforts for controllable growth of high-quality 2D materials via ALD.

INTRODUCTION

Two-dimensional (2D) layered materials are a single layer or multi-layer of atomically stacked structures, featuring van der Waals interaction between adjacent layers but strong covalent bonding within each layer.^{1–4} Compared with their bulk counterparts, 2D materials exhibit many exotic physical, chemical, and electronic properties,^{5,6} which thereby provide unprecedented opportunities for many emerging applications, such as optical and electronic devices,^{7–9} new energy systems,^{10,11} catalysis,¹² and bioengineering.¹³ In the 20th century, the scientific community was uncertain about the existence of 2D materials and commonly believed that films become thermodynamically unstable with decreased layers.¹⁴ It was not until 2004 when a monolayer graphene sheet was reported for the first time by Novoselov et al.¹⁵ using a mechanical exfoliation method. This milestone work instantly ignited tremendous research interest, and nowadays, a large variety of 2D materials have been reported, including elements (e.g., graphene, silicene, and germanane),^{1,16} nitrides (e.g., h-BN and GaN),^{17,18} oxides (e.g., MoO₃),¹⁹ chalcogenides (e.g., WS₂, MoS₂, and SnS_x),^{3,6} MXene (e.g., Ti₃C₂ and Ti₄N₃),¹¹ layered organic materials,^{20,21} and others.

Beyond the mechanical cleavage of graphene, many new methods have been explored for developing 2D layered materials, such as exfoliation, physical and chemical vapor deposition (PVD and CVD), and solution-based processes, and are well documented in the literature.^{5,22–24} However, an ideal method capable of producing large-scale high-quality 2D materials in a controllable manner is still lacking. In this regard, atomic layer deposition (ALD) has recently emerged as a new synthesis technique, featuring its unique growth mechanism and exceptional characteristics. A variety of 2D materials

Progress and Potential

Atomic two-dimensional (2D) layered materials are a class of advanced materials with numerous exotic properties. There has been new progress made in 2D materials since single-layer graphene was exfoliated with Scotch tape in 2004. Over 150 types of 2D materials have been developed to date by various methods. However, it is still challenging to produce large-scale high-quality 2D layered structures in a controllable manner. In this regard, recent studies on atomic layer deposition (ALD) have shown many encouraging advances and enabled three technical routes for precisely controllable synthesis of 2D materials at the atomic scale. In this review, we provide a comprehensive survey on ALD efforts and highlight the underlying principles and growth characteristics of ALD processes. The findings from these studies have the potential to provide guidance for designing new ALD processes to produce better 2D materials.



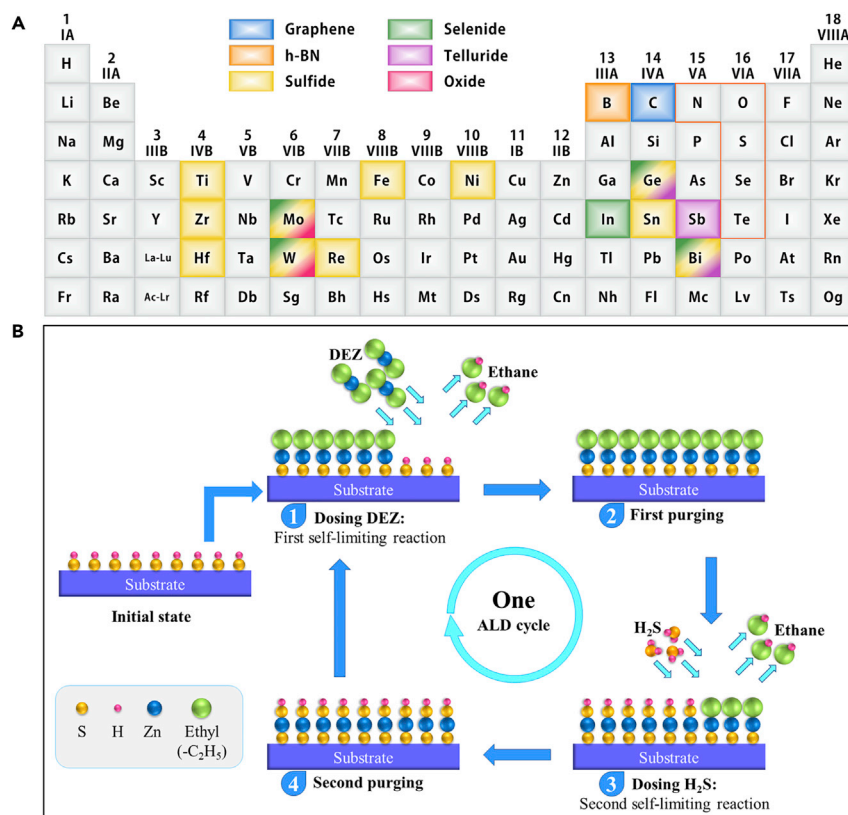


Figure 1. Summary of 2D Materials by ALD and Illustration of ALD Growth Characteristics using ALD ZnS

(A) Schematic summary of 25 different types of 2D layered materials synthesized by ALD, including graphene, h-BN, metal oxides, metal sulfides, metal selenides, and metal tellurides.

(B) Schematic illustration of ALD surface chemistry with a model ALD process of ZnS using DEZ and H_2S as precursors. The surface undergoes two gas-solid surface reactions starting from a surface covered by $-\text{SH}$ ligands: (1) a supply of DEZ initiates the first self-limiting gas-solid reaction, leading to a new layer of $-\text{ZnC}_2\text{H}_5$; (2) a purging procedure is applied to remove the overdosed DEZ and the by-product of ethane; (3) a supply of H_2S prompts the second self-limiting gas-solid reaction, leading to coverage of $-\text{SH}$ ligands; (4) a purging procedure is applied to remove overdosed H_2S and the by-product of ethane. ZnS ALD can proceed with increasing cycles by repeating the four steps, consisting of two gas-solid surface reactions and two purges.

have been reported to date using ALD. In contrast to traditional methods, ALD enables atomically precise controllability over the growth of 2D materials and proceeds at low temperatures. Thus, ALD is a promising tool to achieve high-quality 2D materials in a controllable manner. To this end, we have been motivated to conduct this survey to summarize the new progress on ALD for 2D materials. To provide readers with a comprehensive overview, we have included 25 different 2D layered materials and over 80 ALD processes in this review, including graphene, h-BN, 21 chalcogenides (12 sulfides: MoS_2 , WS_2 , TiS_2 , ZrS_2 , HfS_2 , ReS_2 , SnS_2 , FeS , NiS , GeS , SnS , and Bi_2S_3 ; 6 selenides: MoSe_2 , WSe_2 , InSe , In_2Se_3 , Bi_2Se_3 , and GeSe ; and 3 tellurides: GeTe , Sb_2Te_3 , and Bi_2Te_3), and 2 oxides (MoO_3 and WO_3), as summarized in Figure 1A. Most of these studies have been reported in the past 5 years. We have striven to clarify the underlying growth mechanisms and outline the growth characteristics of these ALD processes. Following this introductory section, we briefly discuss the fundamentals of ALD and then summarize new growth characteristics of 2D materials. In the third section, we provide details of the ALD processes, categorized into six main classes: graphene, h-BN, transition metal

¹Department of Mechanical Engineering, University of Arkansas, Fayetteville, AR 72701, USA

²Center for Advanced Surface Engineering, University of Arkansas, Fayetteville, AR 72701, USA

³These authors contributed equally

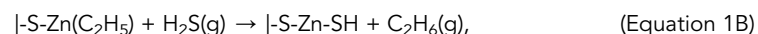
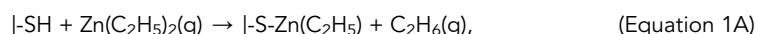
*Correspondence: xbmeng@uark.edu

<https://doi.org/10.1016/j.matt.2019.12.026>

dichalcogenides (TMDs), transition metal monochalcogenides, group IIIA-VA metal chalcogenides, and metal oxides. In the fourth section, we address various ALD strategies and their growth characteristics for 2D materials. We then conclude this work with some discussion on the efforts to date and the outlook for future studies.

ALD FUNDAMENTALS AND STRATEGIES FOR 2D MATERIALS

ALD is a well-recognized vapor-phase thin-film technique, credited to Suntola et al.²⁵ ALD so far has evolved from its initial implication, and its capabilities have been greatly widened for growing a large range of inorganic materials, including elements, oxides, nitrides, sulfides, and ternary and even more complicated compounds.^{26,27} As a surface-controlled process, ALD grows films with atomic-scale precision. This is determined by its unique growth mechanism proceeding with self-limiting surface reactions on substrate surfaces. As a result, ALD can produce films with extremely high uniformity, unrivaled conformality, excellent controllability, and at low temperature.²⁸ To illustrate the ALD growth mechanism, we exemplify the process of ZnS using diethylzinc (DEZ, $\text{Zn}(\text{C}_2\text{H}_5)_2$) and hydrogen sulfide (H_2S) as precursors (Figure 1B),²⁹ in which two half reactions (Equations 1A and 1B) proceed alternatively. The two half reactions are as follows:



where “|” denotes the surface species and “g” signifies the gaseous phase. Starting from a surface covered by –SH ligands in its initial state, in the ALD process for ZnS, a supply of DEZ results in the first half self-limiting gas-solid surface reaction (Equation 1A) between the –SH ligands and DEZ (step 1). During this step, the ligand-exchange reaction generates a new layer of –Zn(C_2H_5) species, and releases ethane (C_2H_6) as the by-product. Once all –SH ligands are depleted, the surface reaction (Equation 1A) completes. In other words, the surface reaction is self-limiting. Then, a purging process (step 2) is applied to clean the overdosed DEZ and the by-product C_2H_6 . Following a thorough purge, a supply of H_2S prompts another half self-limiting gas-solid surface reaction (Equation 1B), leading to a new covering of –SH on the surface while releasing C_2H_6 (step 3). Again, a purging procedure is applied after the second ligand-exchange reaction in order to remove the overdosed H_2S and the by-product C_2H_6 (step 4). Apparently, the two self-limiting half surface reactions are surface controlled by the amount of the surface reactive sites (i.e., –SH and – C_2H_5). By repeating steps 1–4, the ZnS film grows with increasing ALD cycles. In general, three ALD parameters are adjustable to accomplish desirable materials, i.e., precursor, temperature, and substrate. These fundamentals have been well documented in the literature, and readers may refer to some excellent reviews^{26–28,30–32} for more details. In addition to the typical thermal ALD process as illustrated in Figure 1B, there are also some special ALD systems, such as plasma-enhanced ALD (PEALD),³³ particle ALD,³⁴ spatial ALD,³⁵ and others.

In developing 2D materials, some new growth characteristics and strategies have been uncovered in recent ALD studies. Here, we categorize them into three distinctive ALD routes, as illustrated in Figure 2. The first route (R1) (Figure 2A) is the conventional ALD process, in which the net adsorption force ($F_{\text{adsorption}}$) between precursors and their substrate surfaces is always positive with the increasing film thickness ($H_{\text{thickness}}$), enabling continuous growth vertically in a layer-by-layer manner. In this case, the lateral growth is relatively slow. The resultant films are

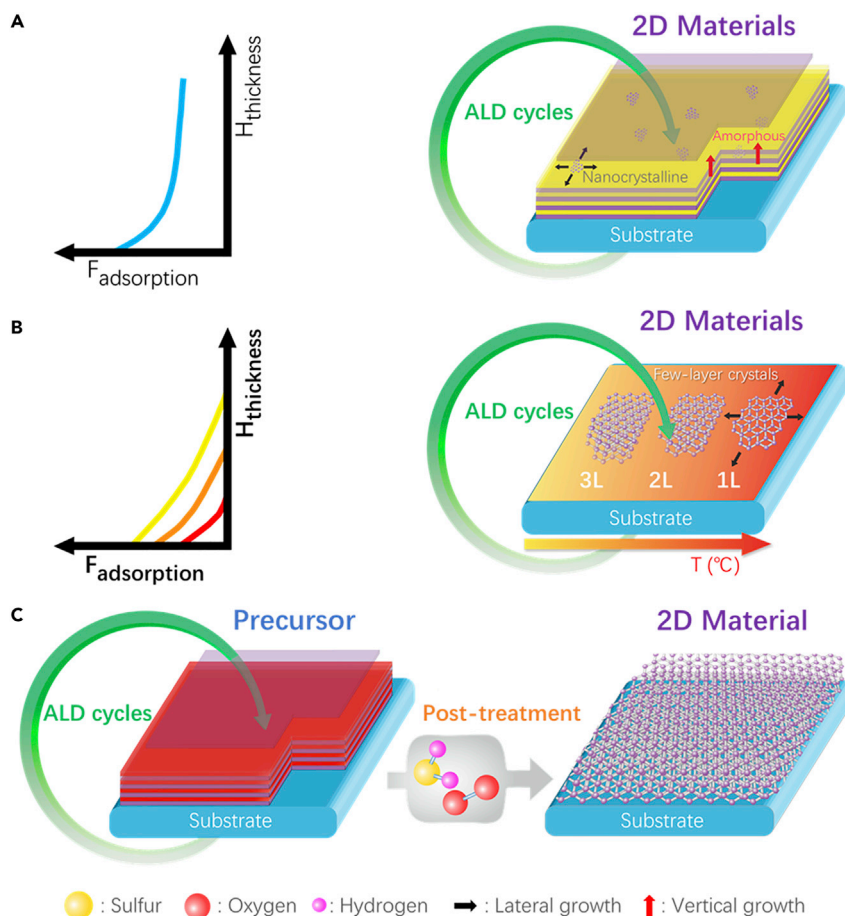


Figure 2. Schematic Illustrations of Three Distinctive Strategies for 2D Materials via ALD

(A) Route 1 (R1): direct growth of amorphous or polycrystalline 2D materials. (B) Route 2 (R2): self-limiting layer synthesis (SLS) process of few-layer crystalline 2D materials. (C) Route 3 (R3): two-step processes for growing crystalline 2D materials, in which a precursor film is first controllably deposited via ALD and then converted into the targeted 2D films via a post-treatment at high temperatures. Yellow, sulfur (S); red, oxygen (O); pink, hydrogen (H).

amorphous, polycrystalline, or a mixture of the two. This R1 route has been seen in most ALD processes of 2D materials. The second route (R2) (Figure 2B) is a self-limiting layer synthesis (SLS) process, in which the number of layers is determined by the temperature rather than ALD cycles.^{36,37} This route has only been used to date for two limited cases: MoS₂³⁶ and WSe₂.³⁷ During the growth of the first layer, precursor molecules chemically adsorb to surface reactive sites on substrates. Once the initial layer is formed over the entire surface, any further chemical adsorption of precursor molecules is hindered due to the absence of suitable adsorption sites on the newly formed 2D atomic layer. Thus, synthesis is forced to proceed with the physical adsorption of precursor molecules. The physical adsorption competes with physical desorption. The net adsorption decreases with increased layers and becomes zero at a critical number of atomic layers, where both the chemical and physical adsorption are disabled. The critical number reduces with increased temperature. This route generally needs high temperatures (e.g., >500°C, depending on the properties of both precursor and substrate) to enable the self-limiting growth of mono- or few-layer crystalline 2D materials. This strategy has three benefits: (1) the precisely controlled layer number (i.e., mono- to trilayer); (2) highly crystalline

2D materials; and (3) 10-cm wafer-scale uniformity. More details are discussed in the following sections. The third route (R3) (Figure 2C) is generally a two-step strategy, in which a precursor film is first developed via ALD at low temperatures and converted into the targeted 2D material via a post-treatment at relatively higher temperatures. For example, metal oxides are first deposited via ALD and then converted into 2D sulfides/selenides via sulfurization/selenization, such as MoS_2 ,^{38,39} MoSe_2 ,⁴⁰ WS_2 ,^{41,42} and Bi_2S_3 .⁴³ Another example is the production of 2D SnS , in which SnS_2 is deposited first via ALD and then reduced in H_2 into 2D SnS at high temperature.⁴⁴ This R3 route has also been used for h-BN, where a polymer was first deposited via ALD at low temperature and then converted into h-BN in NH_3 at high temperature.^{45,46} This route has multiple benefits: (1) ease of generalization for more 2D materials; (2) feasible control over the number of layers in the 2D materials; and (3) wafer-sized uniformity. All these growth strategies of ALD offer tremendous opportunities for growing large-scale 2D materials with high quality and high controllability, which are discussed in the following sections in detail.

ALD PROCESSES AND GROWTH CHARACTERISTICS OF 2D MATERIALS

Graphene

Graphene is an sp^2 -hybridized carbon allotrope with a 2D honeycomb network.¹⁶ So far, it has been fabricated using many approaches, such as exfoliation, CVD, and PVD.^{47,48} Zhang et al.⁴⁹ deposited graphene on Cu foils via PEALD at 400°C , using benzene (C_6H_6) and plasma H_2 (PH_2) as precursors, as summarized in Table 1. The six-carbon atomic rings of benzene might have been connected to form graphene via a dehydrogenation process. In the PEALD process, deposition of less than ten ALD cycles resulted in inhomogeneous graphene monolayers, whereas deposition of more than ten cycles led to multilayered graphene sheets. High-resolution transmission electron microscopy (HR-TEM) has been used to observe the as-deposited films with ten ALD cycles, showing a well-ordered graphene structure (Figure 3A). Raman analyses on the films also revealed a G band at $1,585\text{ cm}^{-1}$ and a 2D band at $2,687\text{ cm}^{-1}$ (Figure 3B). The large 2D/G ratio of ~ 1.7 indicated the high quality of the resulting graphene sheets. Moreover, a single-layered graphene within ten ALD cycles has also been determined by selected-area electron diffraction (SAED). However, the single-layered graphene from this work only has a grain size of $<10\text{ nm}$.

h-BN

Boron nitride (BN) can exist in several polymorphs, such as graphene-like hexagonal (h-BN), diamond-like cubic (c-BN), rhombohedral (r-BN), and wurtzite (w-BN) crystals.⁷⁴ h-BN is an analog of graphene, in which B and N atoms are arranged alternately in a 2D plane, and van der Waals forces hold the h-BN layers together. h-BN has many excellent properties, such as wide band gap, high stability, and excellent thermal conductivity.¹⁸ Thus, h-BN films are promising in a wide range of applications¹⁸ and have been synthesized by various methods.¹⁸

The first ALD effort for BN was reported by Marlid et al.⁵⁰ in 2002 in a process using BBr_3 and NH_3 as precursors. Since then, seven other ALD processes for BN have been developed, as summarized in Table 1. In the first study, the ALD BN was deposited nearly stoichiometrically at both 400°C and 750°C .⁵⁰ X-ray diffraction (XRD) analyses revealed that the BN films were less ordered at 400°C but had improved crystallinity at 750°C , showing a turbostratic BN structure (t-BN). This BBr_3/NH_3 process of ALD BN has been practiced on various substrates at 750°C .^{52–57} Aimed at improving this ALD process at a lower temperature and with a higher growth rate, a subsequent study used a laser to assist the ALD in a wider range of 250°C – 750°C .⁵¹ The laser helped create more reactive fragments, and

Table 1. Summary of Graphene and BN Films Synthesized via Various ALD Processes

Material	Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	GPC (Å/Cycle)	Substrates	Crystallinity ^a	Reference
	Formula	Temperature (°C)						
Graphene	C ₆ H ₆	RT ^b	P ₂ H ₂	400, R1	–	Cu	h	Zhang et al. ⁴⁹
BN	BBr ₃	RT	NH ₃	400, 700, R1	~0.45	silica	t	Mårlid et al. ⁵⁰
				250–750, R1	0.45–1.2	SiO ₂	t	Olander et al. ⁵¹
				750, R1	0.8	silica, AAO, ZnO, Si ₃ N ₄ , carbon fibers	t	Weber et al. ^{52–57}
	BCl ₃	RT	NH ₃	~227, R1	~0.96	ZrO ₂	a	Ferguson et al. ⁵⁸ , Weimer et al. ⁵⁹
				~277, R1	–	Ru(0001)	h ^c	Bjelkevig et al. ⁶⁰
				~277, R1	–	Co(0001)	h ^c	Beatty et al. ⁶¹ , Driver et al. ⁶²
				~327, R1	~0.7	RuO ₂ (110)	h ^c	Jones et al. ⁶³
				~230, R1	–	Al	–	Kilbury et al. ⁶⁴
				630, R1	0.42	Si(100)	–	Consiglio et al. ⁶⁵
				~327, R1	1.65	Ni(111)	h	Jones et al. ⁶⁶
				~327, R1	~1.4	B ₂ O ₃	–	Pilli et al. ⁶⁷
			N ₂ H ₄	350, R1	~0.8	SiGe, Cu, HOPG	a	Wolf et al. ⁶⁸
	TEB	RT	NH ₃	150–350, R1	–	Zn	–	Kilbury et al. ⁶⁴
				600–900, R1	0.7	sapphire or Si	a (600°C), h (900°C) ^d	Snure et al. ⁶⁹
			P ₂ N ₂ /P ₂ H ₂	250–350, R1	0.15–0.47	Si, quartz	h	Haider et al. ⁷⁰
	TEMAB	RT	P ₂ NH ₃	250–350, R1	1.1	Si	h	Park et al. ⁷¹
	B ₃ N ₃ H ₆	RT	e [–] e ⁺	27, 100, R1	~3.2	Si	t and h	Sprenger et al. ⁷²
	B ₂ H ₆ ^f	RT	NH ₃	300–400, R1	–	Si	–	Kovalgin et al. ⁷³
	TCB	70°C	HDMS	80, R3	1.8	Si	h ^g	Hao et al. ^{45,46}

^aFor crystallinity, a, t, and h stand for amorphous, turbostratic, and hexagonal, respectively.

^bRT indicates room temperature.

^cAnnealing in a vacuum at 727°C,⁶⁰ 427°C,^{61,62} and 527°C,⁶³ respectively.

^dThe films grown at 900°C belongs to the precursor-dependent characteristics of CVD growth, instead of ALD growth.

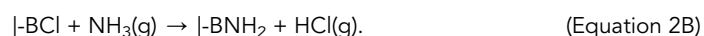
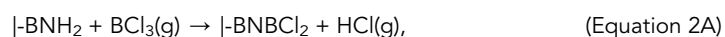
^eThis deposition process applied electron-enhanced atomic layer deposition (EEALD).

^fThis is the hot wire ALD (HWAALD).

^gPost-treatments first in Ar at 600°C and then in NH₃ at 1,400°C.

this laser-assisted ALD (LALD) process resulted in higher growth per cycle (GPC), two times higher than that in thermal ALD. At a temperature higher than 750°C, however, the GPC for both the LALD and conventional ALD processes decreased, probably due to thermal desorption of precursors. The as-deposited BN films via LALD were still a t-BN structure but had improved crystallinity with larger grain sizes.

Among the other ALD processes, Ferguson et al.⁵⁸ used BCl₃ and NH₃ to deposit BN at 227°C, achieving a GPC of 1 Å/cycle. Based on their investigation using *in situ* Fourier transform infrared (FTIR) spectroscopy, Ferguson et al.⁵⁸ clarified the growth mechanism proceeding with two-half reactions (Equations 2A and 2B).



TEM analyses revealed that the as-deposited and post-treated (~727°C in a vacuum) BN films on ZrO₂ substrates (see Figure 3C) were amorphous. Interestingly,

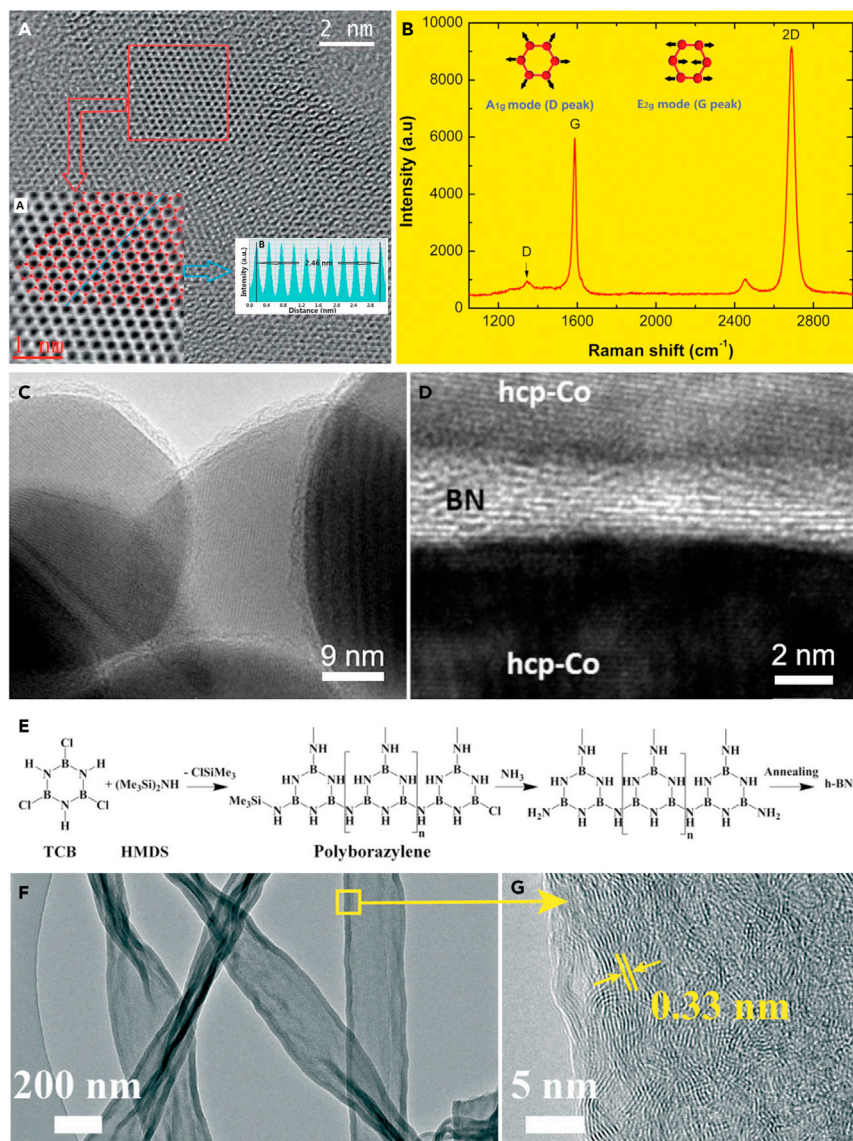


Figure 3. 2D Layered Graphene and h-BN Synthesized by ALD

(A) High-resolution TEM images of graphene deposited by ten ALD cycles.⁴⁹ Insets: (A) the area within the red rectangular frame; (B) the pixel intensity profile of the graphene along the blue line in inset (A) for calculating the distance between the second-nearest neighboring carbon atoms in the hexagonal carbon rings.

(B) Raman spectrum of the graphene grown by ten ALD cycles, showing both G and D bands in the graphene structure.⁴⁹ Reprinted (adapted) with permission from Hu et al.⁴⁹ Copyright (2014) The Royal Society of Chemistry.

(C) TEM image of amorphous 2.5 nm BN films coated on ZrO₂ particles at 227°C.⁵⁸ Reprinted (adapted) with permission from Ferguson et al.⁵⁸ Copyright (2002) Elsevier.

(D) High-resolution TEM image of the Co/BN/Co interface, showing the layered BN films.⁶² Reprinted (adapted) with permission from Driver et al.⁶² Copyright (2016) American Chemical Society.

(E) General reaction mechanism of TCB with HMDS for a two-step ALD process of h-BN.⁴⁵ Reprinted (adapted) with permission from Hao et al.⁴⁵ Copyright (2017) John Wiley and Sons.

(F and G) (F) TEM image and (G) HR-TEM image of h-BN nanotubes.⁴⁶ Reprinted (adapted) with permission from Hao et al.⁴⁶ Copyright (2017) The Royal Society of Chemistry.

Bjelkevig et al.⁶⁰ later reported a crystalline h-BN monolayer on Ru(0001) substrate using the same ALD process. They deposited the BN films at 277°C and then treated them with post-annealing at 727°C in a vacuum. In addition, Co(0001) (Figure 3D),^{61,62} RuO₂(110),⁶³ and Ni(111)⁶⁶ substrates were also favorable for achieving crystalline h-BN. In all cases, post-treatment higher than 427°C in a vacuum was applied. Recently, BCl₃ has been used to couple with N₂H₄ for growing BN via ALD.⁶⁸ The anhydrous N₂H₄ precursor reacted with BCl₃ at lower temperature and enabled high BN stoichiometry with low oxygen and carbon impurities. However, the as-grown films were amorphous at 350°C, which might need a post-treatment for better crystallinity.

Another ALD process of BN used triethylboron (TEB, (C₂H₅)₃B) and NH₃ as precursors. In comparison with the processes mentioned above, this new process was not very successful at low temperature ($\leq 350^\circ\text{C}$)⁶⁴ but realized self-limiting growth (~ 0.7 Å/cycle) using H₂ as the carrier gas at 600°C.⁶⁹ TEB has also been used to couple with N₂/H₂ plasma (^PN₂/^PH₂) for depositing h-BN.⁷⁰ This PEALD study reported appreciable GPCs at temperatures higher than 350°C. However, the GPCs did not saturate with increased TEB doses, indicating thermal decomposition of TEB. XRD and TEM revealed that the as-deposited BN thin films were polycrystalline with a hexagonal structure in the range of 350°C–450°C. In addition, post-annealing slightly improved the crystallinity as well as crystallite size.

More recently, several new energy-enhanced processes of BN have been reported with the assistance of plasma, electrons, and hot wires. Park et al.⁷¹ demonstrated a new PEALD process using tris(ethylmethylamino) borane (TEMAB, B(N(CH₃)₂)₃) and plasma NH₃ (^PNH₃) in the range of 200°C–350°C. XRD confirmed the hexagonal structure of the as-deposited films, and HR-TEM revealed that the h-BN stacks were randomly distributed in the as-deposited films without any specific orientation. Creatively, Sprenger et al.⁷² developed an electron-enhanced ALD (EEALD) process using borazine (B₃N₃H₆) and electrons as precursors. This EEALD enabled BN to be deposited at room temperature and 100°C. Electrons stimulated desorption of hydrogen surface species to create reactive dangling bonds, and the dangling bonds facilitated borazine adsorption at low temperatures to form a h-BN layer horizontally. After a h-BN monolayer was formed, more borazine could be adsorbed on the h-BN surface for vertical growth. The electron energy range from 80 to 160 eV was favorable to achieve the maximum GPC of ~ 3.2 Å/cycle. The films grown at 100°C were polycrystalline h-BN and t-BN. However, this EEALD is line-of-sight and therefore not suitable for conformal coatings. Another energy-enhanced ALD was recently reported using a hot wire (tungsten filament), in which B₂H₆ and NH₃ were used as precursors.⁷³ This hot wire ALD enabled formation of atomic hydrogen and nitrogen-containing radicals by cracking NH₃ and deposited B_xN_y layers in the temperature range of 300°C–400°C.

Interestingly but differently, a recent effort reported a two-step ammonia-free ALD for growing BN, in which a pre-ceramic polymer layer was first deposited using trichloroborazine (TCB, H₃B₃Cl₃N₃) and hexamethyldisilazane (HMDS, (CH₃)₃SiNHSi(CH₃)₃) as precursors and then converted into dense h-BN (Figure 3E).⁴⁵ The ALD deposition of BN-based polymer was performed at 80°C with a GPC of 1.8 Å/cycle, and post-treatment proceeded at 600°C under Ar for 1 h and then at 1,400°C under NH₃ for 5 h. The resulting films were confirmed as h-BN with the removal of the organic template. This two-step process has been used to produce structures of h-BN with a high-aspect ratio, such as BN nanotubes (Figures 3F and 3G).⁴⁶

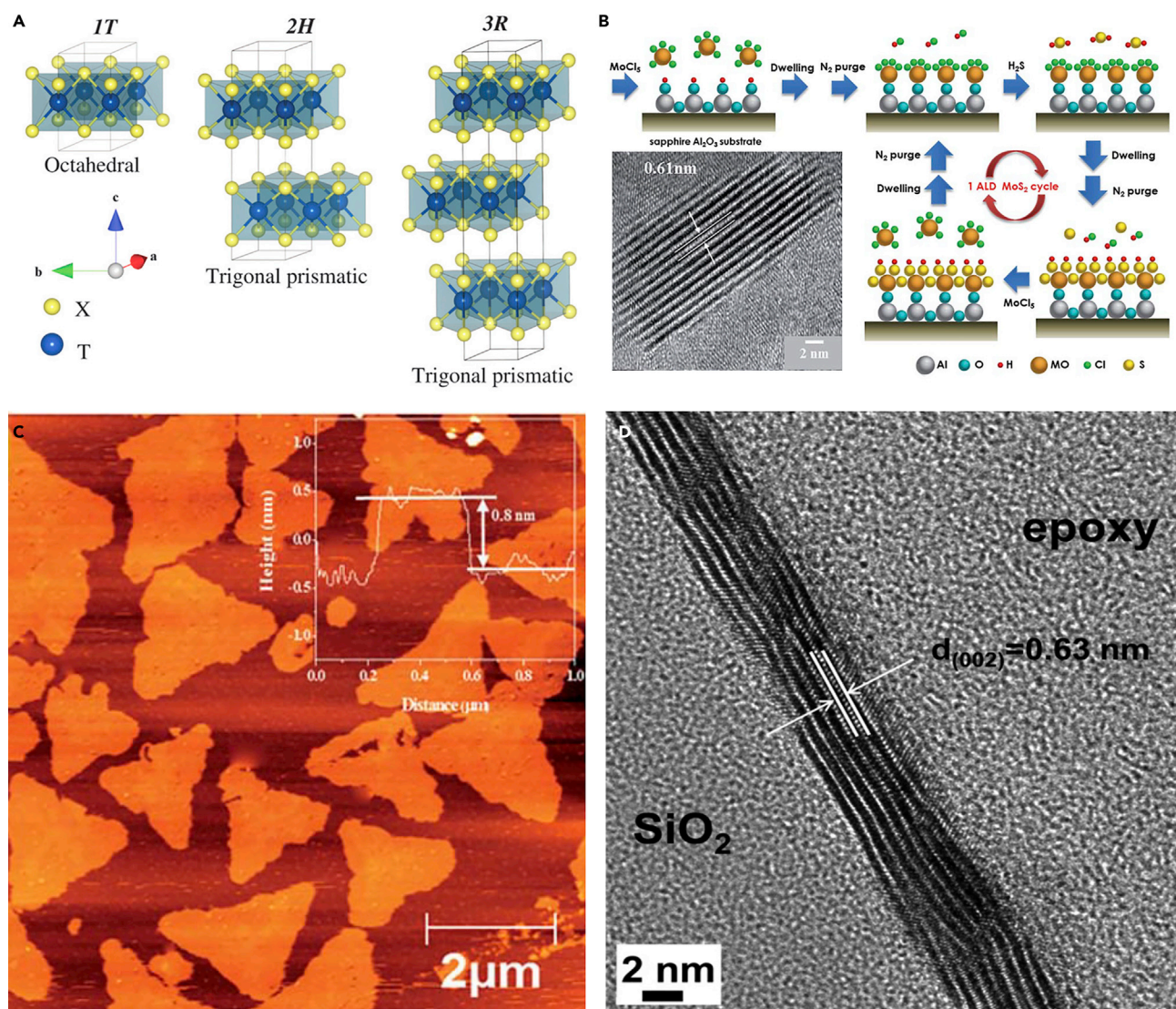


Figure 4. 2D Layered MoS₂ via ALD Processes

(A) Structure representation of 1T, 2H, and 3R TMD polytypes together with their metal atom coordination. The numbers indicate the number of layers in the unit cell, and the letters stand for trigonal, hexagonal, and rhombohedral, respectively.⁷⁶ Reprinted (adapted) with permission from Kuc and Heine.⁷⁶ Copyright (2015) The Royal Society of Chemistry.

(B) Schematic illustration of the ALD MoS₂ process using the precursor pair of MoCl₅/H₂S.⁷⁷ Inset: lattice spacing from a cross-sectional TEM view.⁷⁷ (C) AFM images of an ALD MoS₂ monolayer annealed at 800°C. The inset shows the height profile of the triangular flakes.⁷⁷ Reprinted (adapted) with permission from Tan et al.⁷⁷ Copyright (2014) The Royal Society of Chemistry.

(D) Cross-sectional HR-TEM image of ALD MoS₂ films annealed at 900°C for 5 min under a H₂S atmosphere.⁷⁸ Reprinted (adapted) with permission from Pyeon et al.⁷⁸ Copyright (2016) The Royal Society of Chemistry.

Transition Metal Dichalcogenides

TMDs have a MX₂ stoichiometry, and they are stacked layers of transition metal (M) atoms sandwiched between two layers of chalcogen (X is typically S, Se, or Te) atoms.⁷⁵ The adjacent layers of TMDs are held together by van der Waals forces. TMDs are commonly found in three polymorphs, i.e., 1T, 2H, and 3R, where the letters stand for trigonal, hexagonal, and rhombohedral, respectively, and the numbers indicate the number of layers in the unit cell (Figure 4A).^{76,79} Due to their unique structural characteristics, TMDs have attracted great attention for electronic and optical devices, energy storage, energy conversion, and solid-state lubrication.^{2,80}

Table 2. Summary of MoS₂ Films Synthesized via Various ALD Processes

Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	GPC (Å/Cycle)	Crystallinity	Reference
Formula	Temperature (°C)					
MoCl ₅	120	H ₂ S	300, R1	~1.6	2H ^a	Tan et al. ⁷⁷
	105		350–450, R1	0.87	2H	Browning et al. ⁸⁵
	70		375–475, R1	–	2H ^a	Valdivia et al. ⁸⁶
	200		430–470, R1	~4.0	2H	Huang et al. ⁸⁷
	200		450, R1	6.9	2H	Liu et al. ⁸⁸
	90		500–900, R2	0.07–0.17	2H	Kim et al. ³⁶
Mo(CO) ₆	RT	H ₂ S	155–170, R1	~2.5	2H	Nandi et al. ⁸⁹
			155–175, R1	0.74	2H ^b	Pyeon et al. ⁷⁸
		^p H ₂ S	175–225, R1	~0.5	2H	Jang et al. ⁹⁰
		CH ₃ S ₂ CH ₃	100–120, R1	1.1	2H ^c	Jin et al. ⁹¹ ; Shin et al. ⁹² ; Kwon et al. ⁹³
		O ₃	160, R3	~0.8	2H and 3R ^d	Martella et al. ³⁸
Mo(NMe ₂) ₄	RT	H ₂ S	60, R1	1.2	2H ^a	Jurca et al. ⁹⁴
		HS(CH ₂) ₂ SH	50, R1	–	2H ^c	Cadot et al. ⁹⁵
Mo(the) ₃	115	H ₂ S	300, R1	0.25	2H	Mattinen et al. ⁹⁶
MoF ₆	RT	H ₂ S	200, R1	0.46	2H ^c	Mane et al. ⁹⁷
(^t abun) ₂ (NMe ₂) ₂ Mo	RT	^p O ₂	150, R3	~2.0	2H ^d	Keller et al. ³⁹

^aPost-treatment in sulfur vapor at 800°C for 30 min⁷⁷ or at 900°C for 30 min⁸⁶ or at 1,000°C for 5 h.⁹⁴

^bPost-treatment in H₂S/Ar gas at 900°C for 5 min.⁷⁸

^cPost-treatment in Ar either at 900°C for 5 min^{91–93} or 800°C for 30 min,⁹⁵ or in H₂ either at 350°C for 15 min⁹⁷ or 450°C for 30 min.⁹⁵

^dSulfurization in sulfur vapor at 850°C³⁸ or 900°C for 30 min.³⁹

MoX₂

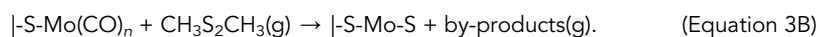
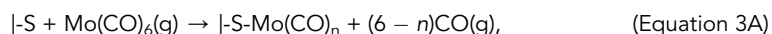
Both MoS₂ and MoSe₂ are group VIB TMDs, sharing similar crystal structures with three polymorphs, i.e., 1T, 2H, and 3R.^{81–84} They are among the most promising TMDs and have many exceptional properties for emerging applications.

MoS₂. Recently, MoS₂ has been actively investigated in ten ALD processes with different precursors, as summarized in Table 2. Among these ALD processes, Tan et al.⁷⁷ reported using MoCl₅ and H₂S as precursors at 300°C. The proposed surface chemistry is illustrated in Figure 4B. The as-grown MoS₂ films were crystalline (inset in Figure 4B), and a post-treatment in sulfur vapor improved their crystallinity. It was found that annealing at 800°C helped reconstruct the MoS₂ films with thinner thickness and higher crystallinity. For example, annealing changed the film thickness from 1.7 nm with ten-cycle ALD MoS₂ to 0.8 nm and from 3.2 nm with 20-cycle ALD MoS₂ to 1.3 nm. The annealed films of ten-cycle ALD MoS₂ show 2-μm triangular crystals (Figure 4C), whereas thicker ALD-deposited films helped produce continuous films after annealing. Using the same precursors, several subsequent studies also confirmed the growth of MoS₂ films in the range of 350°C–475°C.^{85–88} Interestingly but differently, Kim et al.³⁶ even applied much higher growth temperatures in the range of 500°C–900°C. After 120 ALD cycles at growth temperatures of 500°C, 700°C, and 900°C, the as-deposited films were 2, 1.4, and 0.8 nm thick (corresponding to the thickness of tri-, bi-, and monolayer MoS₂),⁹⁸ respectively. In this work, Kim et al. revealed that the layers of the ALD MoS₂ films were primarily determined by the high deposition temperatures rather than the ALD cycles,³⁶ as illustrated as R2 in Figure 2B. The researchers elucidated that, during the growth of the first layer, MoCl₅ was chemically adsorbed to all reactive sites over the entire surface of the substrate and then formed a layer of MoS₂ by reacting with a following supply of

H₂S. The first MoS₂ layer was inherently inactive and lacked suitable adsorption sites on its surface. Thus, synthesis was forced to proceed through the physical adsorption of MoCl₅ molecules on the MoS₂ layer. In this regard, the adsorption/desorption of precursor molecules was determined by growth temperatures. Once a critical number of MoS₂ layers were formed at a given growth temperature, precursor molecules could not be further adsorbed onto the MoS₂ basal plane, and no more layers were deposited. The success of this ALD process is highly related to the high thermal stability of the precursors. In other words, this special ALD process can only be realized for limited precursors. A similar ALD process has been reported by the same group for growing WSe₂,³⁷ which is introduced in a following section.

Another well-investigated precursor pair for ALD MoS₂ is Mo(CO)₆/H₂S. This ALD process features the high vapor of Mo(CO)₆ at room temperature and relatively low growth temperature. With this ALD process, Nandi et al.⁸⁹ synthesized amorphous MoS₂ films at 170°C. Using *in situ* quartz crystal microbalance (QCM), *ex situ* X-ray reflectivity (XRR), and ellipsometry, they revealed that the ALD MoS₂ could realize a GPC of 2.5 Å/cycle. *In situ* FTIR measurements validated the surface chemistry of ligand exchange between Mo(CO)₆ and H₂S, confirming the attachments of the C=O stretch after Mo(CO)₆ doses and its removal after H₂S doses. In a subsequent study, Pyeon et al.⁷⁸ performed this ALD MoS₂ process in the range of 155°C–175°C and confirmed that the as-grown amorphous films could be promoted to their crystalline phase via post-annealing in the range of 500°C–900°C under a H₂S/Ar atmosphere (Figure 4D). In addition, Jang et al.⁹⁰ reported that nanocrystalline 2H-MoS₂ films could be obtained directly at 170°C via a PEALD using precursors of Mo(CO)₆ and H₂S plasma (¹⁸H₂S), with a GPC of 0.5 Å/cycle.

Instead of H₂S, Jin et al.⁹¹ coupled Mo(CO)₆ with dimethyldisulfide (CH₃S₂CH₃, DMDS) for growing MoS₂ via ALD. In the ALD process, the surface first experienced decarbonylation of physisorbed Mo(CO)₆ to form chemisorbed subcarbonyls (Mo(CO)_{*n*}, *n* ≤ 5) on the surface (Equation 3A). Then, DMDS dissociated to form a chemisorbed methylthiolate intermediate (CH₃S–) through S–S bond cleavage. The surface methylthiolates further formed sulfur adatoms via C–S bond cleavage catalyzed by the chemisorbed Mo(CO)_{*n*} surface species, releasing gaseous molecules such as (CH₃)₂S and C₂H₆ (Equation 3B).

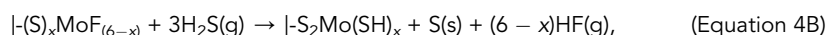
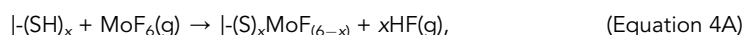


In the study, it was found that the GPC of the MoS₂ films was a function of the deposition temperature, increasing with growth temperature in the range of 60°C–100°C and then stabilizing at 1.1 Å/cycle in the range of 100°C–120°C.⁹¹ At a temperature above 120°C, there was a rapid increase in GPC due to the uncontrolled thermal decomposition of precursors. TEM and XRD analyses revealed that the MoS₂ films at 100°C–120°C were amorphous, and Raman spectra disclosed a locally ordered S-Mo-S unit with an atomic arrangement of 2H-MoS₂. In addition, they also revealed that post-annealing at 900°C for 5 min under Ar could significantly improve the crystallinity of the as-grown MoS₂ films.

In order to avoid corrosive or noxious by-products (i.e., HCl from a MoCl₅ precursor and sulfur contaminants from a DMDS precursor) and reduce growth temperature, Jurca et al.⁹⁴ developed an ALD process for MoS₂ at 60°C, using tetrakis(dimethylamido)-molybdenum(IV) (TDMA-Mo, Mo(NMe₂)₄) and H₂S as precursors. *In situ* QCM

measurements revealed self-limiting ALD growth at 60°C but CVD growth at 120°C or higher. The as-deposited MoS₂ films had a GPC of 1.2 Å/cycle and were amorphous. However, post-treatment at 1,000°C under sulfur vapor for 5 h could improve the crystallinity of the MoS₂ films. In a recent study, TDMA-Mo has also been used to couple with 1,2-ethanedithiol (HS(CH₂)₂SH) to constitute a new ALD process for MoS₂, enabling deposition at 50°C.⁹⁵ The as-grown films were stoichiometric but amorphous. The amorphous MoS₂ films could be converted into the 2H phase via post-treatment either in H₂ at 450°C or in Ar at 800°C.

An additional effort by Mattinen et al.⁹⁶ used tris(2,2,6,6-tetramethylheptane-3,5-dionato)molybdenum (Mo(thd)₃) and H₂S to grow crystalline MoS₂ films in the range of 250°C–400°C via ALD. The deposition below 300°C was self-limiting and uniform, but had a low GPC of ≤0.028 Å/cycle. The low GPC can possibly be ascribed to two factors: steric hindrance and low density of reactive sites. XRD and Raman analyses confirmed good crystallinity of the as-deposited MoS₂ films at 300°C, and thermogravimetric analysis (TGA) revealed that Mo(thd)₃ was prone to decompose at a temperature >350°C. In another study, Mane et al.⁹⁷ used MoF₆ and H₂S for depositing MoS₂ at 200°C and proposed the following ALD surface reactions (Equations 4A and 4B):



where “s” denotes the solid phase. *In situ* QCM measurements confirmed self-limiting growth with a GPC of 0.46 Å/cycle. XRD and Raman analyses revealed an amorphous phase of the as-deposited MoS₂ films, which could be improved in crystallinity and stoichiometry via post-treatment at 350°C in an H₂-rich environment.

Different from the studies mentioned above, Martella et al.³⁸ reported an ALD process via R3 for 2H-MoS₂ films, in which ALD-deposited molybdenum oxides were sulfurized into MoS₂. The molybdenum oxides were deposited at 160°C via an ALD using Mo(CO)₆ and ozone (O₃), containing 70% MoO₃ and 30% Mo₂O₅. In this study, a sulfurization process at 850°C under sulfur vapor successfully transformed the molybdenum oxides into a majority of 2H-MoS₂ phase with some minor 3R phase. In another study, Keller et al.³⁹ also tried to synthesize 2H-MoS₂ films by sulfurizing MoO_x films. MoO₃ films were deposited via two ALD processes: one thermal ALD process using bis(tert-butylimido)bis(dimethylamido)-molybdenum ((^tBuN)₂(NMe₂)₂Mo) and O₃ at 300°C, and another PEALD using ((^tBuN)₂(NMe₂)₂Mo) and O₂ plasma (^pO₂) at 150°C. Compared with the thermal ALD, the PEALD could increase the reactivity of the precursors at relatively low temperatures and led to higher nucleation density. The as-deposited MoO₃ films were first sulfurized under sulfur vapor at 300°C, resulting in amorphous MoS₂ films. Then, post-annealing under sulfur vapor at 900°C or higher temperature transformed the amorphous MoS₂ films into 2H-MoS₂ films with high crystallinity and stoichiometry.

MoSe₂. MoSe₂ has recently been reported in ALD studies, as summarized in Table 3. In this regard, Krbal et al.⁹⁹ demonstrated two ALD processes for depositing MoSe₂ films, using either Mo(CO)₆ or MoCl₅ to couple with ((CH₃)₃Si)₂Se. The deposition temperatures were 167°C and 300°C, respectively. The resultant films from MoCl₅ were nearly stoichiometric and crystalline 2H-MoSe₂ nano-flakes, whereas the films from Mo(CO)₆ had a significant amount of MoO_x and were a mixture of amorphous MoO_x

Table 3. Summary of MoSe₂ and WSe₂ Films Synthesized via Various ALD Processes

Materials	Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	GPC (Å/Cycle)	Crystallinity	Reference
	Formula	Temperature (°C)					
MoSe ₂	MoCl ₅	120	[(CH ₃) ₃ Si] ₂ Se	300, R1	–	2H	Krbal et al. ⁹⁹
	Mo(CO) ₆	55	[(CH ₃) ₃ Si] ₂ Se	167, R1	–	2H	Krbal et al. ⁹⁹
		RT	P ₂ O ₅	162, R3	–	2H ^a	Dai et al. ⁴⁰
WSe ₂	WCl ₆	85	DESe	600–800, R2	–	2H	Park et al. ³⁷
		110	H ₂ Se	390, R1	5	2H	Browning et al. ¹⁰⁰

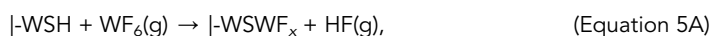
^aSelenization of MoO₃ with Se vapor under H₂/Ar at 820°C for 30 min.

and crystalline MoSe₂. In another work, Dai et al.⁴⁰ demonstrated a two-step approach (i.e., R3 route) to grow wafer-scale MoSe₂ films, in which MoO₃ films were first deposited at 162°C via a PEALD process using Mo(CO)₆ and P₂O₅ as precursors and were then selenized at a high temperature (820°C) under H₂/Ar (Figure 5A). XRD confirmed that the synthesized MoSe₂ films had a hexagonal crystal structure. In particular, this two-step method made 2H-MoSe₂ films with the thickness controllable by controlling the ALD-deposited MoO₃. In their study, Dai et al. confirmed that the as-received MoSe₂ films were 1.7 nm (2 layers), 2.5 nm (3 layers), and 6.2 nm (8 layers) by selenizing 60, 70, and 120 ALD MoO₃ cycles, respectively. Raman (Figure 5B) was used to probe the structures of 3-layer MoSe₂ nanosheets spatially, exhibiting three characteristic peaks, and HR-TEM (Figure 5C) revealed a polycrystalline crystal with the expected honeycomb-like structure.

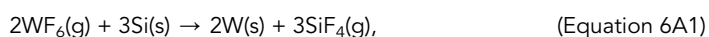
WX₂

Both WS₂ and WSe₂ are also group VIB TMDs with three polymorphs, i.e., 1T, 2H, and 3R.^{101–103} They both have many compelling properties and are promising for a number of applications.

WS₂. Eight ALD processes have been developed so far for depositing W₂S films, as summarized in Table 4. Scharf et al.¹⁰⁴ first reported an ALD process of 2H-WS₂ thin films at 300°C, using WF₆ and H₂S as precursors. Several subsequent studies studied its growth mechanism.^{105,106} These studies revealed that this ALD process only deposited WS₂ on certain substrates, and an ALD ZnS sub-layer promoted the growth of WS₂ films, leading to a GPC of ~1.1 Å/cycle (Figure 6A).^{104,106} However, the WS₂ growth gradually slowed down with increased cycles and finally stabilized at ~0.1 Å/cycle. The researchers speculated that ZnS might have served as a reducing agent to WF₆ and thereby enhanced the surface adsorption of WF₆. The surface reactions for this ALD process are suggested in Equations 5A and 5B.¹⁰⁶



Raman, XRD, and HR-TEM confirmed that the as-deposited films had a 2H crystalline nature.^{104,106} Nonetheless, there were impurities of Zn and F in the as-grown WS₂ thin films. To eliminate the undesirable Zn and F contaminants for semiconductor applications, Delabie et al.¹⁰⁷ used semiconductor-compatible Si thin layers and H₂ plasma (P₂H₂) as reducing agents for depositing WS₂ films in the range of 300°C–450°C, using WF₆ and H₂S as precursors. Si thin layers react with WF₆ in the range of 250°C–450°C, as described in Equations 6A1 and 6A2:



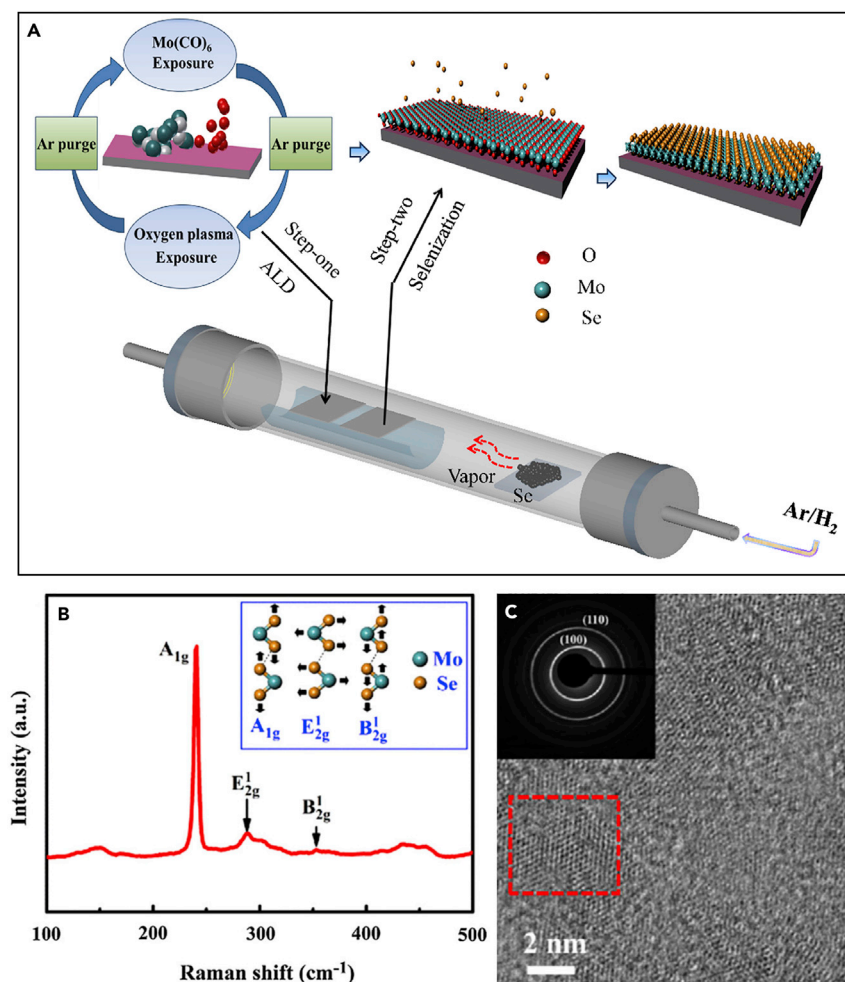
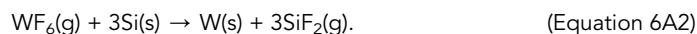


Figure 5. A Two-Step Process for Growing MoSe₂ Films

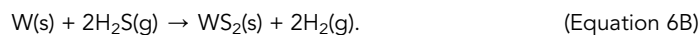
(A) Schematic illustration of the two-step growth process.

(B) Raman spectra of the as-grown three-layer MoSe₂ nanosheet (inset: schematic of Raman active vibration modes).

(C) HR-TEM image with inset of the SAED pattern for the three-layer MoSe₂ nanosheet.⁴⁰ Reprinted (adapted) with permission from Dai et al.⁴⁰ Copyright (2018) Springer.



Metallic W and volatile SiF_x species were continuously produced by the reactions until the sacrificial Si layers were completely consumed. The WS₂ thin films were then formed through the sulfurization of metallic W using H₂S (Equation 6B).



However, only six layers of WS₂ have been fully sulfurized at 450°C. Higher temperatures above 450°C were suggested to achieve thicker WS₂ films. In the case of H₂ plasma as a reducing agent, the PEALD process applied three sequential doses of WF₆, H₂ plasma, and H₂S. Both the plasma power and exposure time had evident influences on the growth and composition of WS₂ thin films. Raman and TEM analyses confirmed the 2H crystallinity of the as-grown WS₂ films using either of these two reducing agents. The resultant polycrystalline grain sizes of WS₂ were ~5 nm and ~10 nm for the reducing agents of

Table 4. Summary of WS₂ Films Synthesized via Various ALD Processes

Precursor A		Precursor B	Reducing Agent	Growth Temperature (°C) and Mechanism	GPC (Å/cycle)	Crystallinity	Reference
Formula	Temperature (°C)						
WF ₆	RT	H ₂ S	DEZ	300, R1	0.1–1.1	2H ^a	Scharf et al. ¹⁰⁴
			DEZ	300–350, R1	0.1–1.1	2H	Scharf et al. ^{105,106}
			Si/ ^p H ₂	300–450, R1	0.2	2H	Delabie et al. ¹⁰⁷
			^p H ₂	300, R1	0.17–0.42	2H	Groven et al. ¹⁰⁸
			^p H ₂	300, R1	~1	2H	Groven et al. ¹⁰⁹
			^p H ₂	300–450, R1	~1	2H	Groven et al. ¹¹⁰
W(CO) ₆	–	H ₂ S	–	150–240, R1	0.2	a ^b	Nandi et al. ¹¹¹
	60		–	400, R1	~3.6	2H	Sun et al. ¹¹²
	40	^p H ₂ S	–	350, R1	~1	2H	Yeo et al. ¹¹³
WCl ₅	110	H ₂ S	–	390, R1	1.25	2H	Browning et al. ¹¹⁴
WH ₂ (iPrCp) ₂	–	^p O ₂	–	300, R3	~1	2H ^c	Song et al. ⁴¹
(N ⁱ Bu) ₂ (NMe ₂) ₂ W	100	^p O ₂	–	160–220, R3	1	2H ^d	Zeng et al. ⁴²

^aPost-annealing in a vacuum at 500°C for 1 h.

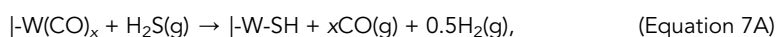
^ba represents amorphous.

^cSulfurization in H₂S/Ar gas at 1,000°C for 30 min.

^dSulfurization in S/H₂/Ar gas at 700°C–900°C for 30 min.

Si and H₂ plasma, respectively. In several following studies, Groven et al.^{108–110} explored the nucleation mechanism of WS₂ films at 300°C using WF₆, H₂ plasma and H₂S as precursors (Figure 6B). Groven et al.¹⁰⁸ revealed a redox chemistry model for the gas-solid reactions caused by WF₆, H₂S, and H₂ plasma. During WF₆ exposure, WF₆ can react with either –SH groups or surface S atoms to form –WF_x surface species, which are reduced later by H₂ plasma into metallic W or W with a lower oxidation state. Subsequent H₂S exposure sulfurized the metallic W and the intermediate oxidation state W into WS₂. However, it was found that the S/W ratio was not sufficient, $\sim 1.9 \pm 0.2$, and there was no metallic W, implying that there were S vacancies near the surface.¹⁰⁸ In the work, the lateral crystal growth of WS₂ was disclosed, and three steps were suggested:¹¹⁰ (1) the adsorption of precursors on substrates, (2) the diffusion of adsorbed precursors toward the lateral reactive sites, and (3) the surface reaction of precursors that led to the growth of 2D materials (as illustrated in Figure 6C). Since the preferential growth of WS₂ was along its crystal edges in the lateral direction, it was believed that larger crystal grain size could be achieved by reducing the nucleation density and promoting lateral growth. To this end, it was preferred to have a starting surface with less reactive sites (i.e., SiO₂), lower H₂ plasma power, higher deposition temperature (e.g., 450°C), and increased reactor pressure.^{109,110}

A new ALD process was later proposed by Nandi et al.¹¹¹ using W(CO)₆ and H₂S as precursors in the range of 155°C–230°C. *In situ* FTIR and residual gas analysis (RGA) were used to determine the surface species and the gaseous by-products for each half reaction, respectively. The ALD half reactions were then established as follows (Equations 7A and 7B):



FTIR measurements verified the existence of W(CO)_x surface groups after W(CO)₆ doses and its disappearance after H₂S doses, indicated by the peak at 2,000 cm^{–1}

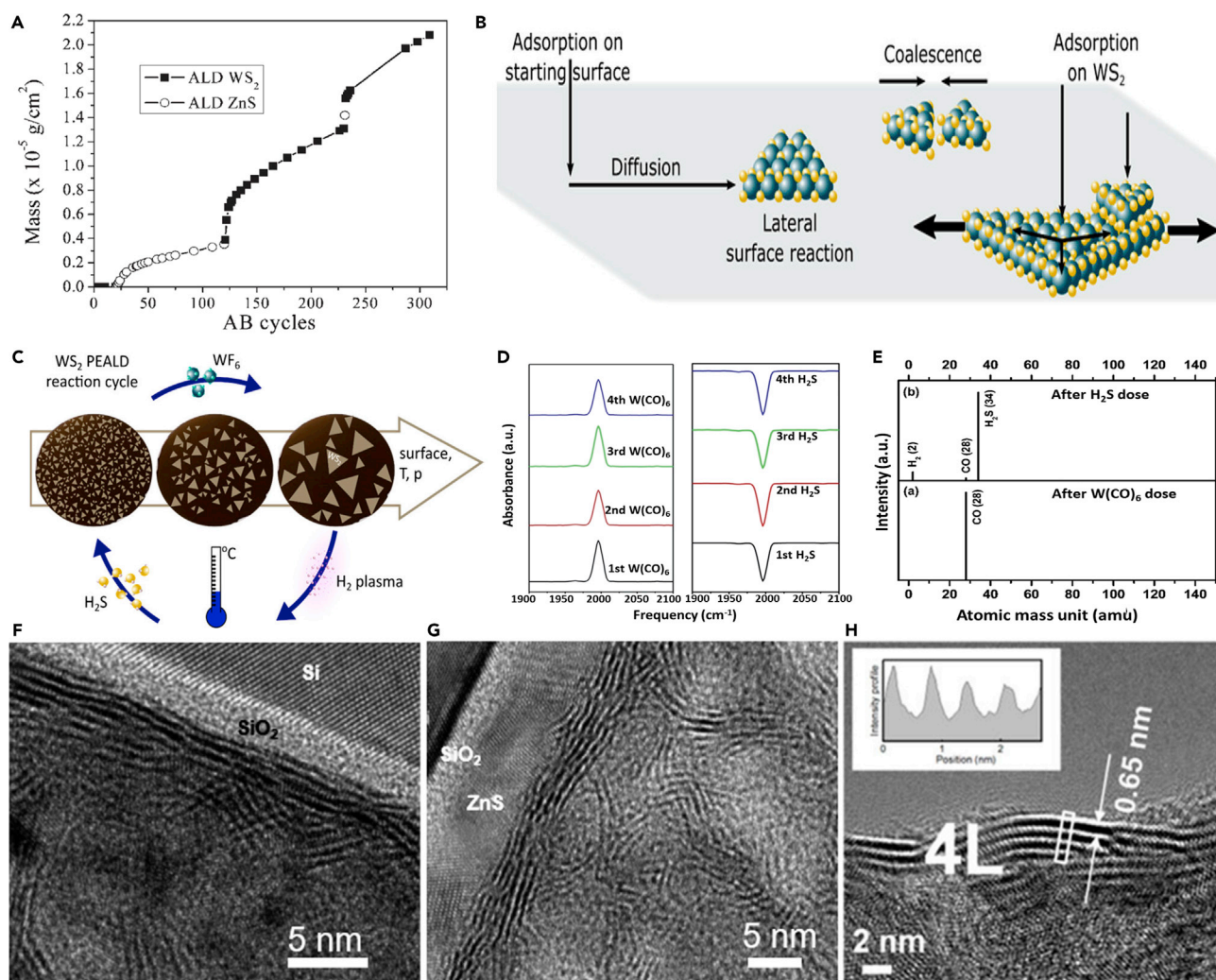


Figure 6. 2D Layered WS_2 Materials via ALD Processes

(A) QCM mass gain of ALD WS_2 and ZnS that pulsed sequentially on a SiO_2 -coated crystal.¹⁰⁴ Reprinted with permission from Scharf et al.¹⁰⁴ Copyright (2004) Cambridge University Press.

(B) Schematic of the ternary WS_2 PEALD reaction cycle, composed of a WF_6 reaction, H_2 plasma exposure, and H_2S reaction.¹¹⁰

(C) Schematic of the proposed qualitative model for the nucleation behavior of WS_2 .¹¹⁰ Reprinted (adapted) with permission from Groven et al.¹¹⁰ Copyright (2018) American Chemical Society.

(D) Difference FTIR spectra recorded for several ALD cycles before and after dosing of W(CO)_6 into the ALD reactor.¹¹¹

(E) RGA spectrum recorded just after the W(CO)_6 dose and H_2S dose.¹¹¹ Reprinted (adapted) with permission from Nandi et al.¹¹¹ Copyright (2016) The Royal Society of Chemistry.

(F and G) Cross-sectional HR-TEM images of the WS_2 films grown on (F) the Si substrate and (G) the ZnS-coated Si substrate.¹¹² Reprinted (adapted) with permission from Sun et al.¹¹² Copyright (2016) Springer Nature.

(H) HR-TEM image of four layers of WS_2 films deposited by the indirect ALD process.⁴¹ Reprinted (adapted) with permission from Song et al.⁴¹ Copyright (2013) American Chemical Society.

(corresponding to $\text{W}=\text{C}=\text{O}$ stretching) (Figure 6D). However, the signature of the $-\text{SH}$ surface group was not observed, probably due to the limited sensitivity of FTIR. RGA confirmed the by-products of CO and H_2 (Figure 6E). In addition, QCM revealed a temperature-dependent growth of WS_2 films. Below 150°C , no growth was observed due to the thermodynamic limit of W(CO)_6 . In the range of 175°C – 205°C , the GPC was stable at $0.2 \text{ \AA/cycle}$. Above 205°C , GPC increased linearly, probably due to thermal decomposition of W(CO)_6 . However, two subsequent studies used W(CO)_6 precursor at 350°C ¹¹³ and 400°C .¹¹² Sun et al.¹¹² reported

successful deposition of crystalline 2H-WS₂ films via ALD at 400°C using W(CO)₆ and H₂S as precursors. This apparently conflicts with the claim of Nandi et al.¹¹¹ that W(CO)₆ decomposed at temperatures above 205°C. Sun et al.¹¹² also investigated the growth of WS₂ films on two types of substrates: Si(100) wafers with a 4-nm native oxide SiO₂ layer (SiO₂/Si) and 8-nm ZnS-coated Si(100) wafers (ZnS/Si). They found that, compared with the SiO₂/Si substrate, the ZnS/Si substrate had little effect on the GPC of ALD WS₂, but there was slight enhancement of the crystallinity of WS₂ films (Figures 6F and 6G). In their work, the highest GPC of 3.6 Å/cycle was achieved. In another work, Yeo et al.¹¹³ replaced H₂S with H₂S plasma (^PH₂S) to couple with the W(CO)₆ precursor. This PEALD deposited WS₂ films at 350°C and had a GPC of ~1.0 Å/cycle. Density functional theory (DFT) calculations showed that H₂S plasma could create various defects on the surface of WS₂, such as S vacancy, W vacancy, and WS₂ anti-site defects. All these defects were favorable to the adsorption of W(CO)₆ and -W(CO)₅ ligands.

Other precursors have been developed for ALD WS₂ films. Paired WCl₅/H₂S was reported by Browning et al.¹¹⁴ to produce WS₂ in a heterojunction with SnS. WS₂ films were deposited on *p*- or *n*-type Si wafers at 390°C. Raman and TEM analyses revealed that they were a few (ten or less) atomic layers of 2H crystals. AFM disclosed a GPC of 1.25 Å/cycle. In contrast, Song et al.⁴¹ applied an indirect R3 growth strategy via sulfurization of WO₃ for WS₂. WO₃ thin films were first deposited via a PEALD process using WH₂(iPrCp)₂ and ^PO₂ as precursors at 300°C, followed by sulfurization conducted in a H₂S-rich environment at 1,000°C (Figure 6H). The GPC of the WO₃ films was 1.0 Å/cycle. It was found that every ten cycles of WO₃ films can be transformed into one layer of WS₂ films. As a result, the number of WS₂ layers can be controlled precisely by adjusting the number of PEALD WO₃ cycles. Raman analysis confirmed the formation of WS₂ films, and SAED patterns revealed the polycrystalline WS₂ nanosheets with sizes in the range of 10–20 nm. A perfect S/W ratio of 2:1 was received by XPS analysis. Recently, Zeng et al.⁴² explored another indirect ALD route for WS₂ films, in which WO₃ films were first deposited via an ALD process (1.2 Å/cycle) in the range of 160°C–220°C, using bis(tertbutylimido)-bis(dimethylamido)-tungsten ((N^tBu)₂(NMe₂)₂W) and ^PO₂ as precursors, and then sulfurized into 2D WS₂ films in a sulfur-rich environment at 700°C–900°C. About every eight cycles of WO₃ films could be converted into one layer of WS₂ via the sulfurization. HR-TEM confirmed that the resulting WS₂ films were structured in layers with crystalline sizes of 5–10 nm, and XPS revealed a nearly stoichiometric S/W ratio of 1.98:1.

WSe₂. Park et al.³⁷ conducted the first ALD work on WSe₂ using WCl₆ and diethyl selenide (DESe, Se(C₂H₅)₂) as precursors. This ALD process was performed at high temperatures of 600°C, 700°C, and 800°C, which actually was a process via R2. The growth of WSe₂ was dependent solely on temperature rather than the number of ALD cycles. The critical thicknesses of WSe₂ films deposited at 600°C, 700°C, and 800°C were 3.9 nm (5 layers), 2.5 nm (3 layers), and 0.8 nm (1 layer) (Figures 7A–7C), respectively. AFM revealed that there was no WSe₂ growth after the first 100 ALD cycles at all three temperatures (Figure 7D). HR-TEM analyses revealed that the ALD-deposited WSe₂ films were crystalline with a grain size of ~200 nm, and their honeycomb-like structure was identifiable (Figure 7E). XPS disclosed a perfect stoichiometric Se/W ratio of 2, with no Cl species detectable. In a subsequent study, Browning et al.¹⁰⁰ reported an ALD process for few-layered WSe₂ films at 390°C, using WCl₅ and H₂Se as precursors. Raman and XPS analyses confirmed the growth of WSe₂, and HR-TEM revealed its layered structure (Figure 7F), with a GPC of 5 Å/cycle. These ALD processes are included in Table 3.

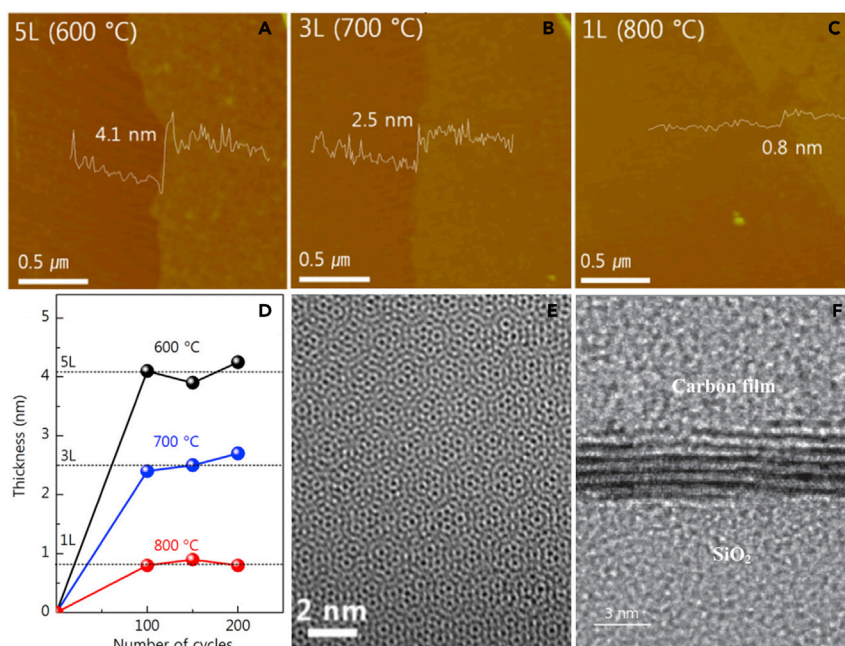


Figure 7. 2D Layered WSe₂ Materials via ALD Processes

(A–D) AFM images and height profiles of (A) 5, (B) 3, and (C) 1 layer WSe₂ after transferring onto a SiO₂ substrate.³⁷ (D) Thickness versus the number of cycles for WSe₂ synthesized at 600°C, 700°C, and 800°C.³⁷

(E) HR-TEM image of a selected region of the WSe₂ film.³⁷ Reprinted (adapted) with permission from Park et al.³⁷ Copyright (2016) IOP Publishing.

(F) Cross-sectional HR-TEM of a few-layered thick WSe₂ film.¹⁰⁰ Reprinted (adapted) with permission from Browning et al.¹⁰⁰ Copyright (2016) IOP Publishing.

TiS₂

TiS₂ is a group IVB TMD material with two crystal structures: 1T and 2H (Figure 8A). 1T is the most stable polytype, and 2H is the metastable phase.¹¹⁵ TiS₂ has many intriguing properties,^{115,120} which make it a promising material for a number of applications.^{121–123} To date, only one ALD process has been reported for growing 2D TiS₂ films, as summarized in Table 5. In this regard, Pore et al.¹¹⁶ investigated an ALD for TiS₂ at 400°C, using TiCl₄ and H₂S as precursors. In the study, they noticed that substrates affected the growth of TiS₂. The TiS₂ films were nearly all plate-like crystallites parallel to the ZnS substrate (Figure 8B), whereas the films were randomly oriented on soda-lime glass (Figure 8C). XRD revealed that the TiS₂ films on ZnS were highly (001) oriented, whereas the ones on soda-lime glass showed multiple peaks. The GPC of TiS₂ was 0.18 Å/cycle on ZnS and 0.26 Å/cycle on soda-lime glass. It was also observed that the TiS₂ crystallites on soda-lime glass were much larger than those on ZnS, and there were some tubular TiS₂ crystals in the films formed on soda-lime glass. In addition, TiS₂ films were also deposited on various metallic films, including Rh, Ir, Pd, Pt, Ru, and TiN. It was noted that the S/Ti ratio in the as-grown TiS_x films varied with different substrates, from 0.5 on Ir to 4.4 on Rh. Three reasons for these variations are suggested: (1) different reactions between H₂S (or TiCl₄) and the substrates, (2) formation of sulfur-containing or sulfur impurities, and (3) oxidation of TiS₂ to TiO₂.¹¹⁶ For example, a S/Ti ratio of 2.2 on Pd resulted from the formation of PdS and the large ratio of 4.4 on Rh was due to large amount of sulfur-rich impurities, which hindered crystallization. Ratios of <2.0 were attributed to a small amount of amorphous TiO_x or TiS_x phases. It was found that the TiS_x films on Pt and Ir substrates were more prone to oxidize. However, it remains unknown

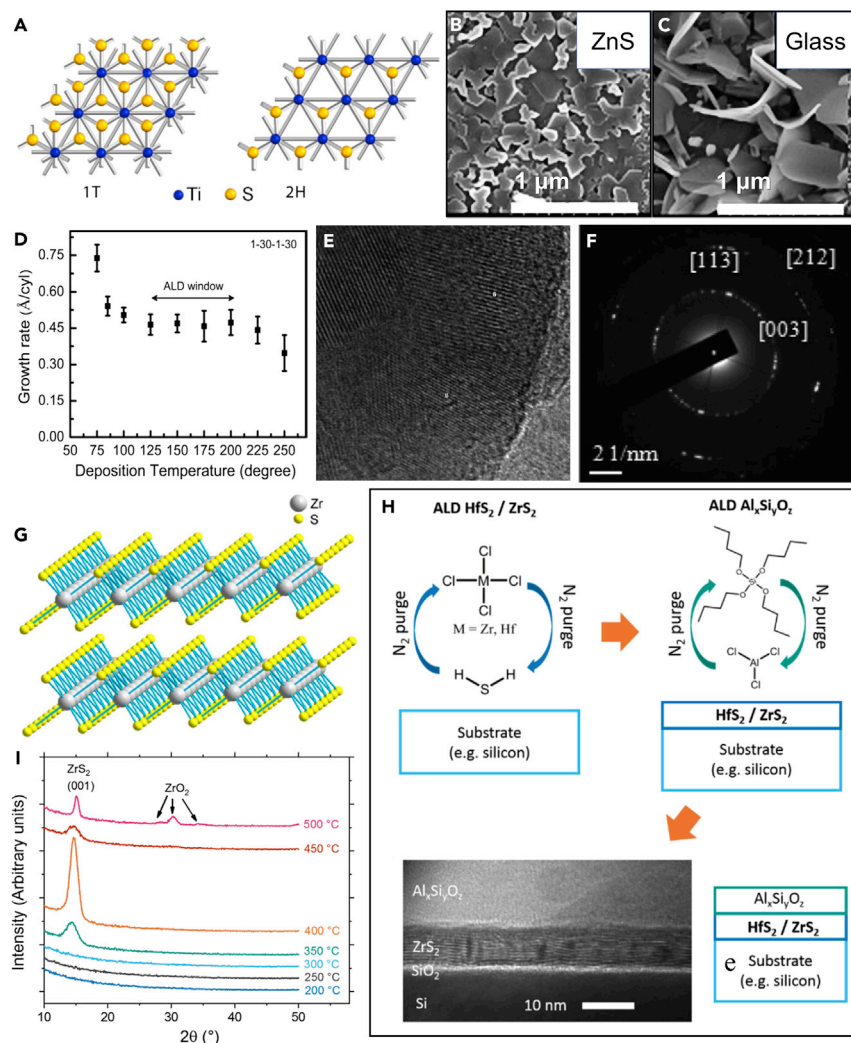


Figure 8. 2D Layered TiS₂ and HfS₂ Materials via ALD Processes

(A) The molecular layer structure of 1T- and 2H-TiS₂ phases.¹¹⁵ Reprinted (adapted) with permission from Putri et al.¹¹⁵ Copyright (2019) Elsevier.

(B and C) Field emission SEM images of TiS₂ films grown by ALD at 400°C on (B) ZnS and (C) soda-lime glass.¹¹⁶ Reprinted (adapted) with permission from Pore et al.¹¹⁶ Copyright (2007) John Wiley and Sons.

(D) GPC for TiS₂ ALD versus deposition temperature by *in situ* QCM measurements.¹¹⁷

(E and F) (E) HR-TEM image and (F) SAED pattern of ALD-deposited TiS₂ films.¹¹⁷ Reprinted with permission from Mahuli and Sarkar.¹¹⁷ Copyright (2015) AIP Publishing.

(G) The chemical structure of 1T ZrS₂.¹¹⁸ Reprinted (adapted) with permission from Xu et al.¹¹⁸ Copyright (2014) Elsevier.

(H) Schematic illustration of the deposition of ZrS₂ or HfS₂ and the protective Al_xSi_{1-x}O₂ layers by ALD at 400°C and the cross-sectional TEM image of an Al_xSi_{1-x}O₂-encapsulated ZrS₂ film on a Si substrate with a native oxide layer.¹¹⁹

(I) Grazing incidence XRD diffractograms of ZrS₂ films deposited at different temperatures.¹¹⁹ Reprinted (adapted) with permission from Mattinen et al.¹¹⁹ Copyright (2019) American Chemical Society (<https://doi.org/10.1021/acs.chemmater.9b01688>).

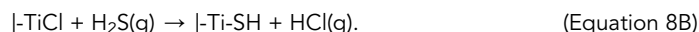
whether the oxidation was caused by trace O₂ and H₂O in the carrier gas or caused by ambient air. Moreover, the researchers found that the deposition at 400°C and 500°C was similar, but there was no deposition at 300°C. Using the same precursors of TiCl₄/H₂S, however, Mahalia et al.¹¹⁷ deposited TiS₂ films in the range

Table 5. Summary of TiS₂, ZrS₂, HfS₂, and ReS₂ Films Synthesized via Various ALD Processes

Materials	Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	Substrate ^a	GPC (Å/Cycle)	Crystallinity ^b	Reference
	Formula	Temperature (°C)						
TiS ₂	TiCl ₄	RT	H ₂ S	400–500, R1	SLG	0.26–0.27	<i>h</i>	Pore et al. ¹¹⁶
					ZnS	0.18	<i>h</i>	
					Rh	0.14	<i>a</i>	
					Ir	0.16	<i>h</i>	
					Pd	0.11	<i>h</i>	
					Pt	0.12	<i>h</i>	
					Ru	0.15	<i>c</i> and/or <i>h</i>	
					TiN	0.26	<i>h</i>	
ZrS ₂	ZrCl ₄	130–135	H ₂ S	200–500, R1	SLG	0.02–0.08	<i>a</i> (<350°C), <i>h</i> (≥350°C)	Mattinen et al. ¹¹⁹
	Zr(NEtMe) ₄	40	H ₂ S	50–200, R1	SLG	0.7–0.8	<i>h</i> (200°C)	
HfS ₂	HfCl ₄	125–130	H ₂ S	200–500, R1	SLG	0.06–0.11	<i>h</i> (≥350°C)	Mattinen et al. ¹¹⁹
	HfI ₄	170	H ₂ S	200–500, R1	SLG	0.1	<i>h</i> (≥300°C)	
	Hf(NEtMe) ₄	40	H ₂ S	50–200, R1	SLG	0.7–0.8	<i>h</i>	
ReS ₂	ReCl ₅	110	H ₂ S	120–500, R1	Si	0.2–0.9	<i>a</i> (<200°C), 1T' (≥200°C)	Hämäläinen et al. ¹²⁴

^aSLG denotes soda-lime glass substrate.^bFor crystallinity, *a*, *h*, *c*, and 1T' represent amorphous, hexagonal, cubic, distorted octahedral phases, respectively.

of 75°C–250°C on porous TiO₂-coated fluorine-doped tin oxide glass. The researchers suggested two surface reactions as follows (Equations 8A and 8B):



In situ QCM measurements verified a steady GPC of 0.46 Å/cycle in the range of 125°C–200°C (Figure 8D). Below 125°C, the GPC was higher, probably due to TiCl₄ condensation. Above 200°C, the GPC gradually decreased, possibly related to desorption of the precursors. HR-TEM images (Figure 8E) and the SAED pattern (Figure 8F) confirmed crystalline TiS₂ films.

ZrS₂

ZrS₂ is also a group IVB TMD material with 1T and 2H phases.¹²⁵ The 1T ZrS₂ (Figure 8G) is semiconducting and has many exotic properties for various applications.^{118,119,126–129} The only study on ALD for ZrS₂ was reported in the latest work by Mattinen et al.,¹¹⁹ in which both ZrS₂ and HfS₂ thin films were investigated due to their analogous crystal structures and lattice parameters. The ALD processes for ZrS₂ and HfS₂ are summarized in Table 5. In the work, two precursor pairs, ZrCl₄/H₂S and Zr(NEtMe)₄/H₂S, were investigated to deposit ZrS₂ films (Figure 8H).¹¹⁹ Using the ZrCl₄/H₂S precursor pair, ZrS₂ films could be grown on a variety of substrates (e.g., soda-lime glass, Si, SiO₂, sapphire, muscovite mica, and Al₂O₃) in the range of 200°C–500°C. The growth was substrate dependent and reflected in the GPC, morphology, and crystallinity. At <350°C, the as-deposited ZrS₂ films were amorphous with chlorine residues (Figure 8I). At higher temperatures of 350°C–500°C, the films were 1T crystalline and (001) oriented. However, XRD also revealed monoclinic ZrO₂ in the

ZrS₂ films at 450°C and higher (Figure 8I), implying oxidization of ZrS₂ during ALD deposition. The oxidation at 450°C might be due to the presence of residual oxidants (e.g., H₂O and O₂) in the ALD reactor.¹¹⁹ XRR measurements showed that the GPCs of ZrS₂ films in the range of 200°C–500°C were quite low; the highest GPC was 0.08 Å/cycle at 400°C. Most of the ZrS₂ samples were capped by a layer of amorphous Al_xSi_yO_z to prevent oxidization of the as-deposited ZrS₂ during characterizations, because ZrS₂ was prone to oxidization at room temperature. The Al_xSi_yO_z layer was also deposited via an ALD process immediately after the ZrS₂ deposition at 400°C. In comparison with the precursor pair of ZrCl₄/H₂S, the Zr(NEtMe)₄/H₂S precursor pair was not very desirable.¹¹⁹ Zr(NEtMe)₄ is prone to decompose at 250°C or higher. Thus, this precursor pair was investigated in the range of 50°C–250°C. At low temperatures (≤200°C), however, the as-deposited films were non-uniform and compositionally inhomogeneous. XRD revealed that the films at 200°C were crystalline with the preferred (001) orientation of 1T ZrS₂ but had two other peaks related to the ZrS₃ phase.¹¹⁹ In addition, the resultant ZrS_x films were more unstable in air than the ZrS₂ films deposited using the ZrCl₄ precursor.

HfS₂

HfS₂ is another group IVB TMD analogous to ZrS₂. The only ALD work on HfS₂ was reported together with the ALD for ZrS₂ discussed above.¹¹⁹ In the work, three precursor pairs, HfCl₄/H₂S, HfI₄/H₂S, and Hf(NEtMe)₄/H₂S, were examined; the HfCl₄/H₂S pair has been mainly investigated.¹¹⁹ Using HfCl₄ and H₂S as precursors (Figure 8H), the HfS₂ films were deposited in the range of 200°C–500°C. It was revealed that the HfS₂ films were amorphous at low deposition temperatures (≤350°C) with chlorine residues, while crystalline 1T HfS₂ films were obtained at high deposition temperatures between 350°C and 500°C. However, XRD peaks of monoclinic HfO₂ were also seen for the HfS₂ films deposited at 450°C or higher. In comparison, the HfS₂ films deposited at 400°C exhibited better crystallinity and purity.

In the work, HfI₄ was also tried in place of HfCl₄ for ALD of HfS₂.¹¹⁹ This ALD process enabled 1T HfS₂ at 300°C, but the HfS₂ surface from HfI₄/H₂S was rougher. Alternatively, Hf(NEtMe)₄ was studied, but the as-deposited HfS₂ films had poor crystallinity, non-uniform coverage, and inhomogeneous composition.

ReS₂

ReS₂ is a 2D TMD with distorted octahedral (1T') monolayer structure.¹³⁰ The distorted 1T' phase structure induces inherent reduced crystal symmetry, leading to strongly in-plane anisotropic optical and transport properties. ReS₂ has also exhibited extraordinary layer-independent properties, which make it attractive in many applications.^{130,131} Only one ALD for ReS₂ has been reported to date using ReCl₅ and H₂S as precursors (Table 5).¹²⁴ In the work, Hämäläinen et al. revealed that the growth of ReS₂ had a strong dependency on temperature. In the range of 120°C–200°C, the GPC of ReS₂ increased from ~0.3 Å/cycle to ~0.9 Å/cycle, the stoichiometry of the films improved from a S/Re ratio of 1–2, and the crystallinity evolved from amorphous to crystalline. At higher temperatures, the GPC was nearly unchanged at 0.8–0.9 Å/cycle at 200°C–300°C but then gradually decreased to ~0.2 Å/cycle at 450°C. The ReS₂ films remained crystalline and stoichiometric above 200°C, while the crystalline orientation varied with temperature. It was observed that between 200°C and 300°C, the ReS₂ crystallites were parallel to the substrate, while above 350°C, the plate-like crystallites had more random orientations toward the substrate. Overall, high-quality crystalline and stoichiometric ReS₂ films can be obtained above 200°C, and the crystal orientation is tunable by adjusting the deposition temperature.

Table 6. Summary of FeS and NiS Films Synthesized via Various ALD Processes

Materials	Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	GPC (Å/Cycle)	Crystallinity ^a	Reference
	Formula	Temperature (°C)					
FeS	Fe(amd) ₂	75	H ₂ S	80–300, R1	0.19–0.4	te at ≤120°C, h or m at ≥160°C	Shao et al. ¹³²
NiS	Ni(thd) ₂	165	H ₂ S	175–350, R1	0.09–0.13	r	Mahuli et al. ¹³³

^aFor crystallinity, h, r, m, and te stand for hexagonal, rhombohedral, monoclinic, and tetragonal, respectively.

Transition Metal Monochalcogenides

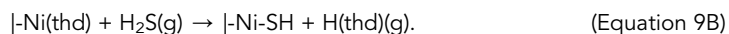
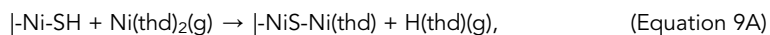
Some transition metal monochalcogenides (MX) are 2D layered materials, such as FeS, NiS, and GaS. In recent studies, both FeS and NiS have been investigated by ALD, as summarized in Table 6. FeS can be tetragonal or hexagonal phase, while NiS has a layered structure.¹³⁴ NiS also has two polymorphs, i.e., hexagonal (α -NiS) and rhombohedral (β -NiS) structures.¹³⁵ The former is metallic with a 3D structure, and the latter is insulating with a 2D millerite structure.¹³⁶

FeS

Shao et al.¹³² developed an ALD process for FeS thin films at 80°C–300°C using bis(*N,N'*-di-tert-butylacetamidinato)iron (Fe(amd)₂) and H₂S as precursors. They revealed that this ALD process could realize self-limiting growth in the range of 80°C–200°C. The GPC decreased from 0.40 Å/cycle at 80°C to 0.19 Å/cycle at 200°C. Above 200°C, Fe(amd)₂ showed some partial decomposition. HR-TEM analyses verified that the ALD FeS_x films deposited at 80°C–250°C were all well-crystalline but with different structures. The films were a 2D layered tetragonal structure at low temperatures (≤120°C) and a hexagonal (or monoclinic) structure at high temperatures (≥160°C). The as-deposited film had a Fe to S ratio of around 1.0, with little C and N impurities of 1.6 at % and 0.5 at %.

NiS

Mahuli et al.¹³³ reported an ALD process for β -NiS using bis(2,2,6,6-tetramethylheptane-3,5-dionate)nickel (Ni(thd)₂) and H₂S as precursors. *In situ* FTIR measurements detected the attachment and removal of -Ni(thd) surface species after Ni(thd)₂ and H₂S dosing, respectively. Thus, two-half ALD reactions were suggested (Equations 9A and 9B):



Using *in situ* QCM and XRR, the GPC was determined as 0.12–0.13 Å/cycle in the range of 175°C–250°C and slightly decreased to 0.09 Å/cycle at 350°C. XRD verified that the as-deposited films at 175°C were polycrystalline rhombohedral β -NiS.

Group IIIA-VA Metal Chalcogenides

InSe and In₂Se₃

InSe is a group IIIA monochalcogenide and consists of Se-In-In-Se layers stacked together by weak van der Waals forces.¹³⁷ Among several InSe polytypes, the hexagonal (β -InSe) and rhombohedral (γ -InSe) phases possess 2D layered structures but show distinct atomic coordination.¹³⁸ The β phase comprises eight atoms extended over two layers, while the γ phase has a rhombohedral unit cell containing only one complex layer of four atoms.^{139,140} To date, only Browning et al.¹³⁷ have deposited γ -InSe films via ALD using InCl₃ and H₂Se at 350°C (Table 7). The γ -InSe films could be uniformly formed on various substrates with a GPC of 0.5 Å/cycle. Moreover, SEM showed that the surface of the InSe films was smooth and featureless.

Table 7. Summary of InSe, In₂Se₃, GeS, GeSe, and GeTe Films Synthesized via Various ALD Processes

Materials	Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	GPC (Å/Cycle)	Crystallinity ^a	Reference
	Formula	Temperature (°C)					
InSe	InCl ₃	110	H ₂ Se	350, R1	0.5	<i>r</i>	Browning et al. ¹³⁷
In ₂ Se ₃	InCl ₃	285	(Et ₃ Si) ₂ Se	295, R1	0.55	–	Hatanpää et al. ¹⁴¹ ; Pore et al. ¹⁴²
GeS	Ge precursor ^b	40	H ₂ S	50–200, R1	0.28–0.02	<i>a</i>	Kim et al. ¹⁴³
GeSe	HGeCl ₃	3	[(CH ₃) ₃ Si] ₂ Se	70–150, R1	1.04–1.15	crystalline ^c	Kim et al. ¹⁴⁴
GeTe	GeCl ₂ •C ₄ H ₈ O ₂	70	(Et ₃ Si) ₂ Te	90, R1	0.42	mixed <i>a</i> and <i>r</i>	Pore et al. ¹⁴²
		65	(Et ₃ Si) ₂ Te	90, R1	–	–	Knapas et al. ¹⁴⁵
			(Et ₃ Si) ₂ Te	90–130, R1	0.11–0.52	<i>r</i> ^d	Sarnet et al. ¹⁴⁶
	GeBr ₂	–	(Et ₃ Si) ₂ Te	140, R1	–	–	Pore et al. ¹⁴²
	Ge(N(Si(CH ₃) ₃) ₂) ₂	50, 70	(Me ₃ Si) ₂ Te	70–120, R1	0.75–2.57	<i>r</i> ^e	Gwon et al. ^{147,148}
	HGeCl ₃	3	(Me ₃ Si) ₂ Te	50–100, R1	–	<i>r</i> ^f	Gwon et al. ¹⁴⁹
				60, R1	~1	<i>a</i>	Cheng et al. ¹⁵⁰

^aFor crystallinity, *a* and *r* represents amorphous and rhombohedral phase, respectively.

^bGe(II) (η²-((NBu^t)CH(CH₃)CH(CH₃)) (NBu^t)).

^cAnnealing at 350°C.

^dAnnealing at 165°C or above.

^eAnnealing at 190°C or above.

^fAnnealing at 170°C.

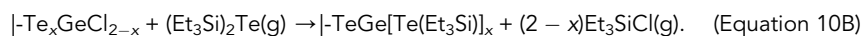
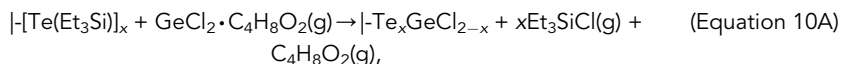
In₂Se₃ has four polymorphs (α , β , δ , γ phase).¹⁵¹ Its 2D structures (α , β , δ phase) are composed of Se-In-Se-In-Se layers, but have different stacking orders. These 2D structures are promising for random access memory, thermoelectrics, and photodetectors. Using ALD, Hatanpää et al.¹⁴¹ and Pore et al.¹⁴² did some preliminary studies on an ALD process of In₂Se₃ using InCl₃ and a selenium alkylsilyl compound ((Et₃Si)₂Se) as precursors at 295°C. EDX revealed the as-deposited films with an In:Se ratio of 41:59.

GeX

Germanium monochalcogenides (GeX: GeS, GeSe, and GeTe) are group IVA compounds. GeS and GeSe have an orthorhombic layered crystal structure.^{152–154} They both are *p*-type semiconductors with narrow band gaps of 1.55–1.65 eV and 1.1–1.2 eV,^{154,155} respectively, showing great potential for optoelectronics. GeTe has two crystalline polymorphs, i.e., a rhombohedral layered structure at low temperature and a cubic rock salt structure above 400°C.¹⁵⁶ Recently, there has much interest in synthesizing 2D GeX via ALD (Table 7).

GeS and GeSe. Kim et al.¹⁴³ reported the synthesis of GeS thin films using rac-1,3-di-tert-butyl-4,5-dimethyl-1,3-diaza-2-germacyclopentane-2-ylidene (Ge(II) (η²-((But)CH(CH₃)CH(CH₃)) (NBu^t))) and H₂S. The GPC was ~0.28 Å/cycle in the range of 50°C–75°C, but dramatically decreased from 0.21 Å/cycle at 100°C to 0.02 Å/cycle in the range of 150°C–200°C. This declining GPC might be due to precursor desorption at high temperature. XRD confirmed that the GeS film deposited at 70°C was amorphous. In another work, GeSe films were deposited via ALD using HGeCl₃ and [(CH₃)₃Si]₂Se as precursors in the range of 70°C–150°C.¹⁴⁴ This ALD process had a decreasing GPC with increasing temperature. The as-deposited films were amorphous with high stoichiometry, and their crystallinity could be improved via annealing at ~350°C. However, the resultant crystal structures were not disclosed.

GeTe. Compared with the afore-mentioned ALD processes for GeS and GeSe, there have more ALD processes for growing GeTe. Pore et al.¹⁴² reported an ALD process using $(\text{Et}_3\text{Si})_2\text{Te}$ and a germanium dichloride dioxane complex ($\text{GeCl}_2 \cdot \text{C}_4\text{H}_8\text{O}_2$) as precursors at 90°C. The as-deposited films exhibited a composition of 54% Ge and 46% Te with mixed amorphous and crystalline rhombohedral phases. The underlying mechanism was later investigated by Knapas et al.¹⁴⁵ They conducted measurements using *in situ* QCM and quadrupole mass spectrometry (QMS), and then proposed surface reactions as follows:



Sarnet et al.¹⁴⁶ investigated the growth characteristics of this GeTe ALD process in the range of 90°C–175°C. They revealed that the GPC decreased from 0.52 Å/cycle at 90°C to 0.11 Å/cycle at 130°C, and there was no film growth at 150°C. The as-deposited films had nearly 1:1 Ge: Te stoichiometry and were amorphous in the range of 90°C–130°C. Via a post-annealing process at 165°C in N_2 , the as-deposited amorphous films could crystallize into the rhombohedral phase. Further annealing at 385°C transformed the films into the cubic phase, which could be recovered to the rhombohedral phase on cooling down to room temperature.

Subsequently, another ALD process of GeTe was reported by Gwon et al.¹⁴⁷ using $\text{Ge}(\text{N}(\text{Si}(\text{CH}_3)_3)_2)_2$ and $((\text{CH}_3)_3\text{Si})_2\text{Te}$ as precursors in the range of 70°C–120°C. In order to improve the reactivity of the two precursors, methanol vapor was co-injected with them to form a reactive Ge intermediate with methoxy ligands (e.g., $\text{Ge}(\text{OCH}_3)_2$) and a Te intermediate with hydrogen ligands (e.g., H_2Te) in gas phases. In the study, it was revealed that, depending on the relative Ge-precursor concentration, various compositional GeTe alloys were formed, including GeTe, GeTe_2 , and Ge_2Te . All the as-deposited films were amorphous but could be crystallized into the rhombohedral phase via post-treatment at 180°C–190°C.

In addition, Gwon et al.¹⁴⁹ developed another ALD process of GeTe using HGeCl_3 and $(\text{Me}_3\text{Si})_2\text{Te}$ as precursors. During the ALD process, HGeCl_3 was cleaved into GeCl_2 and HCl . Then, GeCl_2 reacted with $(\text{Me}_3\text{Si})_2\text{Te}$ to form GeTe films. The GPC decreased with increased temperature in the range of 50°C–100°C. The as-deposited films were amorphous and had stoichiometric GeTe composition. They could be crystallized into the rhombohedral phase via post-treatment at 170°C. Using this process, Cheng et al.¹⁵⁰ later revealed that the as-deposited films at 60°C were moderately Ge-rich ($\text{GeTe}_{0.85}$). It is interesting that these Ge-rich films could be controllably tuned to Te-rich films by incorporating an ALD process of Te.

Other ALD processes have also been explored. Pore et al.¹⁴² investigated several Ge precursors (GeBr_2 , $\text{Ge}(\text{N}(\text{SiMe}_3)_2)_2$, $\text{Ge}(\text{OtBu})_2$, GeCl_4 , and GeBr_4) to couple with $(\text{Et}_3\text{Si})_2\text{Te}$. However, only the $\text{GeBr}_2/(\text{Et}_3\text{Si})_2\text{Te}$ worked at 140°C.

SnS and SnS_2

Tin sulfides (SnS_x) are also group IVA chalcogenides and can exist in two notable 2D layered structures: orthorhombic SnS (Figure 9A) and hexagonal SnS_2 (Figure 10A).¹⁶⁴ SnS is an intrinsic *p*-type semiconductor with an optical band gap of 1.0–1.5 eV,^{165,166} while SnS_2 is an *n*-type semiconductor with an optical band gap of 2.2–2.4 eV.^{157,161,167} Recent ALD studies for SnS_x thin films are summarized in Table 8.

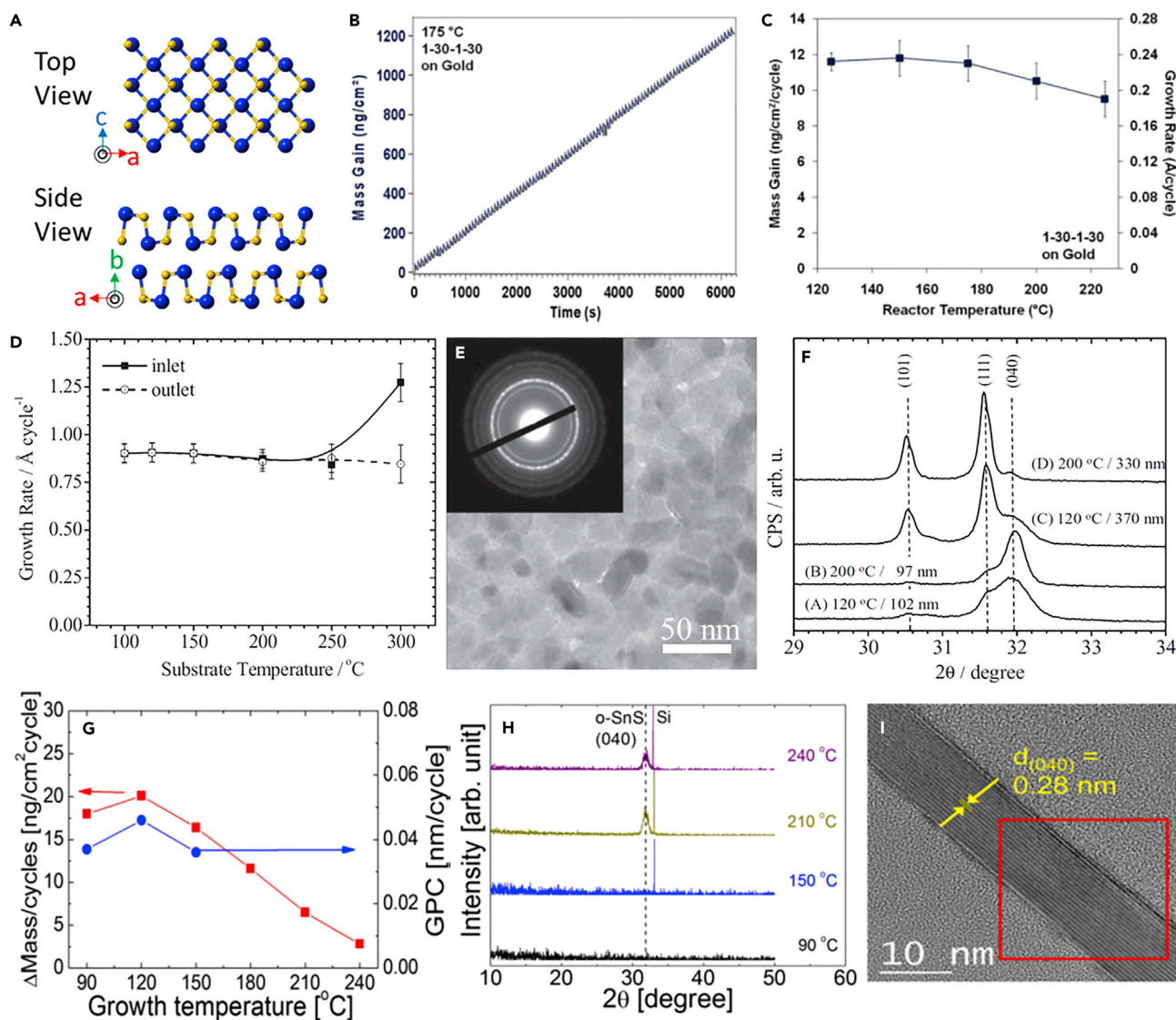


Figure 9. 2D Layered SnS Materials via ALD Processes

- (A) Schematic crystal structure of the layered orthorhombic SnS phase.¹⁵⁷ Reprinted (adapted) with permission from Yang et al.¹⁵⁷ Copyright (2016) American Chemical Society.
- (B) *In situ* QCM data of mass gain versus time for SnS ALD on gold at 175 °C for 100 cycles.¹⁵⁸
- (C) Mass gain per cycle versus reactor temperature for SnS ALD on gold.¹⁵⁸ Reprinted (adapted) with permission from Kim and George.¹⁵⁸ Copyright (2010) American Chemical Society.
- (D) The GPC of SnS films deposited between 100 °C and 300 °C.¹⁵⁹
- (E) Plan-view TEM image of a 25-nm thick SnS film deposited at 120 °C. The inset is an SAED pattern indicating the polycrystalline structure of the film.¹⁵⁹
- (F) XRD spectra of SnS films deposited at 120 °C and 200 °C at two different film thicknesses for each deposition temperature.¹⁵⁹ Reprinted (adapted) with permission from Sinsermsuksakul et al.¹⁵⁹ Copyright (2011) John Wiley and Sons.
- (G) The variation in the GPC and mass gain per cycle of the SnS films as a function of the growth temperature.¹⁶⁰
- (H) θ-2θ XRD of the films grown in the temperature range from 90 °C to 240 °C.¹⁶⁰
- (I) HR-TEM image of the SnS film grown at 240 °C.¹⁶⁰ Reprinted (adapted) with permission from Baek et al.¹⁶⁰ Copyright (2017) American Chemical Society.

SnS. It is challenging to synthesize a single-phase SnS without the other phases (i.e., Sn₂S₃ and SnS₂), because Sn⁴⁺ is more stable than Sn²⁺. In this regard, there have some recent efforts using ALD for SnS thin films with Sn²⁺ precursors. Kim and George¹⁵⁸ first demonstrated an ALD process of SnS using tin(II)

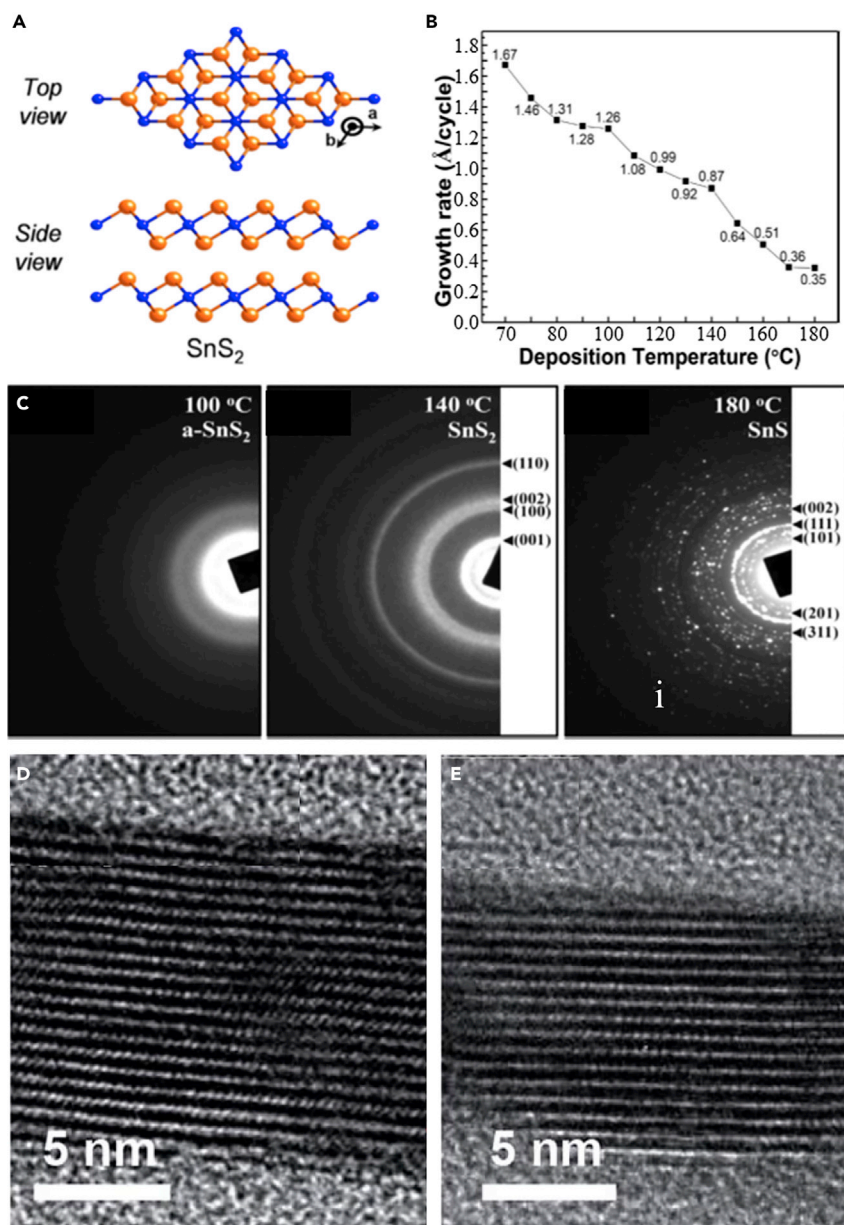


Figure 10. 2D Layered SnS_2 Materials via ALD Processes

(A) Schematic drawing of a hexagonal SnS_2 layered structure.¹⁶¹ Reprinted (adapted) with permission from Ahn et al.¹⁶¹ Copyright (2015) American Chemical Society.

(B) The GPCs of ALD SnS_2 with deposition temperature, using TDMA-Sn and H_2S as precursors.¹⁶²

(C) SAED patterns of 50-nm-thick SnS_x films deposited on Si substrates at (left) 100 $^{\circ}\text{C}$, (middle) 140 $^{\circ}\text{C}$, and (right) 180 $^{\circ}\text{C}$.¹⁶² Reprinted (adapted) with permission from Ham et al.¹⁶² Copyright (2013) American Chemical Society.

(D and E) HR-TEM images of SnS_2 films annealed at (D) 300 $^{\circ}\text{C}$ and (E) 350 $^{\circ}\text{C}$.¹⁶³ Reprinted (adapted) with permission from Ham et al.¹⁶³ Copyright (2016) The Royal Society of Chemistry.

2,4-pentanedionate (also known as tin(II) acetylacetonate or $\text{Sn}(\text{acac})_2$) and H_2S as precursors. The related two half surface reactions are as follows:

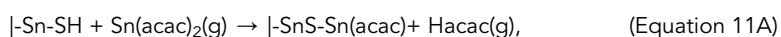


Table 8. Summary of SnS and SnS₂ Films Synthesized via Various ALD Processes

Materials	Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	GPC (Å/Cycle)	Crystallinity ^a	Reference
	Formula	Temperature (°C)					
SnS	Sn(acac) ₂	105	H ₂ S	125–225, R1	0.19–0.23	–	Kim et al. ¹⁵⁸
		95		390, R1	4.7	o	Browning et al. ¹¹⁴
		80		80–200, R1	–	o or c ^b	Bilousov et al. ¹⁶⁸
	Sn(amd) ₂	95	H ₂ S	100–200, R1	0.86–0.9	o	Sinsermsuksakul et al. ¹⁵⁹
	Sn(dmamp) ₂	52	H ₂ S	90–240, R1	0.36 at 150°C	c and o (90–180°C), o (210–240°C)	Baek et al. ¹⁶⁰
	TDMA-Sn	50	H ₂ S	160–180, R1	0.35–0.51	o	Ham et al. ¹⁶²
		40		100, R3	0.75	o ^c	Jang et al. ⁴⁴
	Sn precursor ^d	45	H ₂ S	50–200, R1	0.5–1.2	o (70°C)	Kim et al. ¹⁴³
SnS ₂	TDMA-Sn	55	H ₂ S	135, R1	–	h (on Cu ₂ S), a (on quartz)	Thimsen et al. ¹⁶⁹
		50		60–150, R1	0.64–1.67	a (60–120°C), h (140–150°C)	Ham et al. ¹⁶²
		50		150, R1	~0.27	h ^e	Ham et al. ¹⁶³ ; Lee et al. ¹⁷⁰
		50		150, R1	~0.51	a (<11.2 nm), h (≥ 11.2 nm)	Wondeok et al. ¹⁷¹
		50		150, R1	0.75	h ^f	Lee et al. ¹⁶⁴
	Sn(OAc) ₄	130	H ₂ S	150, R1	0.11	te ^g	Mattinen et al. ¹⁷²

^aFor crystallinity, c, h, o, and te stand for cubic, hexagonal, orthorhombic, and tetragonal, respectively.

^bOrthorhombic phase with unsaturated H₂S at 200°C, and cubic phase with saturated H₂S.

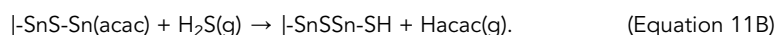
^cAnnealing in H₂ at 360°C to convert SnS₂ into SnS.

^dSn(II) (η²-((NBu⁺)CH(CH₃)CH(CH₃)) (NBu⁺).

^eAnnealing in sulfur at 300°C for 20 min.

^fAnnealing in H₂S at 350°C for 1 h.

^gAnnealing in H₂S/N₂ at 250°C for 1 h.



In situ QCM measurements confirmed the self-limiting growth of SnS (Figure 9B), and XRR measurements revealed that the GPC was almost independent of substrate temperature from 125°C to 225°C (Figure 9C). In a subsequent ALD study, Browning et al.¹¹⁴ deposited SnS using the precursors. They found that the as-deposited films were in mixed phases at temperatures lower than 250°C, while they were the orthorhombic SnS phase at temperatures above 300°C.

Sinsermsuksakul et al.¹⁵⁹ later explored another ALD SnS process using bis(*N,N'*-diisopropylacetamidinato)tin(II) (Sn(MeC(N-*i*Pr)₂)₂ (also referred as Sn(amd)₂) and H₂S. In the range of 100°C–250°C, the GPC of this process was nearly temperature independent, varying from 0.86 to 0.9 Å/cycle (Figure 9D). At 300°C, the GPC showed a dramatic increase, possibly due to the decomposition of Sn(amd)₂. Raman analyses confirmed that the films grown below 200°C were mostly the stoichiometric SnS phase with negligible Sn₂S₃ and SnS₂ phases. XPS detected negligible C, N, and O impurities. TEM and XRD further confirmed that the as-grown SnS films were polycrystalline (Figures 9E and 9F). XRD confirmed the orthorhombic SnS phase for the films grown at 120°C and 200°C. Interestingly, XRD revealed that crystal orientations changed from the (040) plane to the (111) plane with film thickness, instead of growth temperature (Figure 9F). In comparison, the SnS films grown at 200°C exhibited better crystallinity. In a later study from the same group, Kim

et al.¹⁴³ developed a new Sn precursor, Sn(II) (η^2 -((NBu^t)CH(CH₃)CH(CH₃) (NBu^t)). Paired with H₂S, this Sn precursor enabled an ALD process of SnS in the range of 50°C–200°C, which achieved a nearly constant GPC of 1.2 Å/cycle in the range of 50°C–125°C and 0.5 Å/cycle in the range of 150°C–200°C. The drop in GPC was believed to be mainly due to desorption of the Sn precursor at higher temperatures.

In addition, more ALD processes have been reported. In one work, Baek et al.¹⁶⁰ synthesized 2D orthorhombic SnS thin films using bis(1-dimethylamino-2-methyl-2-propoxy)tin(II) (Sn(dmamp)₂) and H₂S as precursors in the range of 90°C–240°C. This ALD process had its maximum GPC at 120°C (Figure 9G). The films grown at temperatures above 150°C became rougher, and this made it difficult to determine film thickness. The θ -2 θ XRD results revealed that the films grown at $\geq 210^\circ\text{C}$ showed the orthorhombic (004) orientation parallel to the substrate (Figure 9H). HR-TEM images confirmed that the SnS films grown at 240°C were layered, showing 0.28-nm interplanar spacing consistent with orthorhombic (040) planes (Figure 9I). XPS also confirmed the high purity of SnS films and detected little impurities. This work also disclosed that the SnS growth was strongly dependent on the substrates. For example, SnS films grown on Al₂O₃ surfaces suffered severe growth retardation at the initial growth stage at 90°C, but showed nearly linear growth on SiO₂ surfaces. Moreover, it was observed that SnS growth on Al₂O₃ ceased after 600 ALD cycles at 240°C.

Besides the Sn²⁺ precursors discussed above, one study used an Sn⁴⁺ precursor, tetrakis(dimethylamido)tin (TDMA-Sn), and H₂S for growing SnS films.¹⁶² In the work, Ham et al.¹⁶² found that the as-deposited films evolved from SnS₂ at $\leq 150^\circ\text{C}$ to SnS at $\geq 160^\circ\text{C}$. However, the SnS films obtained were not highly stoichiometric, due to some minor impurities of SnS₂. More details are discussed in the next section on SnS₂. To overcome SnS₂ impurities, Jang et al.⁴⁴ reported a simple and novel R3 approach to obtain single-phase SnS thin films. The amorphous SnS₂ films were first deposited via ALD using TDMA-Sn and H₂S at 100°C. Then, the films were converted to the orthorhombic SnS phase via a reduction in H₂ at 360°C. This approach enabled high phase purity, good surface smoothness, and good adhesion to SiO₂ substrates.

SnS₂. Thimsen et al.¹⁶⁹ made the first attempt on ALD of SnS₂, which involved a tri-layer stack of binary metal sulfides (Cu₂S, SnS₂, and ZnS). In the work, SnS₂ films were deposited at 135°C using TDMA-Sn and H₂S as precursors. The as-deposited SnS₂ films showed dependence on the substrates. In this regard, it was found that a Cu₂S surface facilitated the growth of (100)-oriented crystalline and stoichiometric SnS₂ films with an S/Sn ratio of ~ 2 . In contrast, the films deposited on a quartz or ZnS surface were amorphous and sulfur deficient. In a later work, Ham et al.¹⁶² conducted a systematic study on ALD SnS₂ films using the same precursors. It was found that the GPC decreased with temperature, i.e., 1.67 Å/cycle at 60°C and 0.35 Å/cycle at 180°C (Figure 10B). The decreasing GPCs were believed to be due to the desorption of surface species on the substrates. The films evolved from an amorphous SnS₂ phase at $\leq 120^\circ\text{C}$ and hexagonal SnS₂ at both 140°C and 150°C to orthorhombic SnS at $\geq 160^\circ\text{C}$ (Figure 10C). Thus, this work clearly demonstrated that ALD is particularly capable of tuning compound composition and crystallinity by adjusting the deposition temperature and varying the substrates.

In order to further improve SnS₂ crystallinity, several studies have invested more effort into the ALD SnS₂ process using TDMA-Sn and H₂S as precursors. Ham et al.¹⁶³ and Lee et al.¹⁷⁰ deposited weak hexagonal SnS₂ films at 150°C but

improved the crystallinity via post-annealing processes in sulfur vapor in the range of 250°C–350°C (Figures 10D and 10E). Ham et al.¹⁶³ found that a 12-nm thick as-deposited SnS₂ film remained almost the same thickness after a 20-min annealing at 300°C (Figure 10D), but changed to ~9.6 nm after a 20-min annealing at 350°C (Figure 10E). This was possibly due to the low melting temperature of SnS₂, leading to volatilization easily at a relatively low temperature (e.g., 350°C). After an annealing at 300°C, the S/Sn ratio was 2.01.¹⁶³ Encouragingly, Lee et al.¹⁶⁴ obtained a better crystallinity of SnS₂ films without surface defects at 350°C using a three-step annealing approach, i.e., annealing sequentially at 250°C, 300°C, and 350°C in H₂S gas for 1 h at each temperature. The initial annealing at 250°C stabilized the surface of SnS₂ and inhibited re-evaporation or reduction. In a recent work, interestingly, Seo et al.¹⁷¹ revealed that the crystallinity of SnS₂ films was thickness related. They deposited films on various substrates (e.g., Si, SiO₂/Si, and glass) at 150°C. They found that the as-deposited films were amorphous when their thicknesses were less than 11.2 nm. However, they were hexagonal SnS₂ when their thickness was larger than 11.2 nm. This indicated a critical thickness of crystallization for the SnS₂ films. This has been confirmed by XRD, Raman, and TEM analyses. In addition, XPS measurements revealed that films less than 11.2 nm had multiphases of SnS_x, while films thicker than 11.2 nm were SnS₂.

A more recent study investigated tin(IV) acetate (Sn(OAc)₄) and H₂S as precursors for ALD SnS₂.¹⁷² In the study, Mattinen et al. performed the ALD at 150°C, with a GPC of ~0.17 Å/cycle, and the as-deposited films were amorphous. Using post-annealing in a H₂S/N₂ atmosphere at 250°C, the films were converted into stoichiometric tetragonal SnS₂ films, one of the layered structures.¹⁷³ The researchers demonstrated high-quality SnS₂ films with precisely controlled thickness by adjusting the ALD cycle number, wafer-scale (5 × 5 cm²) uniformity, and good conformality on 3D trenched structures.

Sb₂Te₃

Sb₂Te₃ is a group VA chalcogenide with cubic and rhombohedral phases.^{174,175} In layered rhombohedral Sb₂Te₃, the unit cell consists of three quintuple layers (QL), and each individual QL has atoms covalently bonded in the following atomic arrangement: Te-Sb-Te-Sb-Te (Figure 11A).^{175,178} Sb₂Te₃ exhibits excellent *p*-type thermoelectric properties at room temperature and is widely used as a topological insulator. On the other hand, Sb₂Te₃ has a crystalline-amorphous phase transition at 140°C, showing promise for phase-change memory devices.¹⁷⁹ Consequently, many efforts have been invested in developing 2D Sb₂Te₃, and multiple ALD processes have been developed to date (Table 9).

Pore et al.¹⁴² investigated an ALD process of Sb₂Te₃ using SbCl₃ and a tellurium alkylsilyl compound ((Et₃Si)₂Te). The GPC was ~0.6 Å/cycle at 60°C but dropped to 0.17 Å/cycle at 90°C and 0.02 Å/cycle at 140°C. The resultant Sb₂Te₃ films were the rhombohedral phase. Similarly, Hatanpää et al.¹⁴¹ and Zastrow et al.¹⁸⁰ reported a GPC of 0.57 Å/cycle at 60°C and 80°C, respectively. Knapas et al.¹⁴⁵ confirmed repeatable growth between the two precursors using *in situ* QCM.

In another ALD process, SbCl₃ was coupled with (trimethylsilyl) telluride ((Me₃Si)₂Te) as precursors.¹⁸¹ Gu et al.¹⁸¹ found that the optimal deposition temperatures were between 60°C and 70°C in order to produce relatively smooth polycrystalline Sb₂Te₃ thin films. Higher temperatures were prone to cause precursor desorption and resulted in discontinuous Sb₂Te₃ films. In addition, substrates had evident effects on the growth of Sb₂Te₃. Gu et al.¹⁸¹ found that, compared with bare Si substrates,

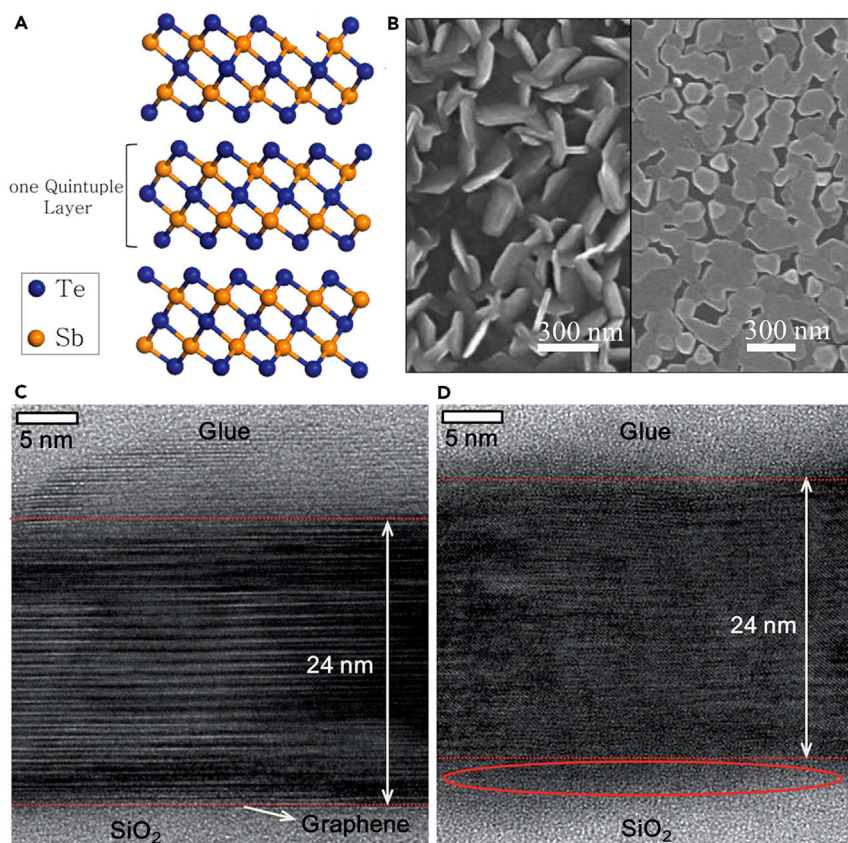


Figure 11. 2D Layered Sb_2Te_3 Materials via ALD Processes

(A) Atomic structure of Sb_2Te_3 with three QLs.¹⁷⁵ Reprinted (adapted) with permission from Kim et al.¹⁷⁵ Copyright (2012) National Academy of Sciences.
 (B) Field emission SEM images of ALD Sb_2Te_3 morphology on (left) dry oxide Si substrate and (right) Si substrate covered with native oxide.¹⁷⁶ Reproduced with permission from Zhang et al.¹⁷⁶ Copyright (2012) The Electrochemical Society.
 (C and D) Cross-sectional HR-TEM images of Sb_2Te_3 on (C) graphene and (D) Si substrate with a native SiO_2 layer.¹⁷⁷ Reprinted (adapted) with permission from Zheng et al.¹⁷⁷ Copyright (2015) The Royal Society of Chemistry.

Si substrates with either native oxide or thermal oxide terminated with $-\text{OH}$ species were favorable to the growth of Sb_2Te_3 films. Zhang et al.¹⁷⁶ and Nminibapiel et al.¹⁸² further observed that Sb_2Te_3 crystallites grew nearly vertically on dry oxide Si substrate (achieved by heating the Si substrate in pure O_2 at 900°C – 1200°C), while Sb_2Te_3 islands were formed on Si substrate with native oxide at 70°C (Figure 11B). Zheng et al.¹⁷⁷ even studied Sb_2Te_3 growth on graphene in the range of 60°C – 100°C . The maximum GPC of $1.2 \text{ \AA/cycle}$ was realized at $\sim 70^\circ\text{C}$. Similar to the growth on Si substrates with a native oxide layer, Zheng et al. also revealed that Sb_2Te_3 grew dominantly in an island-like mode on graphene. The benefits of graphene lie in avoiding the oxidation of Sb_2Te_3 by $-\text{OH}$ sites associated with other substrates. This inhibited the formation of an interfacial layer and retained the crystal structures of Sb_2Te_3 (Figures 11C and 11D).

There have been some other efforts on ALD processes. Pore et al.¹⁴² tried different Sb precursors with $(\text{Et}_3\text{Si})_2\text{Te}$. It was found that, among the Sb precursors tested, antimony triethoxide ($\text{Sb}(\text{OEt})_3$) reacted with $(\text{Et}_3\text{Si})_2\text{Te}$ to grow films at 140°C , but the study lacked more detailed information. Han et al.¹⁸⁴ reported an ALD process

Table 9. Summary of Sb_2Te_3 , Bi_2S_3 , Bi_2Se_3 , and Bi_2Te_3 Films Synthesized via Various ALD Processes

Materials	Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	GPC (Å/Cycle)	Crystallinity ^a	Reference
	Formula	Temperature (°C)					
Sb_2Te_3	SbCl_3	30	$(\text{Et}_3\text{Si})_2\text{Te}$	60–140, R1	0.02–0.6	<i>r</i>	Hatanpää et al. ¹⁴¹ ; Pore et al. ¹⁴²
		55	$(\text{Et}_3\text{Si})_2\text{Te}$	80, R1	0.16	<i>r</i>	Zastrow et al. ¹⁸⁰
		70	$(\text{Me}_3\text{Si})_2\text{Te}$	50–180, R1	–	<i>r</i>	Zhang et al. ¹⁷⁶ ; Gu et al. ¹⁸¹ ; Nminibapiel et al. ¹⁸²
				60–100, R1	1.2	–	Zheng et al. ¹⁷⁷
	$\text{Sb}(\text{OEt})_3$	–	$(\text{Et}_3\text{Si})_2\text{Te}$	140, R1	–	–	Pore et al. ¹⁴²
		40	$(\text{Me}_3\text{Si})_2\text{Te}$	70, R1	0.4	<i>r</i>	Gwon et al. ¹⁴⁹ ; Eom et al. ¹⁸³
	$\text{Sb}(\text{NMe}_2)_3$	40	$\text{Te}(\text{GeMe}_3)_2$	80, R1	–	mixed phases	Han et al. ¹⁸⁴
Bi_2S_3	$\text{Bi}(\text{thd})_3$	200	H_2O	300, R3	–	<i>o</i> ^b	Liu et al. ⁴³
		95	H_2S	125–300, R1	0.34–0.37	<i>o</i>	Mahuli et al. ¹⁸⁵
Bi_2Se_3	BiCl_3	140	$(\text{Et}_3\text{Si})_2\text{Se}$	165, R1	0.97	–	Hatanpää et al. ¹⁴¹ ; Pore et al. ¹⁴²
				160–350, R1	0.1–1.6	major <i>r</i> and minor <i>o</i>	Sarnet et al. ¹⁸⁶
Bi_2Te_3	BiCl_3	140	$(\text{Et}_3\text{Si})_2\text{Te}$	165, R1	~1.2	–	Hatanpää et al. ¹⁴¹ ; Pore et al. ¹⁴²
				160–350, R1	0.05–1.1	<i>r</i>	Sarnet et al. ¹⁸⁷
			$(\text{Me}_3\text{Si})_2\text{Te}$	165–170, R1	–	<i>r</i>	Zhang et al. ¹⁷⁶ ; Nminibapiel et al. ¹⁸²
				100–160, R1	0.4–1.3	<i>r</i>	Bae et al. ¹⁸⁸
	$(\text{Me}_2\text{N})_3\text{Bi}$	RT	$(\text{Et}_3\text{Si})_2\text{Te}$	70–120, R1	–	<i>r</i>	Rusek et al. ¹⁸⁹
	$(\text{MeEtN})_3\text{Bi}$	RT	$(\text{Et}_3\text{Si})_2\text{Te}$	120, R1	–	<i>r</i>	Rusek et al. ¹⁸⁹
	$[(\text{Me}_3\text{Si})_2\text{NBi}-\mu\text{-NSiMe}_3]_2$	100	$(\text{Et}_3\text{Si})_2\text{Te}$	120–170, R1	~1.1	<i>r</i>	Rusek et al. ¹⁸⁹

^aFor crystallinity, *o* and *r* represents orthorhombic and rhombohedral phase, respectively.

^bPost-sulfurization in sulfur vapor at 500°C for 20 min.

using $\text{Sb}(\text{NMe}_2)_3$ and $\text{Te}(\text{GeMe}_3)_2$ at 80°C. In the process, $\text{Te}(\text{GeMe}_3)_2$ precursor was thermally stable up to 120°C, while $\text{Sb}(\text{NMe}_2)_3$ started to thermally decompose at 60°C. The as-deposited films showed a composition of $\text{SbTe}_{1.32}$ with a small fraction of Sb. Raman analysis revealed a mixture of amorphous and crystalline phases. In addition, Eom et al.¹⁸³ demonstrated an ALD process using $\text{Sb}(\text{OC}_2\text{H}_5)_3$ (i.e., $\text{Sb}(\text{OEt})_3$) and $((\text{CH}_3)_3\text{Si})_2\text{Te}$ (i.e., $(\text{Me}_3\text{Si})_2\text{Te}$) as precursors at 70°C. The film growth on the TiN substrates had no nucleation incubation and realized a GPC of 0.4 Å/cycle. The as-deposition films were the rhombohedral Sb_2Te_3 phase with high stoichiometry.

Bi_2X_3

Bi_2S_3 . In group VA, there are three bismuth chalcogenides reported by ALD, i.e., Bi_2S_3 , Bi_2Se_3 , and Bi_2Te_3 . Bi_2S_3 has the layered orthorhombic and rhombohedral structures.¹⁹⁰ The orthorhombic phase is thermodynamically stable under ambient conditions.¹⁹¹ Liu et al.⁴³ first reported an ALD process for growing Bi_2S_3 films via an R3 route. The crystalline Bi_2O_3 films were first deposited via an ALD using $\text{Bi}(\text{thd})_3$ (thd = 2,2,6,6-tetramethyl-3,5-heptanedionato) and H_2O precursors at 300°C and were then converted to orthorhombic Bi_2S_3 through a post-sulfurization process in sulfur vapor at 500°C. Subsequently, Mahuli et al.¹⁸⁵ used $\text{Bi}(\text{thd})_3$ and H_2S to grow Bi_2S_3 films directly via ALD in the range of 125°C–300°C. The GPC was constant

(0.34–0.37 Å/cycle) in the range of 175°C–250°C. Otherwise, the GPC was lower at lower and higher temperatures, caused by either limited reactivity or increased desorption of precursors. The as-deposited Bi₂S₃ films at 200°C were the polycrystalline orthorhombic phase with high purity.

Bi₂Se₃. Bi₂Se₃ has two polytypes: 2D rhombohedral and 1D orthorhombic phases.^{192,193} The layered rhombohedral phase is formed by QLs, i.e., five hexagonal close-packed atomic sublayers (Se-Bi-Se-Bi-Se).¹⁹³ Bi₂Se₃ is a unique semiconductor with a narrow band gap of ~0.3 eV and thus promising in many applications, such as optical recording systems, photoelectrochemical devices, and thermoelectric devices.¹⁹⁴ The early ALD studies were conducted on Bi₂Se₃ by Pore et al.¹⁴² and Hatanpää et al.,¹⁴¹ using BiCl₃ and (Et₃Si)₂Se at 165°C. They revealed that a GPC of 0.97 Å/cycle could be achieved, and the films had a good stoichiometry of 41% Bi and 59% Se. In 2015, Sarnet et al.¹⁸⁶ conducted a thorough study on this process in the range of 160°C–350°C. They found that the film growth had a strong dependence on temperature, dropping from 1.6 Å/cycle at 160°C to 0.1 Å/cycle at 250°C. At 275°C and 300°C, the as-deposited films were discontinuous. In the entire temperature range, the films had high stoichiometry. The films at 160°C were polycrystalline with a major rhombohedral phase and minor metastable orthorhombic (1D) phase.

Bi₂Te₃. Bi₂Te₃ has a layered rhombohedral structure formed by QLs where atoms follow the sequence Te-Bi-Te-Bi-Te.¹⁹⁵ It has high electrical conductivity, low thermal conductivity, and a band gap of 0.16 eV.¹⁹⁶ Pore et al.¹⁴² and Hatanpää et al.¹⁴¹ made the first attempt on ALD Bi₂Te₃ using BiCl₃ and (Et₃Si)₂Te at 165°C. Then, Sarnet et al.¹⁸⁷ carried out a systematic investigation on the process in the range of 160°C–350°C. It was interesting that the Bi₂Te₃ ALD showed much similarity in growth characteristics as the Bi₂Se₃ ALD using BiCl₃ and (Et₃Si)₂Te.¹⁸⁶ This might be attributed to the similar properties of the precursors. The as-deposited films were the polycrystalline rhombohedral phase with high stoichiometry. Another precursor set for ALD Bi₂Te₃ is BiCl₃ and (Me₃Si)₂Te, reported by Zhang et al.¹⁷⁶ and Nminibapiel et al.¹⁸² at 165°C and 170°C, respectively. They commonly disclosed that the initial nucleation of the Bi₂Te₃ thin film was an island growth mode, leading to the vertical growth of islands and clusters rather than lateral growth. Both annealing and longer ALD precursor dose times improved film coverage. In addition, the two studies also reported that the as-deposited Bi₂Te₃ films were the layered rhombohedral structure. In a subsequent work, Bae et al.¹⁸⁸ further clarified the Bi₂Te₃ growth characteristics and film properties in the range of 100°C–160°C. They disclosed that the GPC increased with temperature from 0.4 Å/cycle at 100°C to 1.3 Å/cycle at 160°C. The as-deposited Bi₂Te₃ films were the polycrystalline rhombohedral phase.

In order to overcome the high source temperature (140°C) of BiCl₃ and avoid the formation of corrosive HCl, Rusek et al.¹⁸⁹ explored five homoleptic Bi precursors to pair with (Et₃Si)₂Te for growing Bi₂Te₃ via ALD. Based on the results from differential scanning calorimetry and nuclear magnetic resonance spectroscopy, (Me₂N)₃Bi, (MeEtN)₃Bi, and [(Me₃Si)₂NBi-μ-NSiMe₃]₂ showed high thermal stability and high reactivity with (Et₃Si)₂Te at relatively low temperatures. All the resultant Bi₂Te₃ films using the three Bi precursors yielded high stoichiometry and the polycrystalline rhombohedral structure with perpendicularly oriented crystallites to the substrates. In addition, Bi₂Te₃ films from [(Me₃Si)₂NBi-μ-NSiMe₃]₂ at 170°C exhibited better flatness and smoothness on Si and SiO₂ substrates, and the GPCs were 1.1 and 1.2 Å/cycle on Si and SiO₂ substrates, respectively. All these ALD processes for Bi₂X₃ are summarized in Table 9.

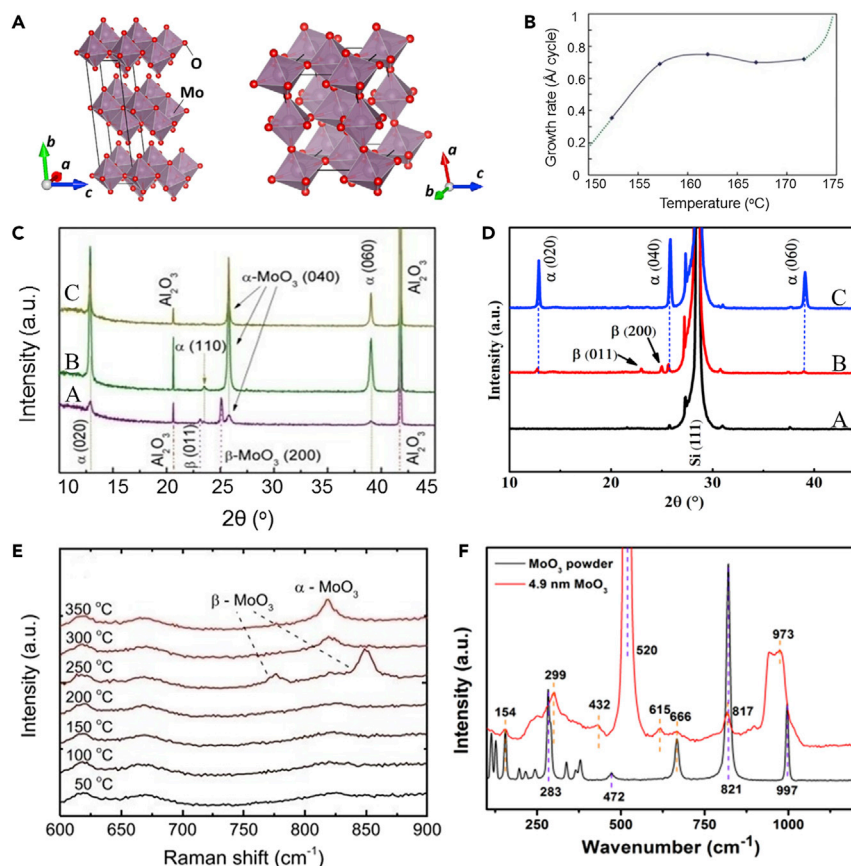


Figure 12. 2D Layered MoO₃ Materials via ALD Processes

(A) Schematic illustration of the crystal structures of (left) orthorhombic layered α -MoO₃ and (right) monoclinic β -MoO₃. The gray and red balls correspond to Mo and O atoms, respectively. The black lines show each unit.¹⁹⁷ Reprinted (adapted) with permission from Shimizu et al.¹⁹⁷ Copyright (2017) Elsevier.

(B) GPCs of MoO₃ as a function of the substrate temperature for the ALD process using Mo(CO)₆ and (O₃ + H₂O) as precursors.¹⁹⁸ Reprinted (adapted) with permission from Diskus et al.¹⁹⁸ Copyright (2011) The Royal Society of Chemistry.

(C) XRD pattern of 40 nm thin films of MoO₃ deposited on Al₂O₃ by ALD, annealed in air: (A) 8 min at 400°C, (B) 4 min at 600°C, and (C) 7.5 min at 600°C.¹⁹⁹ Reprinted (adapted) with permission from Diskus et al.¹⁹⁹ Copyright (2012) AIP Publishing.

(D) XRD patterns of MoO₃ films deposited on Si(111) substrates at 162°C, and formed at various annealing temperatures in air: (A) as-deposited, (B) annealed for 15 min at 300°C, and (C) annealed for 15 min at 400°C.²⁰⁰ Reprinted (adapted) with permission from Dai et al.²⁰⁰ Copyright (2018) Springer Nature.

(E) Raman spectra of the MoO₃ films deposited at temperatures between 50°C and 350°C for films approximately 20 nm thick.²⁰¹ Reprinted (adapted) with permission from Vos et al.²⁰¹ Copyright (2016) AIP Publishing.

(F) Raman spectra of both commercial MoO₃ powder and ALD-deposited 2D-MoO₃ nano-films.²⁰² Reprinted (adapted) with permission from Wei et al.²⁰² Copyright (2018) Elsevier.

Layered Transition Metal Oxides

MoO₃

MoO₃ has two crystal phases, thermodynamically stable orthorhombic α -MoO₃ and metastable monoclinic β -MoO₃ (Figure 12A),¹⁹⁷ showing some unique physical and chemical properties. α -MoO₃ is a 2D layered crystal, while β -MoO₃ is a 3D structure. The β phase can be transformed into the layered α -MoO₃ phase at a temperature of above 350°C.²⁰³ Some recent ALD efforts for MoO₃ films are summarized in Table 10.

Table 10. Summary of MoO₃ and WO₃ Films Synthesized via Various ALD Processes

Materials	Precursor A		Precursor B	Growth Temperature (°C) and Mechanism	GPC (Å/Cycle)	Crystallinity ^a	Reference	
	Formula	Temperature (°C)						
MoO ₃	Mo(CO) ₆	RT	O ₃ +H ₂ O	152–172, R1	0.35–0.7	o ^b	Diskus et al. ¹⁹⁸	
		RT		167, R1	0.7	m ^c , o ^{d,e}	Diskus et al. ¹⁹⁹	
		70	O ₃	120 & 150, R1	0.9–3.4	–	Liu et al. ²⁰⁴	
		RT	P ^o O ₂	130–190, R1	0.47–1.0	m ^f , o ^g	Dai et al. ²⁰⁰	
	(N ^t Bu) ₂ (NMe ₂) ₂ Mo	50	O ₃	100–300, R1	0.35–2.4	–	Bertuch et al. ²⁰⁵	
		50		P ^o O ₂	50–350, R1	0.8–1.88	a (50–200°C), m (≥250°C), o (300–350°C)	Vos et al. ²⁰¹
		–			100–350, R1	–	o ^h	Wei et al. ²⁰²
		–			150 & 250, R1	0.78–1.21	o ⁱ	Xu et al. ²⁰⁶
	MoF ₆	–	Si ₂ H ₆	200, R3 ^j	–	–	Tseng et al. ²⁰⁷	
	WO ₃	W(CO) ₆	RT	O ₃	160–225, R1	0.06–0.28	tc ^k	Malm et al. ²⁰⁸
190, R1					0.21	m ^l	Nandi et al. ²⁰⁹	
67			H ₂ O	150–320, R1	0.2	–	Mamun et al. ^{210,211}	
80			H ₂ O ₂	180–200, R1	~0.1	a ^m	Jackson et al. ²¹²	
(N ^t Bu) ₂ (NMe ₂) ₂ W		75–80	H ₂ O	250–350, R1	0.05–1.0	m ⁿ	Hai et al. ²¹³ , Zhuiykov et al. ²¹⁴ , Hai et al. ²¹⁵	
WO ₂ (^t BuAMD) ₂		–	H ₂ O	200–270, R1	0.13–0.15	mixed m and o	Mouat et al. ²¹⁶	

^aa, o, m, and tc represent amorphous, orthorhombic, monoclinic, triclinic phase, respectively.

^bAnnealing in air at 600°C for 15 min.¹⁹⁸

^cAnnealing in air at 400°C for 8 min.¹⁹⁹

^dAnnealing in air at 600°C for 4 min.¹⁹⁹

^eAnnealing in air at 600°C for 7.5 min.¹⁹⁹

^fAnnealing in air at 300°C for 15 min.²⁰⁰

^gAnnealing in air at 400°C for 15 min.²⁰⁰

^hAnnealing in air at 200°C for 1 h.²⁰²

ⁱAnnealing in air at 250°C for 1 h.²⁰⁶

^jAnnealing in O₃.

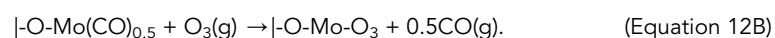
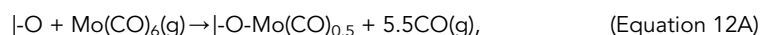
^kPost-annealing in N₂ or O₂ gas at 600–1,000°C.

^lPost-annealing in air at 600°C for 1 h.

^mPost-annealing in air at 500°C and 750°C for 2 h.

ⁿPost-annealing under O₂ flow at 550°C for 1 h.

Diskus et al.¹⁹⁸ first deposited MoO₃ via ALD using Mo(CO)₆, O₃, and H₂O as precursors in the range of 152°C–172°C. They revealed that, in all the possible combinations of H₂O and O₃ as oxygen sources, simultaneously pulsing both O₃ and H₂O produced the highest GPC and therefore was preferred for growing MoO₃ with Mo(CO)₆. The growth mechanism is proposed as follows (Equations 12A and 12B):



The GPC of this process remained constant at ~0.7 Å/cycle in the range of 157°C–172°C but was much lower at 152°C (Figure 12B). All the as-deposited films were amorphous, but annealing could evidently improve their crystallinity. A 12-min annealing at 500°C led to a pure β-MoO₃, a 15-min annealing at 500°C resulted in a mixture of α- and β-MoO₃, and a 15-min annealing at 600°C produced a pure orthorhombic α-MoO₃ phase. In a subsequent study, Diskus et al.¹⁹⁹ further studied

crystallization of the as-deposited amorphous MoO_3 films. An 8-min annealing at 400°C changed the amorphous films into crystalline $\beta\text{-MoO}_3$ with small impurities of $\alpha\text{-MoO}_3$, while 4- and 7.5-min annealing at 600°C changed the amorphous films into polycrystalline $\alpha\text{-MoO}_3$ and fully crystallized orthorhombic $\alpha\text{-MoO}_3$, respectively (Figure 12C).

In another work, Liu et al.²⁰⁴ adopted Mo(CO)_6 and O_3 to synthesize MoO_3 . MoO_3 films had a GPC of $\sim 0.9 \text{ \AA/cycle}$ at 120°C and $\sim 3.4 \text{ \AA/cycle}$ at 150°C . The researchers believed that the higher GPC at 150°C might have been due to pyrolytic breakdown of Mo(CO)_6 , which led to a CVD growth pattern. Therefore, the MoO_3 films grown at 120°C were selected for post-annealing in air in the range of 550°C – 750°C . During annealing, atomic diffusion and surface evaporation resulted in an increase in the interfacial SiO_2 layer and a decrease in the thickness of the MoO_3 film. On the other hand, the film morphology evolved with the annealing temperatures, showing $\sim 50\text{-nm}$ nanoparticles at 500°C , smaller nanoparticles at 650°C , and telephone-cord-like MoO_3 structures at 750°C . These evolutions are related to the dynamic balance between the surface evaporation and the in-plane/substrate interactions. Using the same molybdenum precursor, Dai et al.²⁰⁰ adopted O_2 plasma (PO_2) as the oxygen precursor to deposit MoO_3 ultrathin films via a PEALD in the range of 130°C – 190°C . A constant GPC of $\sim 0.76 \text{ \AA/cycle}$ was found in the range of 157°C – 172°C , which is in a good agreement with Diskus et al.'s study.¹⁹⁸ After annealing at 300°C in air, the as-deposited amorphous MoO_3 films crystallized into $\beta\text{-MoO}_3$ with minor $\alpha\text{-MoO}_3$. After annealing at above 400°C , only $\alpha\text{-MoO}_3$ was obtained (Figure 12D).

Given the fact that Mo(CO)_6 has a relatively poor thermal stability (163°C), Bertuch et al.²⁰⁵ studied $(\text{N}^t\text{Bu})_2(\text{NMe}_2)_2\text{Mo}$ as a Mo precursor to couple with O_3 for depositing MoO_3 in the range of 100°C – 300°C . TGA revealed that this new Mo precursor exhibited excellent volatility and thermal stability with complete evaporation in a single step at 183°C . The study disclosed that the GPC was consistently $0.35 \text{ \AA/cycle}$ at temperatures below 230°C but increased dramatically to $\sim 2.2 \text{ \AA/cycle}$ at 300°C . XPS analyses determined that the films deposited at a temperature above 250°C were stoichiometrically MoO_3 and had less impurities. However, they did not clarify the films' crystallinity. In a subsequent study, Vos et al.²⁰¹ deposited high-purity MoO_3 films over a wider temperature range of 50°C – 350°C via a PEALD using $(\text{N}^t\text{Bu})_2(\text{NMe}_2)_2\text{Mo}$ and O_2 plasma as precursors. Raman and XRD revealed that the deposited MoO_3 via this ALD process was amorphous in the range of 50°C – 200°C , $\beta\text{-MoO}_3$ at 250°C , and $\alpha\text{-MoO}_3$ at temperatures above 300°C (Figure 12E). In addition, a steady GPC of $\sim 0.8 \text{ \AA/cycle}$ was observed in the range of 50°C – 200°C , while there was a gradual increase in GPC from $1.25 \text{ \AA/cycle}$ at 300°C to $1.88 \text{ \AA/cycle}$ at 350°C . The researchers believed that the increased GPC at higher temperatures may be attributed to the transition from amorphous to crystalline phase, because crystal grains could lead to an increase in surface area and thereby an increase in GPC. Furthermore, the high-purity 2D $\alpha\text{-MoO}_3$ phase could also be obtained by annealing the amorphous films in air at 200°C for 1 h (Figure 12F)²⁰² and 250°C for 1 h.²⁰⁶ Xu et al.²⁰⁶ deposited MoO_3 ultrathin films on Au-covered Si wafer using this PEALD process. The GPCs at 150°C and 250°C were 0.78 and $1.21 \text{ \AA/cycle}$, respectively. In comparison, the GPC of MoO_3 films on bare Si substrates with a native oxide layer was $1.61 \text{ \AA/cycle}$ at 250°C . The different GPCs of MoO_3 on Au and SiO_2 surfaces at 250°C were attributed to the passive nature of the Au surface.

In contrast to these studies, Tseng et al.²⁰⁷ developed MoO_3 films using an R3 strategy, in which Mo metal films were first deposited via ALD using MoF_6 and Si_2H_6 as precursors²¹⁷ at 200°C , and then converted into MoO_3 via an oxidization process in O_3 .

WO₃

WO₃ can exhibit five phases related to temperature, i.e., low temperature monoclinic (ϵ -WO₃, -140°C to -50°C), triclinic (δ -WO₃, -50°C to 17°C), room temperature monoclinic (γ -WO₃, 17°C – 330°C), orthorhombic (β -WO₃, 330°C – 740°C), and tetragonal structure (α -WO₃, $>740^{\circ}\text{C}$).^{218–220} γ -WO₃ is the most stable phase at room temperature and has a layered structure. Recent ALD efforts for WO₃ are summarized in Table 10. Liu et al.²²¹ first demonstrated an ALD process using (N^tBu)₂(NMe₂)₂W and H₂O as precursors. The deposition was performed on indium tin oxide-coated glass substrates at 250°C – 350°C with a GPC of 1 Å/cycle. The as-deposited WO₃ films were oxygen deficient and a post-annealing process at 550°C in O₂ helped achieve stoichiometric WO₃ with improved crystallinity. HR-TEM analyses confirmed that the annealed WO₃ films had a well-crystallized monoclinic structure. This process has been used to grow WO₃ thin films for photo-detectors.^{213–215}

W(CO)₆ is another widely used W precursor, with ample vapor and high reactivity. Coupling W(CO)₆ with O₃, Malm et al.²⁰⁸ realized the deposition of WO₃ via ALD in the range of 160°C – 225°C , with a GPC varying from ~ 0.05 to 0.28 Å/cycle. Higher temperatures caused thermal decomposition. In the whole deposition temperature range, the as-deposited WO₃ films were partially crystalline and could crystallized into triclinic structure via annealing at 600°C – $1,000^{\circ}\text{C}$ in an atmosphere of N₂ or O₂. In a subsequent study, Nandi and Sarkar²⁰⁹ studied the surface chemistry during the ALD process of W(CO)₆/O₃ using *in situ* FTIR, confirming the attachment and removal of the carbonyl group (C=O stretching) after W(CO)₆ and O₃ dosing, respectively. The FTIR data indicated a similar surface chemistry as described in Equations 12A and 12B for MoO₃. XRD and SAED verified that the as-deposited films at 190°C were amorphous and could be converted into monoclinic WO₃ after annealing at 600°C for 1 h in atmospheric conditions. Instead of O₃, H₂O and H₂O₂ were investigated as the oxygen sources to couple with W(CO)₆ for growing WO₃. Mamun et al.^{210,211} investigated the ALD process using W(CO)₆ and H₂O in the range of 150°C – 320°C . The as-deposited films were amorphous while post-annealing above 300°C improved their crystallinity. Unfortunately, more details were not provided for this ALD process. The ALD process of WO₃ using W(CO)₆/H₂O₂ was investigated by Jackson et al.²¹² In the range of 180 – 200°C , they found that the precursor pair of W(CO)₆/H₂O₂ could not achieve a pure form of ALD growth, but continuous dosing of W(CO)₆ could always increase the growth rates. They also noticed that W(CO)₆ was prone to decompose even at a temperature of 80°C , and this explained the continuous unsaturated growth of WO₃ with extended dosing of W(CO)₆. In a recent work, WO₂(^tBuAMD) (AMD = acetamidinato) was reported and coupled with H₂O for growing WO₃ via ALD.²¹⁶ However, this ALD process was not well investigated.

CONCLUSIONS AND OUTLOOK

Since the delivery of graphene in 2004, there has been an ever-increasing interest in 2D layered structures, and to date, over 150 different 2D materials have been reported.²²² These atomically thin 2D nanosheets have exhibited numerous exotic properties and therefore are promising in a large variety of emerging applications. In pursuing these 2D materials, various methods have been developed (e.g., exfoliation, CVD, PVD, and wet chemistry)²², determinant to their quality, cost, and applications. So far, however, it is still challenging for the many different methods to produce large-scale high-quality 2D atomic layers in a controllable manner. In this context, ALD has recently been investigated as a new technique for synthesizing 2D materials, especially in the past 5 years, featuring accurate controllability over material composition and crystallinity, atomic-scale precision of material

growth, film uniformity and conformality over any shaped substrates, and at low temperatures even down to room temperature. The recent ALD efforts have exhibited these exceptional capabilities in producing 2D materials.

In this review, we have carried out a comprehensive survey on 2D layered materials synthesized by ALD, including 25 types of 2D materials and over 80 ALD processes. We have categorized the technical strategies of ALD studies into three routes. R1 is the typical ALD process, in which 2D materials are deposited into an amorphous phase, a polycrystalline phase, or a mixture of the two. The film crystallinity is related to the properties of the precursors and the substrates and the temperature. Annealing is a general measure to further improve the crystallinity of the resulting films. Most of the ALD processes reported to date are in this category. R2 is a special ALD phenomenon and has not been well studied. To date, only two studies have been reported for this technical route, i.e., MoS₂ using MoCl₅/H₂S in the range of 500°C–900°C³⁶ and WSe₂ using WCl₆/DESe in the range of 600°C–800°C.³⁷ The two studies were from the same group. This technical route features self-limiting growth of several layers of 2D materials. The number of layers of 2D materials is dependent on the deposition temperature but not the number of ALD cycles. It is different from R1; R2 enables higher quality of 2D layers, and its success is related to the formation of high-quality 2D layers lacking dangling bonds on the layer surface. This technical route has realized excellent layer uniformity up to 95% over large areas at wafer-level scale. Thus, R2 is promising, but its exact mechanism remains unclear. In the two successful cases of MoS₂ and WSe₂ via R2, the two ALD processes both used metal halides (MoCl₅ and WCl₆) as the metal sources and high deposition temperatures above 500°C. The metal halides have high reactivity and high thermal stability. The two coupled precursors (H₂S and DESe, respectively) also have the same characteristics with regard to reactivity and stability. Thus, we postulate that R2 requires precursors with high reactivity and thermal stability and needs to proceed at high deposition temperatures. Based on the two studies, other 2D materials via R2 may be possible by adopting similar precursors (e.g., WCl₆/H₂S for WS₂) and performing at high temperatures (e.g., above 500°C). In this regard, more studies are apparently needed. Of the three technical routes, R3 has some distinctive advantages and has been used for six 2D materials (h-BN,^{45,46} MoS₂,^{38,39} MoSe₂,⁴⁰ WS₂,^{41,42} SnS,⁴⁴ and Bi₂S₃⁴³) with eight ALD processes. In this technical route, ALD processes are used to deposit precursor films, which later convert into targeted 2D materials via a post-treatment (e.g., sulfurization,^{38,39} selenization,⁴⁰ or reduction⁴⁴) at high temperatures. In comparison, R3 is versatile and easy to perform for high-quality 2D films with controlled layer numbers over wafer-scale substrates. These advantages distinguish R3 from R1 and R2. Based on these benefits, R3 has great potential for large-scale high-quality 2D films with controllable layers.

Besides categorizing the different ALD processes into three technical routes, another contribution of this review lies in the systematic summary of all the ALD processes. The data in the ten tables will allow readers to examine the important parameters of the ALD processes for 2D materials. They also are a valuable source for researchers to track ALD studies in this area with ease. The third contribution of this work lies in the summary of the underlying mechanisms and growth characteristics of the different ALD processes. These are expected to help prompt new efforts for developing new precursors and for widening the applications of ALD. In summary, the review offers an express way for researchers to familiarize themselves with the ALD methodologies and processes for 2D materials. For future studies, we expect more excitement and successes in the following aspects.

New ALD Precursors and Processes

Precursors have a vital role in ALD processes. Their properties determine the success of an ALD process, such as reactivity, thermal stability, and volatility. In this regard, new precursors are urgently needed to expand ALD's capabilities for 2D materials. Ideally, they should enable high reactivity, high thermal stability, and high volatility. On the other hand, new precursors that can be produced with ease and cost-effectively are preferred. In this regard, we expect that there will be new precursors enabling low-temperature deposition of large-scale uniform well-crystallized 2D materials.

In Situ and Advanced Characterizations

Many phenomena have not been well understood during ALD processes. To this end, *in situ* lab-based techniques are valuable tools in ALD studies, such as FTIR, QCM, QMS, ellipsometry, and XPS.^{223–225} Combinations of these techniques provide insightful data on materials growth during ALD processes, which will be essential for us to understand the related surface chemistry, chemical composition, and structure of materials. For example, these lab-based techniques can explain the evolution of growth rates with temperature and the changes in surface reactions. In addition to these lab-based techniques, *in situ* synchrotron-based X-ray techniques provide us with a suite of advanced characterizations,²²⁶ including XRD, XRR, XPS, X-ray fluorescence, and X-ray absorption. They are particularly powerful to secure film thickness, roughness, crystallinity, composition, and chemical state, featuring their shorter acquisition time and higher precision than those in the lab-based techniques. They are sensitive to changes on the atomic scale and nanoscale. In this regard, these techniques should be able to provide solutions to explain the evolution of film crystallinity, chemical state, and chemical composition with film thickness, temperature, and substrate. Furthermore, *in situ* TEM is another important tool for characterizing the evolution of materials microstructures with temperature during ALD processes.²²⁷ All these *in situ* tools are particularly significant for ALD studies, and they can greatly help advance our understanding on advanced materials and the fundamentals of the related chemistry.

New Applications

With the expected progress in ALD processes and materials characterization, it can be foreseen that we will be able to produce large-scale high-quality 2D materials in a precise and controllable manner. Therefore, we would further expect that there will be a dramatic increase in applications with ALD-deposited 2D materials, such as in catalysis, energy, biomedicine, sensors, electronics, and semiconductors, ascribed to the excellent capabilities of ALD in producing 2D materials.

With this review, we expect that readers will benefit greatly in the search for high-quality 2D materials with a feasible ALD process. We also expect that there will be more interest sparked among researchers to use ALD as an important tool for developing new 2D materials for emerging applications.

ACKNOWLEDGMENTS

Funding for this research was provided by the Center for Advanced Surface Engineering, under the National Science Foundation grant no. OIA-1457888 and the Arkansas EPSCoR Program, ASSET III.

AUTHOR CONTRIBUTIONS

X.M. proposed, organized, supervised, and edited the entire work, including all the text, figures, and tables. J.C., X.H., and X.W. summarized some parts of this work.

Specifically, J.C. collected information on the ALD processes of graphene, h-BN, MoX_2 , ReS_2 , SnS_x , In_2Se_3 , Sb_2Te_3 , GeX , and B_2X_3 . X.H. collected information on the ALD processes of WX_2 , WO_3 , NiS , FeS , ZrS_2 , and HfS_2 . X.W. collected information on the ALD processes of TiS_2 , InSe , and MoO_3 .

REFERENCES

- Xu, M., Liang, T., Shi, M., and Chen, H. (2013). Graphene-like two-dimensional materials. *Chem. Rev.* 113, 3766–3798.
- Tan, C., and Zhang, H. (2015). Two-dimensional transition metal dichalcogenide nanosheet-based composites. *Chem. Soc. Rev.* 44, 2713–2731.
- Chhowalla, M., Shin, H.S., Eda, G., Li, L.-J., Loh, K.P., and Zhang, H. (2013). The chemistry of two-dimensional layered transition metal dichalcogenide nanosheets. *Nat. Chem.* 5, 263–275.
- Naguib, M., Mochalin, V.N., Barsoum, M.W., and Gogotsi, Y. (2014). 25th anniversary article: MXenes: a new family of two-dimensional materials. *Adv. Mater.* 26, 992–1005.
- Bhimanapati, G.R., Lin, Z., Meunier, V., Jung, Y., Cha, J., Das, S., Xiao, D., Son, Y., Strano, M.S., Cooper, V.R., et al. (2015). Recent advances in two-dimensional materials beyond graphene. *ACS Nano* 9, 11509–11539.
- Butler, S.Z., Hollen, S.M., Cao, L., Cui, Y., Gupta, J.A., Gutiérrez, H.R., Heinz, T.F., Hong, S.S., Huang, J., Ismach, A.F., et al. (2013). Progress, challenges, and opportunities in two-dimensional materials beyond graphene. *ACS Nano* 7, 2898–2926.
- Lin, Y.-M., Dimitrakopoulos, C., Jenkins, K.A., Farmer, D.B., Chiu, H.-Y., Grill, A., and Avouris, P. (2010). 100-GHz transistors from wafer-scale epitaxial graphene. *Science* 327, 662.
- Liu, M., Yin, X., Ulin-Avila, E., Geng, B., Zentgraf, T., Ju, L., Wang, F., and Zhang, X. (2011). A graphene-based broadband optical modulator. *Nature* 474, 64–67.
- Kim, K.S., Zhao, Y., Jang, H., Lee, S.Y., Kim, J.M., Kim, K.S., Ahn, J.-H., Kim, P., Choi, J.-Y., and Hong, B.H. (2009). Large-scale pattern growth of graphene films for stretchable transparent electrodes. *Nature* 457, 706–710.
- Zhu, Y., Murali, S., Stoller, M.D., Ganesh, K.J., Cai, W., Ferreira, P.J., Pirkle, A., Wallace, R.M., Cychosz, K.A., Thommes, M., et al. (2011). Carbon-based supercapacitors produced by activation of graphene. *Science* 332, 1537–1541.
- Anasori, B., Lukatskaya, M.R., and Gogotsi, Y. (2017). 2D metal carbides and nitrides (MXenes) for energy storage. *Nat. Rev. Mater.* 2, 16098.
- Deng, D., Novoselov, K.S., Fu, Q., Zheng, N., Tian, Z., and Bao, X. (2016). Catalysis with two-dimensional materials and their heterostructures. *Nat. Nanotechnol.* 11, 218–230.
- Xu, M., Fujita, D., and Hanagata, N. (2009). Perspectives and challenges of emerging single-molecule DNA sequencing technologies. *Small* 5, 2638–2649.
- Miró, P., Audiffred, M., and Heine, T. (2014). An atlas of two-dimensional materials. *Chem. Soc. Rev.* 43, 6537–6554.
- Novoselov, K.S., Geim, A.K., Morozov, S.V., Jiang, D., Zhang, Y., Dubonos, S.V., Grigorieva, I.V., and Firs, A.A. (2004). Electric field effect in atomically thin carbon films. *Science* 306, 666–669.
- Allen, M.J., Tung, V.C., and Kaner, R.B. (2010). Honeycomb carbon: a review of graphene. *Chem. Rev.* 110, 132–145.
- Al Balushi, Z.Y., Wang, K., Ghosh, R.K., Vila, R.A., Eichfeld, S.M., Caldwell, J.D., Qin, X., Lin, Y.-C., DeSario, P.A., et al. (2016). Two-dimensional gallium nitride realized via graphene encapsulation. *Nat. Mater.* 15, 1166–1171.
- Zhang, K., Feng, Y., Wang, F., Yang, Z., and Wang, J. (2017). Two dimensional hexagonal boron nitride (2D-hBN): synthesis, properties and applications. *J. Mater. Chem. C* 5, 11992–12022.
- Duong, D.L., Yun, S.J., and Lee, Y.H. (2017). van der Waals layered materials: opportunities and challenges. *ACS Nano* 11, 11803–11830.
- Zhuang, X., Mai, Y., Wu, D., Zhang, F., and Feng, X. (2015). Two-dimensional soft nanomaterials: a fascinating world of materials. *Adv. Mater.* 27, 403–427.
- Cai, S.-L., Zhang, W.-G., Zuckermann, R.N., Li, Z.-T., Zhao, X., and Liu, Y. (2015). The organic flatland—recent advances in synthetic 2D organic layers. *Adv. Mater.* 27, 5762–5770.
- Bonaccorso, F., Lombardo, A., Hasan, T., Sun, Z., Colombo, L., and Ferrari, A.C. (2012). Production and processing of graphene and 2d crystals. *Mater. Today* 15, 564–589.
- Nicolosi, V., Chhowalla, M., Anatids, M.G., Strano, M.S., and Coleman, J.N. (2013). Liquid exfoliation of layered materials. *Science* 340, 1226419.
- Zhong, Y.L., Tian, Z., Simon, G.P., and Li, D. (2015). Scalable production of graphene via wet chemistry: progress and challenges. *Mater. Today* 18, 73–78.
- Suntola, T. and Antson, J. (1977). Method for producing compound thin films. US Patent, 4058430, filed November 25, 1975, and granted November 15, 1977.
- Miikkulainen, V., Leskelä, M., Ritala, M., and Puurunen, R.L. (2013). Crystallinity of inorganic films grown by atomic layer deposition: overview and general trends. *J. Appl. Phys.* 113, 021301.
- Puurunen, R.L. (2005). Surface chemistry of atomic layer deposition: a case study for the trimethylaluminum/water process. *J. Appl. Phys.* 97, 121301.
- George, S.M. (2010). Atomic layer deposition: an overview. *Chem. Rev.* 110, 111–131.
- Stuyven, G., De Visschere, P., Hikavvy, A., and Neyts, K. (2002). Atomic layer deposition of ZnS thin films based on diethyl zinc and hydrogen sulfide. *J. Cryst. Growth* 234, 690–698.
- Meng, X., Wang, X., Geng, D., Ozgit-Akgun, C., Schneider, N., and Elam, J.W. (2017). Atomic layer deposition for nanomaterial synthesis and functionalization in energy technology. *Mater. Horiz.* 4, 133–154.
- Meng, X., Yang, X.Q., and Sun, X.L. (2012). Emerging applications of atomic layer deposition for lithium-ion battery studies. *Adv. Mater.* 24, 3589–3615.
- Dasgupta, N.P., Meng, X., Elam, J.W., and Martinson, A.B. (2015). Atomic layer deposition of metal sulfide materials. *Acc. Chem. Res.* 48, 341–348.
- Kim, H., and Oh, I.-K. (2014). Review of plasma-enhanced atomic layer deposition: technical enabler of nanoscale device fabrication. *Jpn. J. Appl. Phys.* 53, 03DA01.
- Weimer, A.W. (2019). Particle atomic layer deposition. *J. Nanopart. Res.* 21, 9.
- Muñoz-Rojas, D., Nguyen, V.H., Masse de la Huerta, C., Aghazadehchors, S., Jiménez, C., and Bellet, D. (2017). Spatial Atomic Layer Deposition (SALD), an emerging tool for energy materials. Application to new-generation photovoltaic devices and transparent conductive materials. *C. R. Phys.* 18, 391–400.
- Kim, Y., Song, J.-G., Park, Y.J., Ryu, G.H., Lee, S.J., Kim, J.S., Jeon, P.J., Lee, C.W., Woo, W.J., Choi, T., et al. (2016). Self-limiting layer synthesis of transition metal dichalcogenides. *Sci. Rep.* 6, 18754.
- Park, K., Kim, Y., Song, J.-G., Jin Kim, S., Wan Lee, C., Hee Ryu, G., Lee, Z., Park, J., and Kim, H. (2016). Uniform, large-area self-limiting layer synthesis of tungsten diselenide. *2D Mater.* 3, 014004.
- Martella, C., Melloni, P., Cinquanta, E., Ciinci, E., Alia, M., Longo, M., Lamperti, A., Vangelista, S., Fanciulli, M., and Molle, A. (2016). Engineering the growth of MoS_2 via atomic layer deposition of molybdenum oxide film precursor. *Adv. Electron. Mater.* 2, 1600330.
- Keller, B.D., Bertuch, A., Provine, J., Sundaram, G., Ferralis, N., and Grossman, J.C. (2017). Process control of atomic layer deposition molybdenum oxide nucleation

- and sulfidation to large-area MoS₂ monolayers. *Chem. Mater.* 29, 2024–2032.
40. Dai, T.-J., Fan, X.-D., Ren, Y.-X., Hou, S., Zhang, Y.-Y., Qian, L.-X., Li, Y.-R., and Liu, X.-Z. (2018). Layer-controlled synthesis of wafer-scale MoSe₂ nanosheets for photodetector arrays. *J. Mater. Sci.* 53, 8436–8444.
 41. Song, J.-G., Park, J., Lee, W., Choi, T., Jung, H., Lee, C.W., Hwang, S.-H., Myoung, J.M., Jung, J.-H., Kim, S.-H., et al. (2013). Layer-controlled, wafer-scale, and conformal synthesis of tungsten disulfide nanosheets using atomic layer deposition. *ACS Nano* 7, 11333–11340.
 42. Zeng, W., Feng, L.-P., Su, J., Pan, H.-X., and Liu, Z.-T. (2018). Layer-controlled and atomically thin WS₂ films prepared by sulfurization of atomic-layer-deposited WO₃ films. *J. Alloys Compd.* 745, 834–839.
 43. Liu, H.F., Antwi, K.K.A., Wang, Y.D., Ong, L.T., Chua, S.J., and Chi, D.Z. (2014). Atomic layer deposition of crystalline Bi₂O₃ thin films and their conversion into Bi₂S₃ by thermal vapor sulfurization. *RSC Adv.* 4, 58724–58731.
 44. Jang, B., Yeo, S., Kim, H., Shin, B., and Kim, S.-H. (2017). Fabrication of single-phase SnS film by H₂ annealing of amorphous SnS_x prepared by atomic layer deposition. *J. Vac. Sci. Technol. A* 35, 031506.
 45. Hao, W., Marichy, C., Journet, C., and Brioude, A. (2017). A novel two-step ammonia-free atomic layer deposition approach for boron nitride. *ChemNanoMat* 3, 656–663.
 46. Hao, W., Marichy, C., and Brioude, A. (2017). Promising properties of ALD boron nitride nanotube mats for water purification. *Environ. Sci. Nano* 4, 2311–2320.
 47. Edwards, R.S., and Coleman, K.S. (2013). Graphene synthesis: relationship to applications. *Nanoscale* 5, 38–51.
 48. Hu, Y.J., Jin, J.A., Zhang, H., Wu, P., and Cai, C.X. (2010). Graphene: synthesis, functionalization and applications in chemistry. *Acta Phys. Chim. Sin.* 26, 2073–2086.
 49. Zhang, Y., Ren, W., Jiang, Z., Yang, S., Jing, W., Shi, P., Wu, X., and Ye, Z.-G. (2014). Low-temperature remote plasma-enhanced atomic layer deposition of graphene and characterization of its atomic-level structure. *J. Mater. Chem. C* 2, 7570–7574.
 50. Mårlid, B., Ottosson, M., Pettersson, U., Larsson, K., and Carlsson, J.-O. (2002). Atomic layer deposition of BN thin films. *Thin Solid Films* 402, 167–171.
 51. Olander, J., Ottosson, L.M., Heszler, P., Carlsson, J.-O., and Larsson, K.M.E. (2005). Laser-assisted atomic layer deposition of boron nitride thin films. *Chem. Vap. Depos.* 11, 330–337.
 52. Weber, M., Koonkaew, B., Balme, S., Utke, I., Picaut, F., Iatsunskyi, I., Coy, E., Miele, P., and Bechelany, M. (2017). Boron nitride nanoporous membranes with high surface charge by atomic layer deposition. *ACS Appl. Mater. Interfaces* 9, 16669–16678.
 53. Weber, M., Coy, E., Iatsunskyi, I., Yate, L., Miele, P., and Bechelany, M. (2017). Mechanical properties of boron nitride thin films prepared by atomic layer deposition. *CrystEngComm* 19, 6089–6094.
 54. Weber, M., Lamboux, C., Navarra, B., Miele, P., Zanna, S., Dufond, M.E., Santinacci, L., and Bechelany, M. (2018). Boron nitride as a novel support for highly stable palladium nanocatalysts by atomic layer deposition. *Nanomaterials* 8, 849.
 55. Weber, M., Iatsunskyi, I., Coy, E., Miele, P., Cornu, D., and Bechelany, M. (2018). Novel and facile route for the synthesis of tunable boron nitride nanotubes combining atomic layer deposition and annealing processes for water purification. *Adv. Mater. Interfaces* 5, 1800056.
 56. Weber, M., Tuleushova, N., Zgheib, J., Lamboux, C., Iatsunskyi, I., Coy, E., Flaud, V., Tingry, S., Cornu, D., Miele, P., et al. (2019). Enhanced electrocatalytic performance triggered by atomically bridged boron nitride between palladium nanoparticles and carbon fibers in gas-diffusion electrodes. *Appl. Catal. B Environ.* 257, 117917.
 57. Weber, M., Kim, J.-Y., Lee, J.-H., Kim, J.-H., Iatsunskyi, I., Coy, E., Miele, P., Bechelany, M., and Kim, S.S. (2019). Highly efficient hydrogen sensors based on Pd nanoparticles supported on boron nitride coated ZnO nanowires. *J. Mater. Chem. A* 7, 8107–8116.
 58. Ferguson, J.D., Weimer, A.W., and George, S.M. (2002). Atomic layer deposition of boron nitride using sequential exposures of BCl₃ and NH₃. *Thin Solid Films* 413, 16–25.
 59. Weimer, M.A., Weimer, A.W., and Park, W. (2008). Theory of conduction in ultrafast metal-insulator varistors. *J. Appl. Phys.* 104, 114516.
 60. Bjelkevig, C., Mi, Z., Xiao, J., Dowben, P.A., Wang, L., Mei, W.-N., and Kelber, J.A. (2010). Electronic structure of a graphene/hexagonal-BN heterostructure grown on Ru(0001) by chemical vapor deposition and atomic layer deposition: extrinsically doped graphene. *J. Phys. Condens. Matter* 22, 302002.
 61. Beatty, J., Cao, Y., Tanabe, I., Sky Driver, M., Dowben, P.A., and Kelber, J.A. (2014). Atomic layer-by-layer deposition of h-BN(0001) on cobalt: a building block for spintronics and graphene electronics. *Mater. Res. Express* 1, 046410.
 62. Driver, M.S., Beatty, J.D., Olanipekun, O., Reid, K., Rath, A., Voyles, P.M., and Kelber, J.A. (2016). Atomic layer epitaxy of h-BN(0001) multilayers on Co(0001) and molecular beam epitaxy growth of graphene on h-BN(0001)/Co(0001). *Langmuir* 32, 2601–2607.
 63. Jones, J., Beauclair, B., Olanipekun, O., Lightbourne, S., Zhang, M., Pollok, B., Pilli, A., and Kelber, J. (2017). Atomic layer deposition of h-BN(0001) on RuO₂(110)/Ru(0001). *J. Vac. Sci. Technol. A* 35, 01B139.
 64. Kilbury, O.J., Barrett, K.S., Fu, X., Yin, J., Dinari, D.S., Gump, C.J., Weimer, A.W., and King, D.M. (2012). Atomic layer deposition of solid lubricating coatings on particles. *Powder Technol.* 221, 26–35.
 65. Consiglio, S., Clark, R.D., O'Meara, D., Wajda, C.S., Tapily, K., and Leusink, G.J. (2016). Comparison of B₂O₃ and BN deposited by atomic layer deposition for forming ultrashallow dopant regions by solid state diffusion. *J. Vac. Sci. Technol. A* 34, 01A102.
 66. Jones, J., Pilli, A., Lee, V., Beatty, J., Beauclair, B., Chugh, N., and Kelber, J. (2019). Atomic layer deposition of h-BN(0001) multilayers on Ni(111) and chemical vapor deposition of graphene on h-BN(0001)/Ni(111). *J. Vac. Sci. Technol. A* 37, 060903.
 67. Pilli, A., Jones, J., Chugh, N., Kelber, J., Pasquale, F., and LaVoie, A. (2019). Atomic layer deposition of BN as a novel capping barrier for B₂O₃. *J. Vac. Sci. Technol. A* 37, 041505.
 68. Wolf, S., Edmonds, M., Sardashti, K., Clemons, M., Park, J.H., Yoshida, N., Dong, L., Nemani, S., Yieh, E., Holmes, R., et al. (2018). Low-temperature amorphous boron nitride on Si_{0.7}Ge_{0.3}(001), Cu, and HOPG from sequential exposures of N₂H₄ and BCl₃. *Appl. Surf. Sci.* 439, 689–696.
 69. Snure, M., Paduano, Q., Hamilton, M., Shoaf, J., and Mann, J.M. (2014). Optical characterization of nanocrystalline boron nitride thin films grown by atomic layer deposition. *Thin Solid Films* 571, 51–55.
 70. Haider, A., Ozgit-Akgun, C., Goldenberg, E., Okay, A.K., and Biyikli, N. (2014). Low-temperature deposition of hexagonal boron nitride via sequential injection of triethylboron and N₂/H₂ plasma. *J. Am. Ceram. Soc.* 97, 4052–4059.
 71. Park, H., Kim, T.K., Cho, S.W., Jang, H.S., Lee, S.I., and Choi, S.-Y. (2017). Large-scale synthesis of uniform hexagonal boron nitride films by plasma-enhanced atomic layer deposition. *Sci. Rep.* 7, 40091.
 72. Sprenger, J.K., Sun, H., Cavanagh, A.S., Roshko, A., Blanchard, P.T., and George, S.M. (2018). Electron-enhanced atomic layer deposition of boron nitride thin films at room temperature and 100°C. *J. Phys. Chem. C* 122, 9455–9464.
 73. Kovalgin, A.Y., Yang, M., Banerjee, S., Apaydin, R.O., Aarnink, A.A.I., Kinge, S., and Wolters, R.A.M. (2017). Hot-wire assisted ALD: a study powered by in situ spectroscopic ellipsometry. *Adv. Mater. Interfaces* 4, 1700058.
 74. Izyumskaya, N., Demchenko, D.O., Das, S., Özgür, Ü., Avrutin, V., and Morkoç, H. (2017). Recent development of boron nitride towards electronic applications. *Adv. Electron. Mater.* 3, 1600485.
 75. Manzeli, S., Ovchinnikov, D., Pasquier, D., Yaziev, O.V., and Kis, A. (2017). 2D transition metal dichalcogenides. *Nat. Rev. Mater.* 2, 17033.
 76. Kuc, A., and Heine, T. (2015). The electronic structure calculations of two-dimensional transition-metal dichalcogenides in the presence of external electric and magnetic fields. *Chem. Soc. Rev.* 44, 2603–2614.
 77. Tan, L.K., Liu, B., Teng, J.H., Guo, S., Low, H.Y., and Loh, K.P. (2014). Atomic layer deposition of a MoS₂ film. *Nanoscale* 6, 10584–10588.

78. Pyeon, J.J., Kim, S.H., Jeong, D.S., Baek, S.-H., Kang, C.-Y., Kim, J.-S., and Kim, S.K. (2016). Wafer-scale growth of MoS₂ thin films by atomic layer deposition. *Nanoscale* 8, 10792–10798.
79. Kuc, A. (2015). Low-dimensional transition-metal dichalcogenides. In *Chemical Modelling: Volume 11* (The Royal Society of Chemistry), pp. 1–29.
80. Kumar, P., Abuhimad, H., Wahyudi, W., Li, M., Ming, J., and Li, L.-J. (2016). Review—two-dimensional layered materials for energy storage applications. *ECS J. Solid State Sci. Technol.* 5, Q3021–Q3025.
81. Song, I., Park, C., and Choi, H.C. (2015). Synthesis and properties of molybdenum disulfide: from bulk to atomic layers. *RSC Adv.* 5, 7495–7514.
82. Jayabal, S., Wu, J., Chen, J., Geng, D., and Meng, X. (2018). Metallic 1T-MoS₂ nanosheets and their composite materials: preparation, properties and emerging applications. *Mater. Today Energy* 10, 264–279.
83. Jayabal, S., Saranya, G., Wu, J., Liu, Y., Geng, D., and Meng, X. (2017). Understanding the high-electrocatalytic performance of two-dimensional MoS₂ nanosheets and their composite materials. *J. Mater. Chem. A* 5, 24540–24563.
84. Eftekhari, A. (2017). Molybdenum diselenide (MoSe₂) for energy storage, catalysis, and optoelectronics. *Appl. Mater. Today* 8, 1–17.
85. Browning, R., Padigi, P., Solanki, R., Tweet, D.J., Schuele, P., and Evans, D. (2015). Atomic layer deposition of MoS₂ thin films. *Mater. Res. Express* 2, 035006.
86. Valdivia, A., Tweet, D.J., Jr., and Conley, J.F. (2016). Atomic layer deposition of two dimensional MoS₂ on 150 m006D substrates. *J. Vac. Sci. Technol. A* 34, 021515.
87. Huang, Y., Liu, L., Zhao, W., and Chen, Y. (2017). Preparation and characterization of molybdenum disulfide films obtained by one-step atomic layer deposition method. *Thin Solid Films* 624, 101–105.
88. Liu, L., Huang, Y., Sha, J., and Chen, Y. (2017). Layer-controlled precise fabrication of ultrathin MoS₂ films by atomic layer deposition. *Nanotechnology* 28, 195605.
89. Nandi, D.K., Sen, U.K., Choudhury, D., Mitra, S., and Sarkar, S.K. (2014). Atomic layer deposited MoS₂ as a carbon and binder free anode in Li-ion battery. *Electrochim. Acta* 146, 706–713.
90. Jang, Y., Yeo, S., Lee, H.-B.-R., Kim, H., and Kim, S.-H. (2016). Wafer-scale, conformal and direct growth of MoS₂ thin films by atomic layer deposition. *Appl. Surf. Sci.* 365, 160–165.
91. Jin, Z., Shin, S., Kwon, D.H., Han, S.-J., and Min, Y.-S. (2014). Novel chemical route for atomic layer deposition of MoS₂ thin film on SiO₂/Si substrate. *Nanoscale* 6, 14453–14458.
92. Shin, S., Jin, Z., Kwon, D.H., Bose, R., and Min, Y.-S. (2015). High turnover frequency of hydrogen evolution reaction on amorphous MoS₂ thin film directly grown by atomic layer deposition. *Langmuir* 31, 1196–1202.
93. Kwon, D.H., Jin, Z., Shin, S., Lee, W.-S., and Min, Y.-S. (2016). A comprehensive study on atomic layer deposition of molybdenum sulfide for electrochemical hydrogen evolution. *Nanoscale* 8, 7180–7188.
94. Jurca, T., Moody, M.J., Henning, A., Emery, J.D., Wang, B., Tan, J.M., Lohr, T.L., Lauhon, L.J., and Marks, T.J. (2017). Low-temperature atomic layer deposition of MoS₂ films. *Angew. Chem. Int. Ed.* 56, 4991–4995.
95. Cadot, S., Renault, O., Frégnaux, M., Rouchon, D., Nolot, E., Szeto, K., Thieuleux, C., Veyre, L., Okuno, H., Martin, F., and Quadrelli, E.A. (2017). A novel 2-step ALD route to ultra-thin MoS₂ films on SiO₂ through a surface organometallic intermediate. *Nanoscale* 9, 538–546.
96. Mattinen, M., Hatanpää, T., Sarnet, T., Mizohata, K., Meinander, K., King, P.J., Khriachtchev, L., Räisänen, J., Ritala, M., and Leskelä, M. (2017). Atomic layer deposition of crystalline MoS₂ thin films: new molybdenum precursor for low-temperature film growth. *Adv. Mater. Interfaces* 4, 1700123.
97. Mane, A.U., Letourneau, S., Mandia, D.J., Liu, J., Libera, J.A., Lei, Y., Peng, Q., Graugnard, E., and Elam, J.W. (2018). Atomic layer deposition of molybdenum disulfide films using MoF₆ and H₂S. *J. Vac. Sci. Technol. A* 36, 01A125.
98. Yu, Y., Li, C., Liu, Y., Su, L., Zhang, Y., and Cao, L. (2013). Controlled scalable synthesis of uniform, high-quality monolayer and few-layer MoS₂ films. *Sci. Rep.* 3, 1866.
99. Krbal, M., Prikryl, J., Zazpe, R., Dvorak, F., Bures, F., and Macak, J.M. (2018). 2D MoSe₂ structures prepared by atomic layer deposition. *Phys. Status Solidi RRL* 12, 1800023.
100. Browning, R., Kuperman, N., Solanki, R., Kanzyuba, V., and Rouvimov, S. (2016). Large area growth of layered WSe₂ films. *Semicond. Sci. Technol.* 31, 095002.
101. Eftekhari, A. (2017). Tungsten dichalcogenides (WS₂, WSe₂, and WTe₂): materials chemistry and applications. *J. Mater. Chem. A* 5, 18299–18325.
102. Gong, Y., Lei, S., Ye, G., Li, B., He, Y., Keyshar, K., Zhang, X., Wang, Q., Lou, J., Liu, Z., et al. (2015). Two-step growth of two-dimensional WSe₂/MoSe₂ heterostructures. *Nano Lett.* 15, 6135–6141.
103. Schutte, W.J., De Boer, J.L., and Jellinek, F. (1987). Crystal structures of tungsten disulfide and diselenide. *J. Solid State Chem.* 70, 207–209.
104. Scharf, T., Prasad, S.V., Mayer, T., Goeke, R., and Dugger, M. (2004). Atomic layer deposition of tungsten disulfide solid lubricant thin films. *J. Mater. Res.* 19, 3443–3446.
105. Scharf, T.W., Diercks, D.R., Gorman, B.P., Prasad, S.V., and Dugger, M.T. (2009). Atomic layer deposition of tungsten disulfide solid lubricant nanocomposite coatings on rolling element bearings. *Tribol. Trans.* 52, 284–292.
106. Scharf, T.W., Prasad, S.V., Dugger, M.T., Kotula, P.G., Goeke, R.S., and Grubbs, R.K. (2006). Growth, structure, and tribological behavior of atomic layer-deposited tungsten disulfide solid lubricant coatings with applications to MEMS. *Acta Mater.* 54, 4731–4743.
107. Delabie, A., Caymax, M., Groven, B., Heyne, M., Haesevoets, K., Meersschaert, J., Nuytten, T., Bender, H., Conard, T., Verdonck, P., et al. (2015). Low temperature deposition of 2D WS₂ layers from WF₆ and H₂S precursors: impact of reducing agents. *Chem. Commun.* 51, 15692–15695.
108. Groven, B., Heyne, M., Nalin Mehta, A., Bender, H., Nuytten, T., Meersschaert, J., Conard, T., Verdonck, P., Van Elshocht, S., Vandervorst, W., et al. (2017). Plasma-enhanced atomic layer deposition of two-dimensional WS₂ from WF₆, H₂ plasma, and H₂S. *Chem. Mater.* 29, 2927–2938.
109. Groven, B., Mehta, A.N., Bender, H., Smets, Q., Meersschaert, J., Franquet, A., Conard, T., Nuytten, T., Verdonck, P., Vandervorst, W., et al. (2018). Nucleation mechanism during WS₂ plasma enhanced atomic layer deposition on amorphous Al₂O₃ and sapphire substrates. *J. Vac. Sci. Technol. A* 36, 01A105.
110. Groven, B., Nalin Mehta, A., Bender, H., Meersschaert, J., Nuytten, T., Verdonck, P., Conard, T., Smets, Q., Schram, T., Schoenaers, B., et al. (2018). Two-dimensional crystal grain size tuning in WS₂ atomic layer deposition: an insight in the nucleation mechanism. *Chem. Mater.* 30, 7648–7663.
111. Nandi, D.K., Sen, U.K., Dhara, A., Mitra, S., and Sarkar, S.K. (2016). Intercalation based tungsten disulfide (WS₂) Li-ion battery anode grown by atomic layer deposition. *RSC Adv.* 6, 38024–38032.
112. Sun, Y., Chai, Z., Lu, X., and He, D. (2017). A direct atomic layer deposition method for growth of ultra-thin lubricant tungsten disulfide films. *Sci. China Technol. Sci.* 60, 51–57.
113. Yeo, S., Nandi, D.K., Rahul, R., Kim, T.H., Shong, B., Jang, Y., Bae, J.-S., Han, J.W., Kim, S.-H., and Kim, H. (2018). Low-temperature direct synthesis of high quality WS₂ thin films by plasma-enhanced atomic layer deposition for energy related applications. *Appl. Surf. Sci.* 459, 596–605.
114. Browning, R., Plachinda, P., Padigi, P., Solanki, R., and Rouvimov, S. (2016). Growth of multiple WS₂/SnS layered semiconductor heterojunctions. *Nanoscale* 8, 2143–2148.
115. Putri, Y.E., Said, S.M., and Diantoro, M. (2019). Nanoarchitected titanium complexes for thermal mitigation in thermoelectric materials. *Renew. Sust. Energy Rev.* 101, 346–360.
116. Pore, V., Ritala, M., and Leskelä, M. (2007). Atomic layer deposition of titanium disulfide thin films. *Chem. Vap. Depos.* 13, 163–168.
117. Mahuli, N., and Sarkar, S.K. (2015). Atomic layer deposition of titanium sulfide and its application in extremely thin absorber solar cells. *J. Vac. Sci. Technol. A* 33, 01A150.
118. Xu, X., Liu, W., Kim, Y., and Cho, J. (2014). Nanostructured transition metal sulfides for lithium ion batteries: progress and challenges. *Nano Today* 9, 604–630.

119. Mattinen, M., Popov, G., Vehkamäki, M., King, P.J., Mizohata, K., Jalkanen, P., Räisänen, J., Leskelä, M., and Ritala, M. (2019). Atomic layer deposition of emerging 2D semiconductors, HfS_2 and ZrS_2 , for optoelectronics. *Chem. Mater.* 31, 5713–5724.
120. Xu, C., Brown, P.A., and Shuford, K.L. (2015). Strain-induced semimetal-to-semiconductor transition and indirect-to-direct band gap transition in monolayer 1T- TiS_2 . *RSC Adv.* 5, 83876–83879.
121. Conroy, L.E., and Park, K.C. (1968). Electrical properties of the Group IV disulfides, titanium disulfide, zirconium disulfide, hafnium disulfide and tin disulfide. *Inorg. Chem.* 7, 459–463.
122. Let, A.L., Mainwaring, D.E., Rix, C., and Murugaraj, P. (2008). Thio sol–gel synthesis of titanium disulfide thin films and powders using titanium alkoxide precursors. *J. Non Cryst. Solids* 354, 1801–1807.
123. Umebayashi, T., Yamaki, T., Itoh, H., and Asai, K. (2002). Band gap narrowing of titanium dioxide by sulfur doping. *Appl. Phys. Lett.* 81, 454–456.
124. Hämäläinen, J., Mattinen, M., Mizohata, K., Meinander, K., Vehkamäki, M., Räisänen, J., Ritala, M., and Leskelä, M. (2018). Atomic layer deposition of rhenium disulfide. *Adv. Mater.* 30, 1703622.
125. Zhen, Y.-X., Yang, M., Zhang, H., Fu, G.-S., Wang, J.-L., Wang, S.-F., and Wang, R.-N. (2017). Ultrahigh power factors in P-type 1T- ZrX_2 ($\text{X}=\text{S}, \text{Se}$) single layers. *Sci. Bull.* 62, 1530–1537.
126. Abdulsalam, M., and Joubert, D.P. (2016). Optical spectrum and excitons in bulk and monolayer MX_2 ($\text{M}=\text{Zr}, \text{Hf}$; $\text{X}=\text{S}, \text{Se}$). *Phys. Status Solidi B* 253, 705–711.
127. Tributsch, H. (1981). Photoelectrochemistry of layer-type zirconium disulfide. *J. Electrochem. Soc.* 128, 1261–1269.
128. Zhang, W., Huang, Z., Zhang, W., and Li, Y. (2014). Two-dimensional semiconductors with possible high room temperature mobility. *Nano Res.* 7, 1731–1737.
129. Li, Y., Kang, J., and Li, J. (2014). Indirect-to-direct band gap transition of the ZrS_2 monolayer by strain: first-principles calculations. *RSC Adv.* 4, 7396–7401.
130. Rahman, M., Davey, K., and Qiao, S.-Z. (2017). Advent of 2D rhenium disulfide (ReS_2): fundamentals to applications. *Adv. Funct. Mater.* 27, 1606129.
131. Tongay, S., Sahin, H., Ko, C., Luce, A., Fan, W., Liu, K., Zhou, J., Huang, Y.-S., Ho, C.-H., Yan, J., et al. (2014). Monolayer behaviour in bulk ReS_2 due to electronic and vibrational decoupling. *Nat. Commun.* 5, 3252.
132. Shao, Y., Guo, Z., Li, H., Su, Y., and Wang, X. (2017). Atomic layer deposition of iron sulfide and its application as a catalyst in the hydrogenation of azobenzenes. *Angew. Chem. Int. Ed.* 56, 3226–3231.
133. Mahuli, N., and Sarkar, S.K. (2016). Atomic layer deposition of NiS and its application as cathode material in dye sensitized solar cell. *J. Vac. Sci. Technol. A* 34, 01A142.
134. Kuhn, S.J., Kidder, M.K., Parker, D.S., dela Cruz, C., McGuire, M.A., Chance, W.M., Li, L., Debeer-Schmitt, L., Ermentrout, J., Littrell, K.C., et al. (2017). Structure and property correlations in FeS . *Phys. C Supercond.* 534, 29–36.
135. Liu, S., Shi, Q., Tong, J., Li, S., and Li, M. (2014). Controlled synthesis of spherical α - NiS and urchin-like β - NiS microstructures. *J. Exp. Nanosci.* 9, 475–481.
136. Krishnakumar, S.R., Shanthi, N., and Sarma, D.D. (2002). Electronic structure of millerite NiS . *Phys. Rev. B* 66, 115105.
137. Browning, R., Kuperman, N., Moon, B., and Solanki, R. (2017). Atomic layer growth of InSe and Sb_2Se_3 layered semiconductors and their heterostructure. *Electronics* 6, 27.
138. Amory, C., Bernède, J.C., and Marsillac, S. (2003). Study of a growth instability of γ - In_2Se_3 . *J. Appl. Phys.* 94, 6945–6948.
139. Gomes da Costa, P., Dandrea, R.G., Wallis, R.F., and Balkanski, M. (1993). First-principles study of the electronic structure of γ - InSe and β - InSe . *Phys. Rev. B* 48, 14135–14141.
140. Huang, W., Gan, L., Li, H., Ma, Y., and Zhai, T. (2016). 2D layered group IIIA metal chalcogenides: synthesis, properties and applications in electronics and optoelectronics. *CrystEngComm* 18, 3968–3984.
141. Hatanpää, T., Pore, V., Ritala, M., and Leskelä, M. (2009). Alkylsilyl compounds of selenium and tellurium: new precursors for ALD. *ECS Trans.* 25, 609–616.
142. Pore, V., Hatanpää, T., Ritala, M., and Leskelä, M. (2009). Atomic layer deposition of metal tellurides and selenides using alkylsilyl compounds of tellurium and selenium. *J. Am. Chem. Soc.* 131, 3478–3480.
143. Kim, S.B., Sinsermsuksakul, P., Hock, A.S., Pike, R.D., and Gordon, R.G. (2014). Synthesis of N-heterocyclic stannylene (Sn(II)) and germylene (Ge(II)) and a Sn(II) amidinate and their application as precursors for atomic layer deposition. *Chem. Mater.* 26, 3065–3073.
144. Kim, W., Yoo, S., Yoo, C., Park, E.-S., Jeon, J., Kwon, Y.-J., Woo, K.S., Kim, H.-J., Lee, Y.-K., and Hwang, C.S. (2018). Atomic layer deposition of GeSe films using HGeCl_3 and $[(\text{CH}_3)_3\text{Si}]_2\text{Se}$ with the discrete feeding method for the ovonic threshold switch. *Nanotechnology* 29, 365202.
145. Knapas, K., Hatanpää, T., Ritala, M., and Leskelä, M. (2010). In situ reaction mechanism studies on atomic layer deposition of Sb_2Te_3 and GeTe from $(\text{Et}_3\text{Si})_2\text{Te}$ and chlorides. *Chem. Mater.* 22, 1386–1391.
146. Sarnet, T., Pore, V., Hatanpää, T., Ritala, M., Leskelä, M., Schrott, A., Zhu, Y., Raoux, S., and Cheng, H.-Y. (2011). Atomic layer deposition and characterization of GeTe thin films. *J. Electrochem. Soc.* 158, D694–D697.
147. Gwon, T., Eom, T., Yoo, S., Lee, H.-K., Cho, D.-Y., Kim, M.-S., Buchanan, I., Xiao, M., Ivanov, S., and Hwang, C.S. (2016). Atomic layer deposition of GeTe films using $\text{Ge[N(Si(CH}_3)_3)_2]_2$, $[(\text{CH}_3)_3\text{Si}]_2\text{Te}$, and methanol. *Chem. Mater.* 28, 7158–7166.
148. Gwon, T., Mohamed, A.Y., Yoo, C., Park, E.-s., Kim, S., Yoo, S., Lee, H.-K., Cho, D.-Y., and Hwang, C.S. (2017). Structural analyses of phase stability in amorphous and partially crystallized Ge -rich GeTe films prepared by atomic layer deposition. *ACS Appl. Mater. Interfaces* 9, 41387–41396.
149. Gwon, T., Eom, T., Yoo, S., Yoo, C., Park, E.-S., Kim, S., Kim, M.-S., Buchanan, I., Xiao, M., Ivanov, S., and Hwang, C.S. (2017). Atomic layer deposition of GeTe and Ge-Sb-Te films using HGeCl_3 , $\text{Sb(OC}_2\text{H}_5)_3$, and $[(\text{CH}_3)_3\text{Si}]_2\text{Te}$ and their reaction mechanisms. *Chem. Mater.* 29, 8065–8072.
150. Cheng, L., Adinolfi, V., Weeks, S.L., Barabash, S.V., and Littau, K.A. (2019). Conformal deposition of GeTe films with tunable Te composition by atomic layer deposition. *J. Vac. Sci. Technol. A* 37, 020907.
151. Li, W., Sabino, F.P., Crasto de Lima, F., Wang, T., Miwa, R.H., and Janotti, A. (2018). Large disparity between optical and fundamental band gaps in layered In_2Se_3 . *Phys. Rev. B* 98, 165134.
152. Ramasamy, P., Kwak, D., Lim, D.-H., Ra, H.-S., and Lee, J.-S. (2016). Solution synthesis of GeS and GeSe nanosheets for high-sensitivity photodetectors. *J. Mater. Chem. C* 4, 479–485.
153. Zhang, S., Wang, N., Liu, S., Huang, S., Zhou, W., Cai, B., Xie, M., Yang, Q., Chen, X., and Zeng, H. (2016). Two-dimensional GeS with tunable electronic properties via external electric field and strain. *Nanotechnology* 27, 274001.
154. Zhou, X., Zhang, Q., Gan, L., Li, H., Xiong, J., and Zhai, T. (2016). Booming development of Group IV–VI semiconductors: fresh blood of 2D family. *Adv. Sci.* 3, 1600177.
155. Makinistian, L., and Albanesi, E.A. (2007). Ab initio calculations of the electronic and optical properties of germanium selenide. *J. Phys. Condens. Matter* 19, 186211.
156. Chopra, K.L., and Bahl, S.K. (1969). Amorphous versus crystalline GeTe films. I. Growth and structural behavior. *J. Appl. Phys.* 40, 4171–4178.
157. Yang, Y.B., Dash, J.K., Xiang, Y., Wang, Y., Shi, J., Dinolfo, P.H., Lu, T.M., and Wang, G.C. (2016). Tuning the phase and optical properties of ultrathin SnS_2 films. *J. Phys. Chem. C* 120, 13199–13214.
158. Kim, J.-Y., and George, S.M. (2010). Tin monosulfide thin films grown by atomic layer deposition using tin 2,4-pentanedionate and hydrogen sulfide. *J. Phys. Chem. C* 114, 17597–17603.
159. Sinsermsuksakul, P., Heo, J., Noh, W., Hock, A.S., and Gordon, R.G. (2011). Atomic layer deposition of tin monosulfide thin films. *Adv. Energy Mater.* 1, 1116–1125.
160. Baek, I.-H., Pyeon, J.-J., Song, Y.G., Chung, T.-M., Kim, H.-R., Baek, S.-H., Kim, J.-S., Kang, C.-Y., Choi, J.-W., Hwang, C.S., et al. (2017). Synthesis of SnS thin films by atomic layer deposition at low temperatures. *Chem. Mater.* 29, 8100–8110.
161. Ahn, J.-H., Lee, M.-J., Heo, H., Sung, J.-H., Kim, K., Hwang, H., and Jo, M.-H. (2015).

- Deterministic two-dimensional polymorphism growth of hexagonal n-type SnS_2 and orthorhombic p-type SnS crystals. *Nano Lett.* 15, 3703–3708.
162. Ham, G., Shin, S., Park, J., Choi, H., Kim, J., Lee, Y.-A., Seo, H., and Jeon, H. (2013). Tuning the electronic structure of tin sulfides grown by atomic layer deposition. *ACS Appl. Mater. Interfaces* 5, 8889–8896.
 163. Ham, G., Shin, S., Park, J., Lee, J., Choi, H., Lee, S., and Jeon, H. (2016). Engineering the crystallinity of tin disulfide deposited at low temperatures. *RSC Adv.* 6, 54069–54075.
 164. Lee, S., Shin, S., Ham, G., Lee, J., Choi, H., Park, H., and Jeon, H. (2017). Characteristics of layered tin disulfide deposited by atomic layer deposition with H_2S annealing. *AIP Adv.* 7, 045307.
 165. An, C., Tang, K., Shen, G., Wang, C., Yang, Q., Hai, B., and Qian, Y. (2002). Growth of belt-like SnS crystals from ethylenediamine solution. *J. Cryst. Growth* 244, 333–338.
 166. Ettema, A.R.H.F., de Groot, R.A., Haas, C., and Turner, T.S. (1992). Electronic structure of SnS deduced from photoelectron spectra and band-structure calculations. *Phys. Rev. B* 46, 7363–7373.
 167. Deshpande, N.G., Sagade, A.A., Gudage, Y.G., Lokhande, C.D., and Sharma, R. (2007). Growth and characterization of tin disulfide (SnS_2) thin film deposited by successive ionic layer adsorption and reaction (SILAR) technique. *J. Alloys Compd.* 436, 421–426.
 168. Bilousov, O.V., Ren, Y., Törndahl, T., Donzel-Gargand, O., Ericson, T., Platzer-Björkman, C., Edöf, M., and Häggglund, C. (2017). Atomic layer deposition of cubic and orthorhombic phase tin monosulfide. *Chem. Mater.* 29, 2969–2978.
 169. Thimsen, E., Riha, S.C., Baryshev, S.V., Martinson, A.B.F., Elam, J.W., and Pellin, M.J. (2012). Atomic layer deposition of the quaternary chalcogenide $\text{Cu}_2\text{ZnSnS}_4$. *Chem. Mater.* 24, 3188–3196.
 170. Lee, J., Lee, J., Ham, G., Shin, S., Park, J., Choi, H., Lee, S., Kim, J., Sul, O., Lee, S., and Jeon, H. (2017). Improved electrical properties of atomic layer deposited tin disulfide at low temperatures using ZrO_2 layer. *AIP Adv.* 7, 025311.
 171. Wondeok, S., Seokyeon, S., Giyul, H., Juhyun, L., Seungjin, L., Hyeongsu, C., and Hyeongtag, J. (2017). Thickness-dependent structure and properties of SnS_2 thin films prepared by atomic layer deposition. *Jpn. J. Appl. Phys.* 56, 031201.
 172. Mattinen, M., King, P.J., Khriachtchev, L., Meinander, K., Gibbon, J.T., Dhanak, V.R., Räisänen, J., Ritala, M., and Leskelä, M. (2018). Low-temperature wafer-scale deposition of continuous 2D SnS_2 films. *Small* 14, 1800547.
 173. Bacaksiz, C., Cahangirov, S., Rubio, A., Senger, R.T., Peeters, F.M., and Sahin, H. (2016). Bilayer SnS_2 : tunable stacking sequence by charging and loading pressure. *Phys. Rev. B* 93, 125403.
 174. Zheng, Y., Xia, M., Cheng, Y., Rao, F., Ding, K., Liu, W., Jia, Y., Song, Z., and Feng, S. (2016). Direct observation of metastable face-centered cubic Sb_2Te_3 crystal. *Nano Res.* 9, 3453–3462.
 175. Kim, M., Kim, C.H., Kim, H.-S., and Ihm, J. (2012). Topological quantum phase transitions driven by external electric fields in Sb_2Te_3 thin films. *Proc. Natl. Acad. Sci. U S A* 109, 671–674.
 176. Zhang, K., Nminibapiel, D., Tangirala, M., Baumgart, H., and Kochergin, V. (2013). Fabrication of Sb_2Te_3 and Bi_2Te_3 multilayer composite films by atomic layer deposition. *ECS Trans.* 50, 3–9.
 177. Zheng, L., Cheng, X., Cao, D., Wang, Q., Wang, Z., Xia, C., Shen, L., Yu, Y., and Shen, D. (2015). Direct growth of Sb_2Te_3 on graphene by atomic layer deposition. *RSC Adv.* 5, 40007–40011.
 178. Gusmão, R., Sofer, Z., Luxa, J., and Pummer, M. (2019). Antimony chalcogenide van der Waals nanostructures for energy conversion and storage. *ACS Sustain. Chem. Eng.* 7, 15790–15798.
 179. Liu, B., Song, Z., Feng, S., and Chen, B. (2005). Characteristics of chalcogenide nonvolatile memory nano-cell-element based on Sb_2Te_3 material. *Microelectron. Eng.* 82, 168–174.
 180. Zastrow, S., Gooth, J., Boehnert, T., Heiderich, S., Toellner, W., Heimann, S., Schulz, S., and Nielsch, K. (2013). Thermoelectric transport and Hall measurements of low defect Sb_2Te_3 thin films grown by atomic layer deposition. *Semicond. Sci. Technol.* 28, 035010.
 181. Gu, D., Nminibapiel, D., Baumgart, H., Robinson, H., and Kochergin, V. (2011). Atomic layer deposition of antimony telluride thin films using $(\text{Me}_3\text{Si})_2\text{Te}$ with SbCl_3 as precursors. *ECS Trans.* 58, 59–66.
 182. Nminibapiel, D., Zhang, K., Tangirala, M., Baumgart, H., Chakravadhanula, V.S., Kübel, C., and Kochergin, V. (2013). Microstructure analysis of ALD $\text{Bi}_2\text{Te}_3/\text{Sb}_2\text{Te}_3$ thermoelectric nanolaminates. *ECS Trans.* 58, 59–66.
 183. Eom, T., Choi, S., Choi, B.J., Lee, M.H., Gwon, T., Rha, S.H., Lee, W., Kim, M.-S., Xiao, M., Buchanan, I., et al. (2012). Conformal formation of $(\text{GeTe})_{1-x}(\text{Sb}_2\text{Te}_3)_x$ layers by atomic layer deposition for nanoscale phase change memories. *Chem. Mater.* 24, 2099–2110.
 184. Han, B., Kim, Y.-J., Park, J.-M., Yusup, L.L., Ishii, H., Lansalot-Matras, C., and Lee, W.-J. (2016). Reaction mechanism underlying atomic layer deposition of antimony telluride thin films. *J. Nanosci. Nanotechnol.* 16, 4924–4928.
 185. Mahuli, N., Saha, D., and Sarkar, S.K. (2017). Atomic layer deposition of p-type Bi_2S_3 . *J. Phys. Chem. C* 121, 8136–8144.
 186. Sarnet, T., Hatanpää, T., Vehkamäki, M., Flyktman, T., Ahopelto, J., Mizohata, K., Ritala, M., and Leskelä, M. (2015). $(\text{Et}_3\text{Si})_2\text{Se}$ as a precursor for atomic layer deposition: growth analysis of thermoelectric Bi_2Se_3 . *J. Mater. Chem. C* 3, 4820–4828.
 187. Sarnet, T., Hatanpää, T., Puukilainen, E., Mattinen, M., Vehkamäki, M., Mizohata, K., Ritala, M., and Leskelä, M. (2015). Atomic layer deposition and characterization of Bi_2Te_3 thin films. *J. Phys. Chem. A* 119, 2298–2306.
 188. Bae, C., Böhnert, T., Gooth, J., Lim, S., Lee, S., Kim, H., Heimann, S., Schulz, S., Shin, H., and Nielsch, K. (2014). Thermopower engineering of Bi_2Te_3 without alloying: the interplay between nanostructuring and defect activation. *Semicond. Sci. Technol.* 29, 064003.
 189. Rusek, M., Komossa, T., Bendt, G., and Schulz, S. (2017). Bismuth amides as promising ALD precursors for Bi_2Te_3 films. *J. Cryst. Growth* 470, 128–134.
 190. Yang, M., Luo, Y.Z., Zeng, M.G., Shen, L., Lu, Y.H., Zhou, J., Wang, S.J., Sou, I.K., and Feng, Y.P. (2017). Pressure induced topological phase transition in layered Bi_2S_3 . *Phys. Chem. Chem. Phys.* 19, 29372–29380.
 191. Efthimiopoulos, I., Kemichick, J., Zhou, X., Khare, S.V., Ikuta, D., and Wang, Y. (2014). High-pressure studies of Bi_2S_3 . *J. Phys. Chem. A* 118, 1713–1720.
 192. Tumelero, M.A., Benetti, L.C., Isoppo, E., Faccio, R., Zangari, G., and Pasa, A.A. (2016). Electrodeposition and ab initio studies of metastable orthorhombic Bi_2Se_3 : a novel semiconductor with bandgap for photovoltaic applications. *J. Phys. Chem. C* 120, 11797–11806.
 193. Vilaplana, R., Santamaría-Pérez, D., Gomis, O., Manjón, F.J., González, J., Segura, A., Muñoz, A., Rodríguez-Hernández, P., Pérez-González, E., Marín-Borrás, V., et al. (2011). Structural and vibrational study of Bi_2Se_3 under high pressure. *Phys. Rev. B* 84, 184110.
 194. Kadel, K., Kumari, L., Li, W.Z., Huang, J.Y., and Provencio, P.P. (2010). Synthesis and thermoelectric properties of Bi_2Se_3 nanostructures. *Nanoscale Res. Lett.* 6, 57.
 195. Mishra, S.K., Satpathy, S., and Jepsen, O. (1997). Electronic structure and thermoelectric properties of bismuth telluride and bismuth selenide. *J. Phys. Condens. Matter* 9, 461–470.
 196. Bahabri, F. (2012). Investigation of the structural and optical properties of bismuth telluride (Bi_2Te_3) thin films. *Life Sci. J.* 9, 290–294.
 197. Shimizu, R., Yamamoto, K., Suzuki, T., Ohsawa, T., Shiraki, S., and Hitosugi, T. (2015). Low-temperature deposition of meta-stable $\beta\text{-MoO}_3(011)$ epitaxial thin films using step-and-terrace substrates. *Thin Solid Films* 595, 153–156.
 198. Diskus, M., Nilsen, O., and Fjellvåg, H. (2011). Growth of thin films of molybdenum oxide by atomic layer deposition. *J. Mater. Chem.* 21, 705–710.
 199. Diskus, M., Nilsen, O., Fjellvåg, H., Diplas, S., Beato, P., Harvey, C., Lantman, E.v.S., and Weckhuysen, B.M. (2012). Combination of characterization techniques for atomic layer deposition MoO_3 coatings: from the amorphous to the orthorhombic $\alpha\text{-MoO}_3$ crystalline phase. *J. Vac. Sci. Technol. A* 30, 01A107.
 200. Dai, T., Ren, Y., Qian, L., and Liu, X. (2018). Characterization of molybdenum oxide thin films grown by atomic layer deposition. *J. Electron. Mater.* 47, 6709–6715.
 201. Vos, M.F.J., Maccio, B., Thissen, N.F.W., Bol, A.A., and Kessels, W.M.M. (2016). Atomic

layer deposition of molybdenum oxide from (NtBu)₂(NMe₂)-Mo and O₂ plasma. *J. Vac. Sci. Technol. A* 34, 01A103.

202. Wei, Z., Hai, Z., Akbari, M.K., Qi, D., Xing, K., Zhao, Q., Verpoort, F., Hu, J., Hyde, L., and Zhuiykov, S. (2018). Atomic layer deposition-developed two-dimensional α -MoO₃ windows excellent hydrogen peroxide electrochemical sensing capabilities. *Sens. Actuators B Chem.* 262, 334–344.
203. Yao, D.D., Ou, J.Z., Latham, K., Zhuiykov, S., O'Mullane, A.P., and Kalantar-zadeh, K. (2012). Electrodeposited α - and β -phase MoO₃ films and investigation of their gasochromic properties. *Cryst. Growth Des.* 12, 1865–1870.
204. Liu, H., Yang, R.B., Yang, W., Jin, Y., and Lee, C.J.J. (2018). Atomic layer deposition and post-growth thermal annealing of ultrathin MoO₃ layers on silicon substrates: formation of surface nanostructures. *Appl. Surf. Sci.* 439, 583–588.
205. Bertuch, A., Sundaram, G., Saly, M., Moser, D., and Kanjolia, R. (2014). Atomic layer deposition of molybdenum oxide using bis(tert-butylimido)bis(dimethylamido) molybdenum. *J. Vac. Sci. Technol. A* 32, 01A119.
206. Xu, H., Akbari, M.K., Hai, Z., Wei, Z., Hyde, L., Verpoort, F., Xue, C., and Zhuiykov, S. (2018). Ultra-thin MoO₃ film goes wafer-scaled nano-architectonics by atomic layer deposition. *Mater. Des.* 149, 135–144.
207. Tseng, Y.-C., Mane, A.U., Elam, J.W., and Darling, S.B. (2012). Ultrathin molybdenum oxide anode buffer layer for organic photovoltaic cells formed using atomic layer deposition. *Sol. Energy Mater. Sol. Cells* 99, 235–239.
208. Malm, J., Sajavaara, T., and Karppinen, M. (2012). Atomic layer deposition of WO₃ thin films using W(CO)₆ and O₃ precursors. *Chem. Vap. Depos.* 18, 245–248.
209. Nandi, D.K., and Sarkar, S.K. (2014). Atomic layer deposition of tungsten oxide for solar cell application. *Energy Procedia* 54, 782–788.
210. Mamun, M.A., Zhang, K., Baumgart, H., and Elmestafa, A.A. (2014). Nanomechanical properties of tungsten trioxide (WO₃) grown by atomic layer deposition. *ECS Trans.* 64, 211–217.
211. Mamun, M.A., Zhang, K., Baumgart, H., and Elmestafa, A.A. (2015). Nanomechanical and morphological characterization of tungsten trioxide (WO₃) thin films grown by atomic layer deposition. *ECS J. Solid State Sci. Technol.* 4, P398–P401.
212. Jackson, D.H.K., Dunn, B.A., Guan, Y., and Kuech, T.F. (2014). Tungsten hexacarbonyl and hydrogen peroxide as precursors for the growth of tungsten oxide thin films on titania nanoparticles. *AIChE J.* 60, 1278–1286.
213. Hai, Z., Akbari, M.K., Xue, C., Xu, H., Depuydt, S., and Zhuiykov, S. (2017). Photodetector with superior functional capabilities based on monolayer WO₃ developed by atomic layer deposition. *Sens. Actuators B Chem.* 245, 954–962.
214. Zhuiykov, S., Hyde, L., Hai, Z., Akbari, M.K., Kats, E., Detavernier, C., Xue, C., and Xu, H. (2017). Atomic layer deposition-enabled single layer of tungsten trioxide across a large area. *Appl. Mater. Today* 6, 44–53.
215. Hai, Z., Akbari, M.K., Xue, C., Xu, H., Hyde, L., and Zhuiykov, S. (2017). Wafer-scaled monolayer WO₃ windows ultra-sensitive, extremely-fast and stable UV-A photodetection. *Appl. Surf. Sci.* 405, 169–177.
216. Mouat, A.R., Mane, A.U., Elam, J.W., Delferro, M., Marks, T.J., and Stair, P.C. (2016). Volatile hexavalent oxo-amidinate complexes: molybdenum and tungsten precursors for atomic layer deposition. *Chem. Mater.* 28, 1907–1919.
217. Seghete, D., Rayner, G.B., Cavanagh, A.S., Anderson, V.R., and George, S.M. (2011). Molybdenum atomic layer deposition using MoF₆ and Si₂H₆ as the reactants. *Chem. Mater.* 23, 1668–1678.
218. Wang, F., Di Valentin, C., and Pacchioni, G. (2011). Electronic and structural properties of WO₃: a systematic hybrid DFT study. *J. Phys. Chem. C* 115, 8345–8353.
219. Wang, J., and Liu, C.-J. (2015). Preparation of 2D WO₃ nanomaterials with enhanced catalytic activities: current status and perspective. *ChemBioEng Rev.* 2, 335–350.
220. Azam, A., Kim, J., Park, J., Novak, T.G., Tiwari, A.P., Song, S.H., Kim, B., and Jeon, S. (2018). Two-dimensional WO₃ nanosheets chemically converted from layered WS₂ for high-performance electrochromic devices. *Nano Lett.* 18, 5646–5651.
221. Liu, R., Lin, Y., Chou, L.-Y., Sheehan, S.W., He, W., Zhang, F., Hou, H.J.M., and Wang, D. (2011). Water splitting by tungsten oxide prepared by atomic layer deposition and decorated with an oxygen-evolving catalyst. *Angew. Chem. Int. Ed.* 50, 499–502.
222. Choi, W., Choudhary, N., Han, G.H., Park, J., Akinwande, D., and Lee, Y.H. (2017). Recent development of two-dimensional transition metal dichalcogenides and their applications. *Mater. Today* 20, 116–130.
223. Meng, X., Libera, J.A., Fister, T.T., Zhou, H., Hedlund, J.K., Fenter, P., and Elam, J.W. (2014). Atomic layer deposition of gallium sulfide films using hexakis(dimethylamido) digallium and hydrogen sulfide. *Chem. Mater.* 26, 1029–1039.
224. Langereis, E., Heil, S.B.S., Knoops, H.C.M., Keuning, W., van de Sanden, M.C.M., and Kessels, W.M.M. (2009). In situ spectroscopic ellipsometry as a versatile tool for studying atomic layer deposition. *J. Phys. D Appl. Phys.* 42, 073001.
225. Temperton, R.H., Gibson, A., and O'Shea, J.N. (2019). In situ XPS analysis of the atomic layer deposition of aluminium oxide on titanium dioxide. *Phys. Chem. Chem. Phys.* 21, 1393–1398.
226. Devloo-Casier, K., Ludwig, K.F., Detavernier, C., and Dendooven, J. (2014). In situ synchrotron based x-ray techniques as monitoring tools for atomic layer deposition. *J. Vac. Sci. Technol. A* 32, 010801.
227. Min, K.H., Sinclair, R., Park, I.S., Kim, S.T., and Chung, U.I. (2005). Crystallization behaviour of ALD-Ta₂O₅ thin films: the application of in-situ TEM. *Philos. Mag.* 85, 2049–2063.